

CHAPTER ONE

THEORIES OF CONSUMPTION

INTRODUCTION

Consumption is the sole end and purpose of all production.

Adam Smith

How do households decide how much of their income to consume today and how much to save for the future?

This is a microeconomic question because it addresses the behavior of individual decision makers. Yet its answer has macroeconomic consequences. As we have seen in previous chapters, households' consumption decisions affect the way the economy as a whole behaves both in the long run and in the short run.

- ❖ Consumption takes **two-thirds of GDP**, so fluctuations in consumption are a key element of booms and recessions. For instance, the IS –LM Model shows that changes in consumers' spending plans can be a source of shocks to the economy.
- ❖ **Then what determines aggregate consumption?**

Since the birth of macroeconomics, many economists have written about the theory of consumer behavior and suggested alternative ways of interpreting the data on consumption and income.

- ❖ This chapter presents the views of **five prominent** economists to show the diverse approaches to explaining consumption.

1.1. John Maynard Keynes and the Consumption Function

- ❖ The consumption function was central to Keynes' theory of economic fluctuations.
 - ❖ **Keynes** made **three conjectures** about the consumption function based on his casual observation.
1. The **marginal propensity to consume (MPC)** — the amount consumed out of an additional dollar of income is between zero and one. He claimed that the “**fundamental psychological law**” is that out of every dollar of earned income, people will consume part of it and save

the rest. I.e., as income increases consumption also increases, but the increase in consumption is not as much as the increase in income.

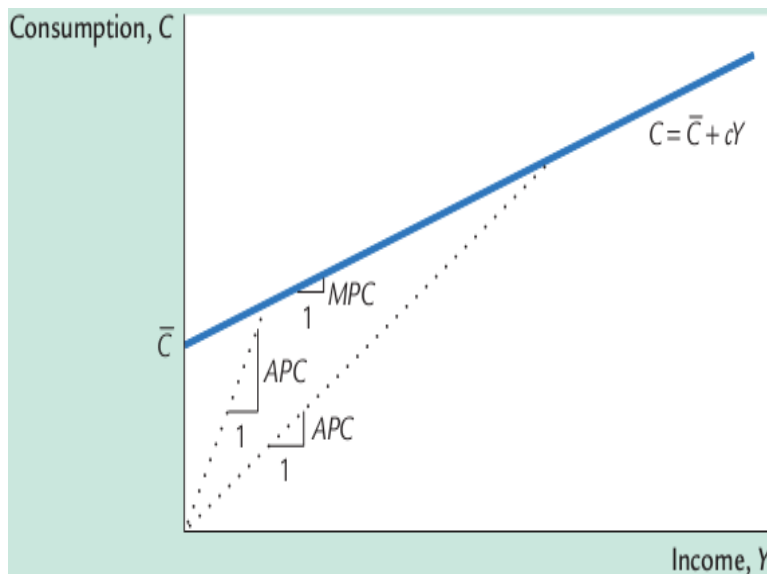
2. The **average propensity to consume (APC)** - the ratio of consumption to income falls as income rises.
 - ✓ Keynes believed that saving was a luxury, so he expected the rich to save a higher proportion of their income than the poor.
3. **Income is the primary determinant of consumption and that the interest rate does not have an important role.**

On the basis of these three conjectures, the Keynesian consumption function is often written as:

$$C = \bar{C} + cY \quad \bar{C} > 0, \quad 0 < c < 1,$$

, where C is consumption, Y is disposable income, $\bar{C}$ is autonomous consumption, and c is the marginal propensity to consume. This consumption function is graphed as follows.

Figure: 1.1: The Keynesian Consumption Function



This figure graphs a consumption function with the three properties that Keynes conjectured.

First, the marginal propensity to consume c is between zero and one. **Second**, the APC falls as income rises.

Third, consumption is determined by current income.

Note: The marginal propensity to consume, MPC, is the slope of the consumption function. The average propensity to consume, $APC = C/Y$, equals the slope of a line drawn from the origin to a point on the consumption function.

Notice that the above consumption function exhibits the three properties that Keynes posited.

- ✓ It satisfies Keynes's first property because the marginal propensity to consume c is between zero and one, so that higher income leads to higher consumption and also to higher saving.
- ✓ This consumption function satisfies Keynes's second property because the average propensity to consume, APC is

$$APC = C/Y = \bar{C}/Y + c.$$

As Y rises, $\bar{C}/Y$ falls, and so the average propensity to consume C/Y falls.

- And finally, Keynes's third property is satisfied because the interest rate is not included in this equation as a determinant of consumption.

1.1.1 The Early Empirical Successes

- ✚ Soon after Keynes proposed the consumption function, economists began collecting and examining data to test his conjectures.
- ✚ The earliest studies indicated that the Keynesian consumption function is a good approximation of how consumers behave.
 - In some of these studies, researchers surveyed **households and collected data** on consumption and income.
 - They found that households with higher income consumed more, which confirms that the marginal propensity to consume is greater than zero.
 - They also found that households with higher income saved more, which confirms that the marginal propensity to consume is less than one.
 - In addition, these researchers found that higher-income households saved a larger fraction of their income, which **confirms** that the **APC falls as income rises**.
 - Thus, these data verified Keynes's conjectures about the marginal and average propensities to consume.
 - In other studies, researchers examined **aggregate data on consumption and income for the period between the two world wars (i.e., short time-series aggregate data)**.
 - These data also supported the Keynesian consumption function.
 - Finally, because the correlation between income and consumption was so strong, no other variable appeared to be important for explaining consumption.

- Thus, the data also confirmed Keynes's third conjecture that income is the primary determinant of how much people choose to consume.

1.1.2 Secular Stagnation, Simon Kuznets, and the Consumption Puzzle

Although the Keynesian consumption function met with early successes, two anomalies soon arose. Both concern Keynes's conjecture that the average propensity to consume falls as income rises.

- **The first anomaly:** During World War II, on the basis of the Keynesian consumption function, economists reasoned that as incomes in the economy grew over time, households would consume a smaller and smaller fraction of their incomes.
- If so, the low consumption would lead to an inadequate demand for goods and services.
- In other words, on the basis of the Keynesian consumption function, these economists predicted that the economy would experience what they called **secular stagnation**--a long depression of indefinite duration--unless fiscal policy was used to expand aggregate demand.
- It turned out that the end of the war did not throw the economy into another depression, but it did suggest that Keynes's conjecture that the average propensity to consume would fall as income rose appeared not to hold.
- **The second anomaly** arose when economist **Simon Kuznets constructed new aggregate data on consumption** and income dating back to 1869 (Long-time series data). His work would later earn him a Nobel Prize.
 - Kuznets discovered that the ratio of consumption to income was remarkably stable over time, despite large increases in income; again, Keynes's conjecture was called into question.
 - This brings us to the puzzle... **Consumption Puzzle**
- **Studies of household data and short time-series confirmed Keynes's conjectures. Yet studies of long time-series found no tendency for the APC to fall as income rises over time.**
- **The failure of the secular-stagnation hypothesis and the findings of Kuznets** both indicated that the **average propensity to consume is fairly constant over time.**

- ❖ This presented a puzzle: Why did Keynes's conjectures hold up well in the studies of household data and in the studies of short time-series, but fail when long-time series were examined?
- ✚ In the 1950s, **Franco Modigliani** and **Milton Friedman** each proposed explanations of these seemingly contradictory findings. Both economists later won Nobel Prizes, partly because of their work on consumption.
- ✚ But before we see how Modigliani and Friedman tried to solve the consumption puzzle, we must discuss Irving Fisher's contribution to consumption theory because both Modigliani's life-cycle hypothesis and Friedman's permanent-income hypothesis rely on the theory of consumer behavior proposed by Irving Fisher.

1.2 Irving Fisher and Intertemporal Choice

- The consumption function introduced by Keynes relates current consumption to current income. This relationship, however, is incomplete.
- When people decide how much to consume and how much to save, they consider both the present and the future. The more consumption they enjoy today, the less they will be able to enjoy tomorrow. In making this tradeoff, households must look ahead to the income they expect to receive in the future and to the consumption of goods and services they hope to be able to afford.
- ✚ The economist Irving Fisher developed the model with which economists analyze how rational, forward-looking consumers make intertemporal choices—that is, choices involving different periods of time.
- ✚ **Fisher's model illuminates the constraints consumers face, the preferences they have, and how these constraints and preferences together determine their choices about consumption and saving.**

1.2.1 The Intertemporal Budget Constraint

- ✚ When consumers are deciding how much to consume today versus how much to consume in the future, they face an **intertemporal budget constraint**, which measures the total resources available for consumption today and in the future.

To develop Fisher's model, let's examine the constraints consumers face. For simplicity, we examine the decision facing a consumer who lives for two periods. Period one represents the consumer's youth and period two represents the consumer's old age. The consumer earns income

Y_1 and consumes C_1 in period one, and earns income Y_2 and consumes C_2 in period two (All variables are real).

✚ Because the consumer has the opportunity to borrow and save, consumption in any single period can be either greater or less than income in that period.

✚ Consider how the consumer's income in the two periods constrains consumption in the two periods.

✚ In the first period, saving equals income minus consumption. That is,

$$S = Y_1 - C_1, \text{ where } S \text{ is saving.}$$

✚ In the second period, consumption equals the accumulated saving, including the interest earned on that saving, plus second-period income. That is,

$$C_2 = (1 + r)S + Y_2, \quad \text{where } r \text{ is the real interest rate}$$

NB: the variable S can represent either saving or borrowing (negative saving) and that these equations hold in both cases, and we assume that the interest rate for borrowing is the same as the interest rate for saving.

✚ To derive the consumer's budget constraint, combine the two preceding equations. Substitute the first equation for S into the second equation to obtain

$$C_2 = (1 + r)(Y_1 - C_1) + Y_2$$

Rearrange terms to obtain:

$$(1 + r)C_1 + C_2 = (1 + r)Y_1 + Y_2$$

Now divide both sides by $1 + r$ to obtain:

$$\boxed{C_1 + \frac{C_2}{1 + r} = Y_1 + \frac{Y_2}{1 + r}} \Rightarrow \text{The consumer's intertemporal budget constraint}$$

This equation relates consumption in the two periods to income in the two periods. The consumer's budget constraint is easily interpreted.

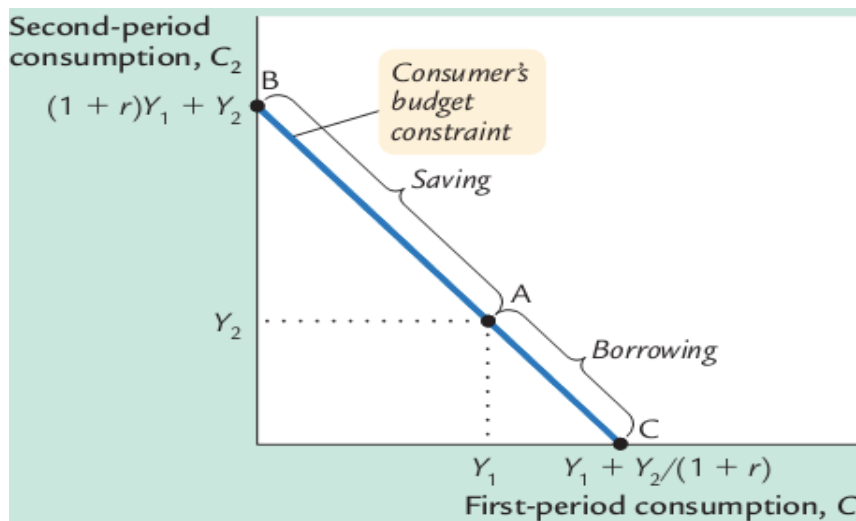
✚ If the interest rate is zero, the budget constraint shows that total consumption in the two periods equals total income in the two periods.

- ✚ In the usual case in which the interest rate is greater than zero, future consumption and future income are discounted by a factor $1 + r$. This **discounting** arises from the interest earned on savings.
- ✚ In essence, because the consumer earns interest on current income that is saved, future income is worth less than current income. Similarly, because future consumption is paid for out of savings that have earned interest, future consumption costs less than current consumption.
- ✚ The factor $1/(1 + r)$ is the price of second-period consumption measured in terms of first-period consumption: it is the amount of first-period consumption that the consumer must forgo to obtain 1 unit of second-period consumption.

Figure 1.2 graphs the consumer's budget constraint. Three points are marked on the figure.

- At point A, the consumer consumes exactly his income in each period ($C_1 = Y_1$ and $C_2 = Y_2$), so there is neither saving nor borrowing between the two periods.
- At point B, the consumer consumes nothing in the first period ($C_1 = 0$) and save all income, so second-period consumption C_2 is $(1 + r)Y_1 + Y_2$.
- At point C, the consumer plans to consume nothing in the second period ($C_2 = 0$) and borrows as much as possible against second-period income, so first period consumption C_1 is $Y_1 + Y_2/(1 + r)$. Of course, these are only three of the many combinations of first- and second-period consumption that the consumer can afford: all the points on the line from B to C are available to the consumer.

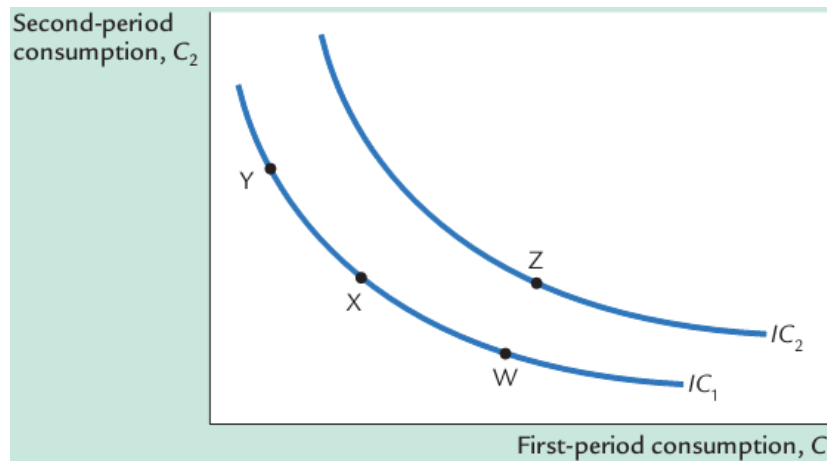
Figure 1.2: The consumer's budget constraint



1.2.2 Consumer Preferences

- ❑ The consumer's preferences regarding consumption in the two periods can be represented by indifference curves, as in figure 6.3.
- ❖ **An indifference curve** shows the combination of 1st-period and 2nd-period consumption that makes the consumer equally happy.
 - ✓ The slope at any point on the indifference curve shows how much 2nd-period consumption the consumer requires in order to be compensated for a 1-unit reduction in 1st-period consumption.
 - ✓ This slope is the **marginal rate of substitution** between 1st-period consumption and 2nd-period consumption. It tells us the rate at which the consumer is willing to substitute 2nd-period consumption for 1st-period consumption.

Figure 1.3: The consumer's Preferences



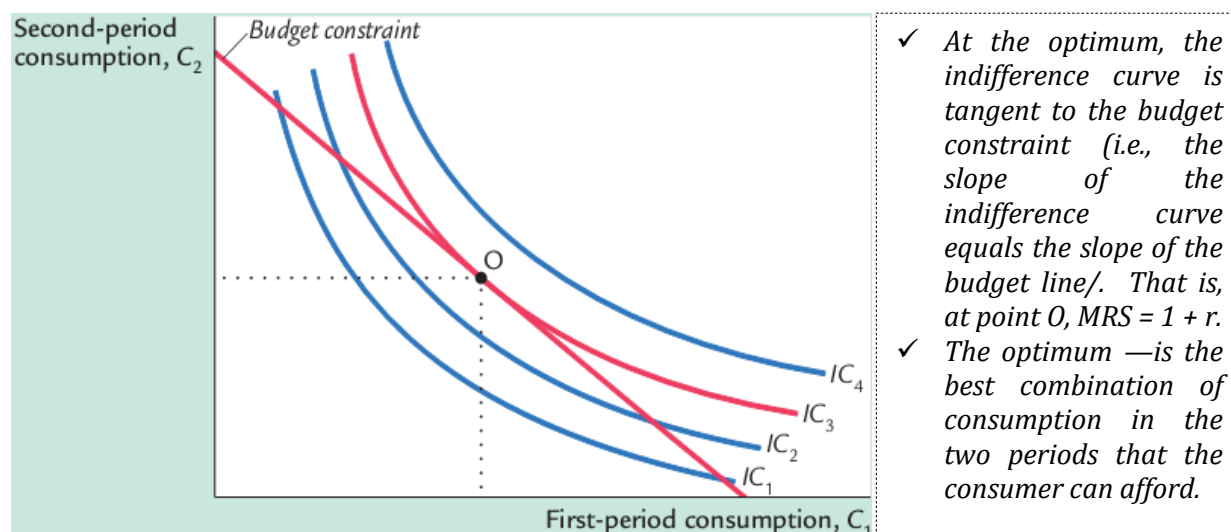
✚ This figure shows two of many indifference curves.

- ✓ Higher indifference curves such as IC_2 are preferred to lower curves such as IC_1 .
- ✓ The consumer is equally happy at points W, X, and Y,

1.2.3 Optimization /Consumer Equilibrium/

- ✚ Having discussed the consumer's budget constraint and preferences, we can consider the decision about how much to consume. The consumer would like to end up with the best possible combination of consumption in the two periods—that is, on the highest possible indifference curve. But the budget constraint requires that the consumer also end up on or below the budget line, because the budget line measures the total resources available to him.
- ✚ The consumer achieves his highest (or optimal) level of satisfaction by choosing the point on the budget constraint that is on the highest indifference curve (point O).

Figure 1.4: The Consumer's Optimum

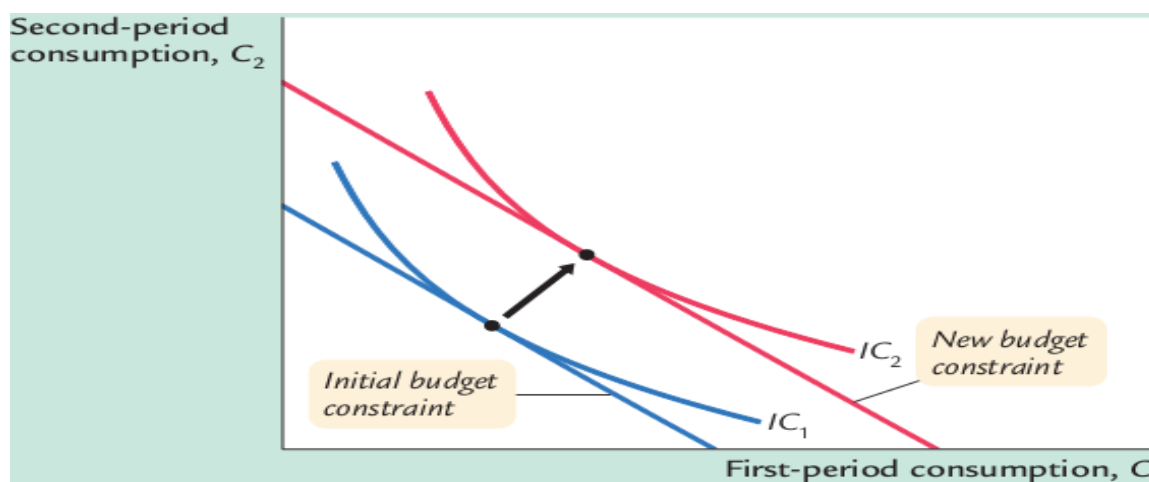


1.2.4 How Changes in Income Affect Consumption

Now let's examine how consumption responds to an increase in income.

- ✓ An increase in either Y_1 or Y_2 shifts the budget constraint outward, as in Figure 16.5.
- ✓ The higher budget constraint allows the consumer to choose a better combination of first- and second-period consumption —that is, the consumer can now reach a higher indifference curve.
- ✓ If consumption in period one and consumption in period two are both **normal goods**—those that are demanded more as income rises, this increase in income raises consumption in both periods.

Figure 1.5: An Increase in Income and consumer optimum



- ✚ The key conclusion from Figure 6.5 is that regardless of whether the increase in income occurs in the first period or the second period, the consumer spreads it over consumption in both periods.
- ✚ This behavior is sometimes called **consumption smoothing**. Because the consumer can borrow and lend between periods, the timing of the income is irrelevant to how much is consumed today (except, of course, that future income is discounted by the interest rate).
- ✚ Notice that this conclusion is quite different from that reached by Keynes.
 - ✓ **Keynes posited that a person's current consumption depends largely on his current income.**
 - ✓ **Fisher's model says, instead, that consumption is based on the resources the consumer expects over his lifetime.**

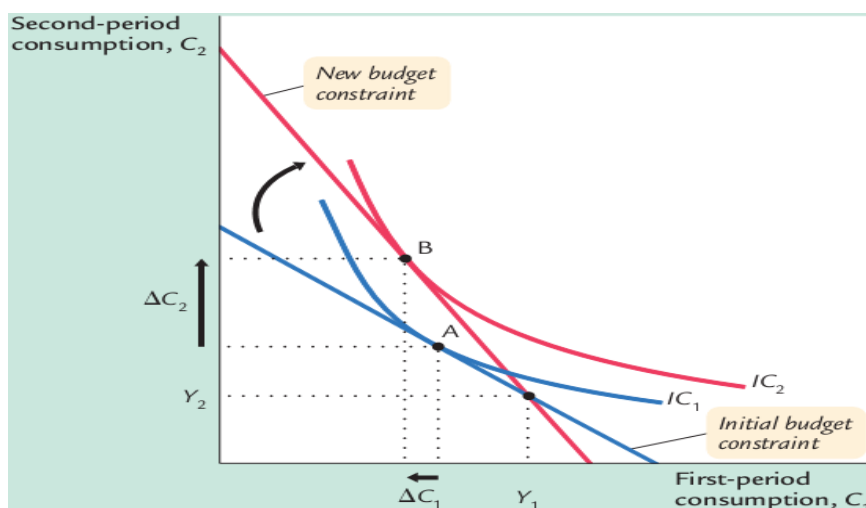
1.2.5 How Changes in the Real Interest Rate Affect Consumption

Let's now use Fisher's model to consider how a change in the real interest rate alters the consumer's choices.

- ✚ Figure 1.6 shows that an increase in the real interest rate rotates the consumer's budget line around the point (Y_1, Y_2) and, thereby, alters the amount of consumption he chooses in both periods.
- ✚ Here, the consumer moves from point A to point B. You can see that for the indifference curves drawn in this figure first-period consumption falls and second-period consumption rises.

- ✚ Economists decompose the impact of an increase in the real interest rate on consumption into **two effects: an income effect and a substitution effect.**
- ✓ **The income effect** is the change in consumption that results from the movement to a higher indifference curve. Because the consumer is a saver rather than a borrower (as indicated by the fact that first-period consumption is less than first-period income), the increase in the interest rate makes him better off (as reflected by the movement to a higher indifference curve). If consumption in period one and consumption in period two are both normal goods, the consumer will want to spread this improvement in his welfare over both periods. This income effect tends to make the consumer want more consumption in both periods.
- ✓ **The substitution effect** is the change in consumption that results from the change in the relative price of consumption in the two periods. In particular, consumption in period two becomes less expensive relative to consumption in period one when the interest rate rises. That is, because the real interest rate earned on saving is higher, the consumer must now give up less first-period consumption to obtain an extra unit of second-period consumption. **This substitution effect tends to make the consumer choose more consumption in period two and less consumption in period one.**
- ✚ The consumer's choice depends on both the income effect and the substitution effect. Both effects act to increase the amount of second-period consumption; hence, we can conclude that **an increase in the real interest rate raises second-period consumption.**
- ✚ But the two effects have opposite impacts on first-period consumption, so the increase in the interest rate could either raise or lower it. Hence, depending on the relative size of the income and substitution effects, an increase in the interest rate could either stimulate or depress saving.

Figure 1.6: Changes in the Real Interest Rate and Consumer optimum



✓ In this figure, the higher interest rate reduces first-period consumption by ΔC_1 and raises second-period consumption by ΔC_2 .

1.2.6 Constraints on Borrowing

- ✚ Fisher's model assumes that the consumer can borrow as well as save. The ability to borrow allows current consumption to exceed current income. In essence, when the consumer borrows, he consumes some of his future income today.
- ✚ Yet for many people such borrowing is impossible. The inability to borrow prevents current consumption from exceeding current income. A constraint on borrowing can therefore be expressed as $C_1 \leq Y_1$.

This inequality states that consumption in period one must be less than or equal to income in period one. This additional constraint on the consumer is called a **borrowing constraint** or, sometimes, a **liquidity constraint**.

Figure 1.7: Borrowing constraint

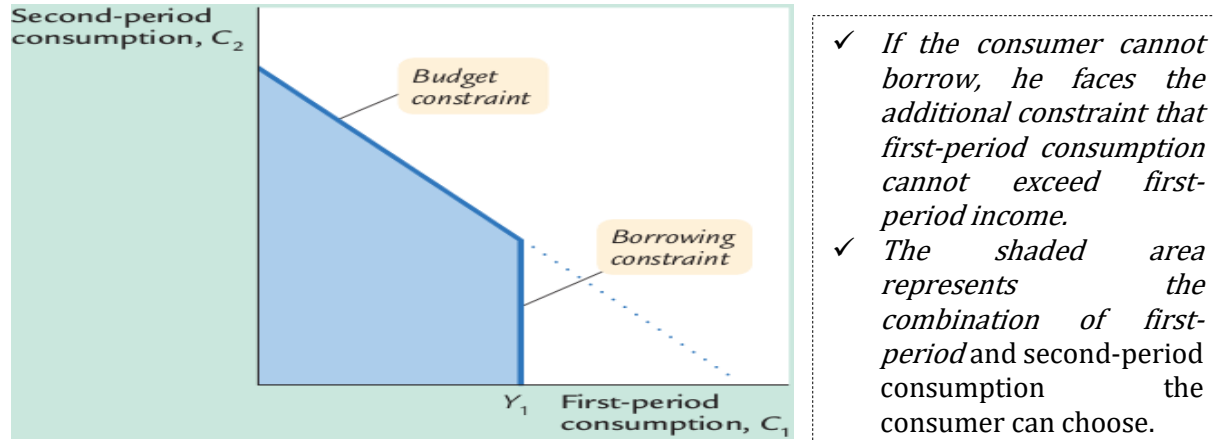
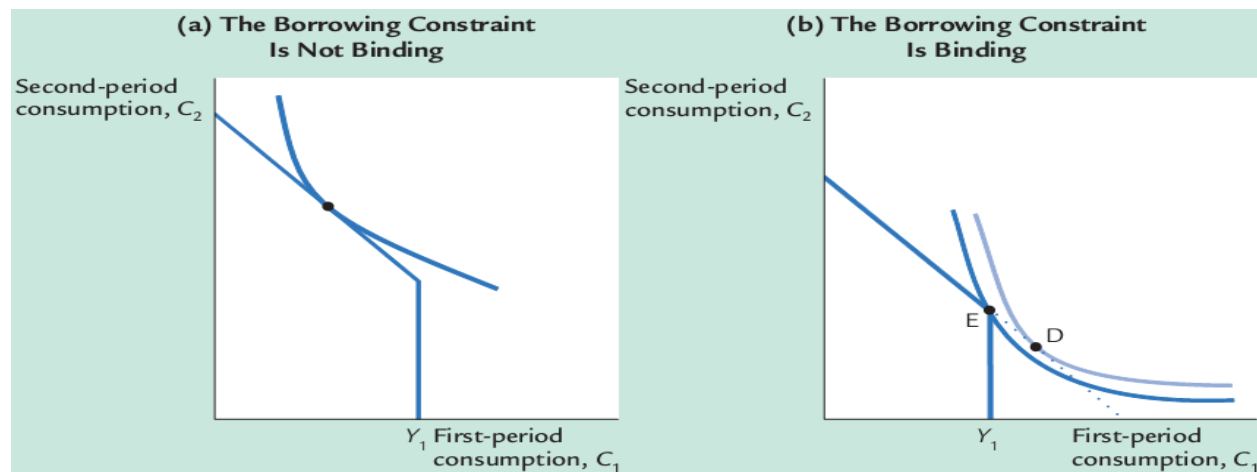


Figure 1.8: The Consumer's Optimum with a Borrowing Constraint



- ✚ When the consumer faces a borrowing constraint, there are two possible situations.
 - ✓ In panel (a), the consumer chooses first-period consumption to be less than first-period income, so the borrowing constraint is not necessary/binding and does not affect consumption in either period.
 - ✓ In panel (b), the borrowing constraint is binding. The consumer would like to borrow and choose point **D**. But because borrowing is not allowed, the best available choice is point **E**. When the borrowing constraint is binding, first-period consumption equals first-period income.
- ✚ To sum up, the analysis of borrowing constraints leads us to conclude that there are two consumption functions.
 - ✓ For some consumers, the borrowing constraint is not binding, and consumption in both periods depends on the present value of lifetime income, $Y_1 + [Y_2/(1 + r)]$.
 - ✓ For other consumers, the borrowing constraint binds, and the consumption function is

$C_1 = Y_1$ and $C_2 = Y_2$. Hence, for those consumers who would like to borrow but cannot, consumption depends only on current income.

1.3 Franco Modigliani and the Life-Cycle Hypothesis

- Modigliani emphasized that income varies systematically over people's lives and that saving allows consumers to move income from those times in life when income is high to those times when it is low. This interpretation of consumer behavior formed the basis for his **life-cycle hypothesis**.
- One important reason that income varies over a person's life is retirement. Most people plan to stop working at about age 65, and they expect their incomes to fall when they retire. Yet they do not want a large drop in their standard of living, as measured by their consumption. To maintain consumption after retirement, people must save during their working years. Let's see what this motive for saving implies for the consumption function.
- Consider a consumer who expects to live another T year, has wealth of W , and expects to earn income Y until she retires R years from now. What level of consumption will the consumer choose if she wishes to maintain a smooth level of consumption over her life?
- The consumer's lifetime resources are composed of initial wealth W and life-time earnings of $R \times Y$. The consumer can divide up her lifetime resources among her T remaining years of life. We assume that she wishes to achieve the smoothest possible path of consumption over her lifetime. Therefore, she divides this total of $W + RY$ equally among the T years and each year consumes.

$$C = (W + RY) / T$$

We can write this person's consumption function as

$$C = (1/T)W + (R/T)Y$$

- ✓ For example, if the consumer expects to live for 50 more years and work for 30 of them, then $T = 50$ and $R = 30$, so her consumption function is

$$C = 0.02 W + 0.6Y$$

- This equation indicates that consumption depends on both income and wealth. An extra \$1 of income per year raises consumption by \$0.60 per year, and an extra \$1 of wealth raises consumption by \$0.02 per year.

- If every individual in the economy plans consumption like this, then the aggregate consumption function is much the same as the individual one. In **particular, aggregate consumption depends on both wealth and income**. That is, the economy's consumption function is:

$$C = \alpha W + \beta Y,$$

where the parameter α is the marginal propensity to consume out of wealth, and the parameter β is the marginal propensity to consume out of income.

This life-cycle model of consumer behavior can solve the consumption puzzle. According to the life-cycle consumption function, the average propensity to consume is

$$C/Y = \alpha(W/Y) + \beta$$

- Because wealth does not vary proportionately with income from person to person or from year to year, we should find that high income corresponds to a low average propensity to consume when looking at data across individuals or over short periods of time.
- But, over long periods of time, wealth and income grow together, resulting in a constant ratio W/Y and thus a constant average propensity to consume.
- **To sum up, Modigliani resolved the consumption puzzle posed by Simon Kuznets's data by saying that:**
 - ✓ **In the short run when wealth is constant, average propensity to consume falls.**
 - ✓ **In the long run, as wealth increases with income, average propensity to consume is constant.**

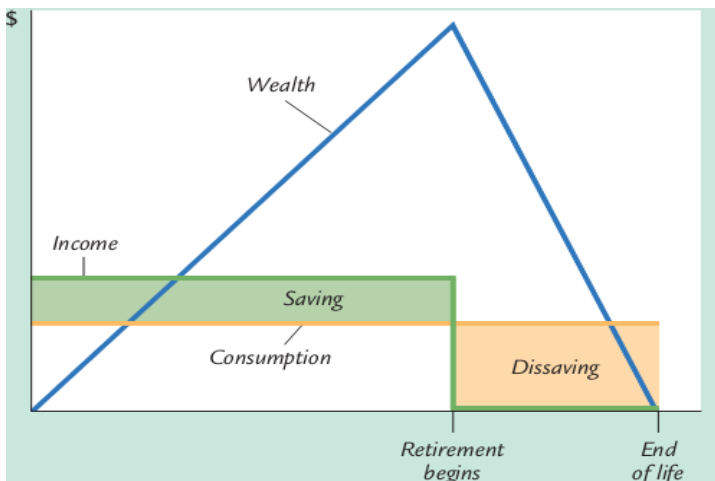


Figure: 1.9: Consumption, Income, and Wealth Over the Life Cycle

✚ If the consumer **smooths consumption** over her life (as indicated by the horizontal consumption line), she will save and accumulate wealth during her working years and then dissave and run down her wealth during retirement.

1.4 Milton Friedman and the Permanent-Income Hypothesis

- ✓ Milton Friedman proposed the permanent-income hypothesis to explain consumer behavior.
- ✓ Friedman's permanent-income hypothesis emphasizes that people experience random and temporary changes in their incomes from year to year.
- ✓ Friedman suggested that we view current income Y as the sum of two components, permanent income Y^P and transitory income Y^T . That is,

$$Y = Y^P + Y^T$$

- ✓ **Permanent income** is the part of income that people expect to persist into the future.
- ✓ **Transitory income** is the part of income that people do not expect to persist, e.g.: lottery income.
- ✓ Friedman reasoned that consumption primarily depend on permanent income, because consumers use saving and borrowing to smooth consumption in response to transitory changes in income.
- ✓ For example, if a person received a permanent raise of \$10,000 per year, his consumption would rise by about as much.
- ✓ Yet if a person won \$10,000 in a lottery, he would not consume it all in one year. Instead, he would spread the extra consumption over the rest of his life. Assuming an interest rate of zero and a remaining life span of 50 years, consumption would rise by only \$200 per year in response to the \$10,000 prize.
- ✓ Thus, consumers spend their permanent income, but they save rather than spend most of their transitory income.
- ✓ Friedman concluded that we should view the consumption function as approximately

$$C = \alpha Y^P,$$

where α is a constant that measures the fraction of permanent income consumed.

- **The permanent-income hypothesis, as expressed by this equation, states that consumption is proportional to permanent income.**
- The permanent-income hypothesis solves the consumption puzzle by suggesting that the standard Keynesian consumption function uses the wrong variable.

- According to the permanent-income hypothesis, consumption depends on permanent income; yet many studies of the consumption function try to relate consumption to current income.
- Friedman argued that this errors-in-variables problem explains the seemingly contradictory findings.

Let's see what Friedman's hypothesis implies for the average propensity to consume. Divide both sides of his consumption function by Y to obtain

$$APC = C/Y = aY^P/Y$$

- ✚ According to the permanent-income hypothesis, the APC depends on the ratio of permanent income to current income. When current income temporarily rises above permanent income, the APC temporarily falls; when current income temporarily falls below permanent income, the APC temporarily rises.

Now consider the studies of household data.

- Friedman reasoned that these data reflect a combination of permanent and transitory income.
 - ✓ Households with high permanent income have proportionately higher consumption.
 - ✓ If all variation in current income came from the permanent component, the average propensity to consume would be the same in all households.
 - ✓ But some of the variation in income comes from the transitory component, and households with high transitory income do not have higher consumption.
 - ✓ Therefore, researchers find that high-income households have, on average, lower average propensities to consume.

Similarly, consider the studies of time-series data.

- ✚ Friedman reasoned that year-to-year fluctuations in income are dominated by transitory income.
 - ✓ Therefore, **years of high income should be years of low APC.**
 - ✓ **But over long periods of time, the variation in income comes from the permanent component.** Hence, **in long time-series, one should observe a constant average propensity to consume**, as in fact Kuznets found.

1.5 Robert Hall and the Random-Walk Hypothesis

- + The economist Robert Hall was the first to derive the implications of rational expectations for consumption.
- + He showed that if the permanent-income hypothesis is correct and if consumers have rational expectations, then changes in consumption over time should be unpredictable.
- + When changes in a variable are unpredictable, the variable is said to follow a **random walk**.
- + According to Hall, the combination of the permanent-income hypothesis and rational expectations implies that **consumption follows a random walk**.
- **Hall reasoned as follows:**
 - + According to the permanent-income hypothesis, consumers face fluctuating income and try their best to smooth their consumption over time.
 - + At any moment, consumers choose consumption based on their current expectations of their lifetime incomes.
 - + Over time, they change their consumption because they receive news that causes them to revise their expectations. For example, a person getting an unexpected promotion increases consumption, whereas a person getting an unexpected demotion decreases consumption.
 - + In other words, **changes in consumption reflect “surprises” about life-time income**. If consumers are optimally using all available information, then they should be surprised only by events that were entirely unpredictable.
- ✓ Therefore, changes in their consumption should be unpredictable as well.

CHAPTER TWO: THE THEORY OF INVESTMENT

This chapter is devoted to the theory of investment. Why? Because investment is an important determinant for:

- ✓ Long run growth (through the role of capital formation)
- ✓ Business cycle dynamics

Chapter Objectives:

At the end of this chapter, you will be able to:

- ✓ *Understand the notion and categories of investment*
- ✓ *Appreciate the r/ship b/n investment spending, capital stock and output/growth*
- ✓ *Identify and elaborate the implications of the leading Theories of investment*
- ✓ *Identify the factors which shifts investment function*
- ✓ *Explain why investment rises during expansion and falls during recessions*

2.1. INTRODUCTION: **Investment versus Capital Stock**

✚ What is investment?

✚ Strictly speaking, investment is the change in capital stock during a period. **It is the formation of real capital, tangible or intangible, that will produce a stream of goods and services in the future.**

✚ Consequently, unlike capital, investment is a flow concept and not a stock concept . This means that while **capital** is measured at **a point in time** (eg: the given dollar value of all the buildings, machines, and inventories at a point in time.), **investment** can only be measured **over a period of time** (i.e., investment is the flow of spending that adds to the physical stock of capital).

○ Investment spending is aimed at providing a higher standard of income in the future. Thus, investment links the present to the future.

➤ **Investment Affects Future Output via Rising the stock of Capital**

○ The capital stock is a very important input used to produce output and the level of investment ultimately determines the level of the capital stock and thus the growth of output.

✓ **The r/ship b/n investment and the capital stock can be expressed in the following way:**

$$K_{t+1} = K_t + I_t - \delta K_t$$

Where

K_t = stock of capital at time t.

I_t = Gross investment at time t.

δ = the depreciation rate.

- ✓ Another way of stating this relationship is looking at the **net change in the capital stock**, expressed as;

$$K_{t+1} - K_t = I_t - \delta K_t = I_{n,t}$$

where $I_{n,t}$ is net investment at time t.

- ✚ This implies that if $I_{n,t}$ increases then K_{t+1} will increase and thus $Y_{t+1} > Y_t$, that is, output should increase.
 - Note that the notation used above assumes that investment undertaken today does not really become productive until the future.
- o Investment can be undertaken not only by firms or households, but also by the government. For example, looking at Business Fixed Investment of different countries, government sector undertakes significant proportion of this category.
- o Investment makes up about 20% of GDP, and although obviously not as large as consumption it is important because investment over long periods determines the size of the stock of capital and thus helps determine long-run growth.
- o Investment spending is very volatile and thus responsible for much of the fluctuations of GDP across the business cycle.
 - As a result, understanding what factors affect investment is crucial not only because it is part of GDP but b/c it is crucial in determining fluctuations and growth in GDP.
- o In this chapter, therefore, we first examine the components of investment and then study what determines investment by referring to the leading theories of investment.

2.2. TYPES /CATEGORIES/ OF INVESTMENT

- ✓ Investment undertaken in an economy is classified according to the following three major categories:
 - ✚ **Business fixed investment** indicates the annual increase in equipment, machineries and structures/ plant, office buildings, etc/ by businesses used in production.
 - ✚ **Residential Construction investment**-investment in new housing units by people to live in and/or by landlords to rent out.
 - ✚ **Inventory investment** includes those goods that businesses put aside in storage, including materials and supplies, work in progress, and finished goods.

2.3. MODELS OF INVESTMENT

We have noted above that investment fluctuates too much, and this fluctuation in investment can be a proximate source of fluctuation in output both in the short-run and in the long-run. In this section, therefore, we build models of each type of investment to explain these fluctuations in investment spending and thereby output.

2.3.1. BUSINESS FIXED INVESTMENT

- The largest piece of investment spending, accounting for about three-quarters of the total is business fixed investment.
- Business fixed investment includes everything from fax machines to factories, computers to company cars.

2.3.1.1. The Neoclassical Model Of Business Fixed Investment

- The Neoclassical model of investment is the standard model of business fixed investment.
- It dates mainly from Jorgensen's papers in the 1960s.

❖ **The model examines the benefits and costs of owning capital goods.**

- It shows how the level of investment is related to
 - The marginal product of capital,
 - The interest rate, &
 - The tax rules affecting firms.

📌 To develop the model, assume that there are **two types of firms**:

- i. **Production firms** rent the capital they use to produce goods and services &
- ii. **Rental firms** own capital, rent it out to production firms.

In this context, **investment is the rental firms' spending on new capital goods.**

A. The Capital Rental Market

📌 Let's first consider a typical production firm. The production firm decides how much capital to rent to maximize its profit which is the excess of revenues over costs.

- ✓ It rents capital at a rental rate, R and sells its output at a price, P ; the **real cost of a unit of capital** to the production firm is R/P .
- ✓ The **real benefit of a unit of capital** is the **marginal product of capital**; MP_K – the extra output produced with one more unit of capital. The Marginal product of capital declines as the amount of capital rises.
- Therefore, competitive firms rent capital to the point where $MP_K = R/P$. The real rental price of capital, R/P adjusts to equilibrate the demand for capital (w/c is determined by the MP_K) and the fixed supply. Note the following figure.

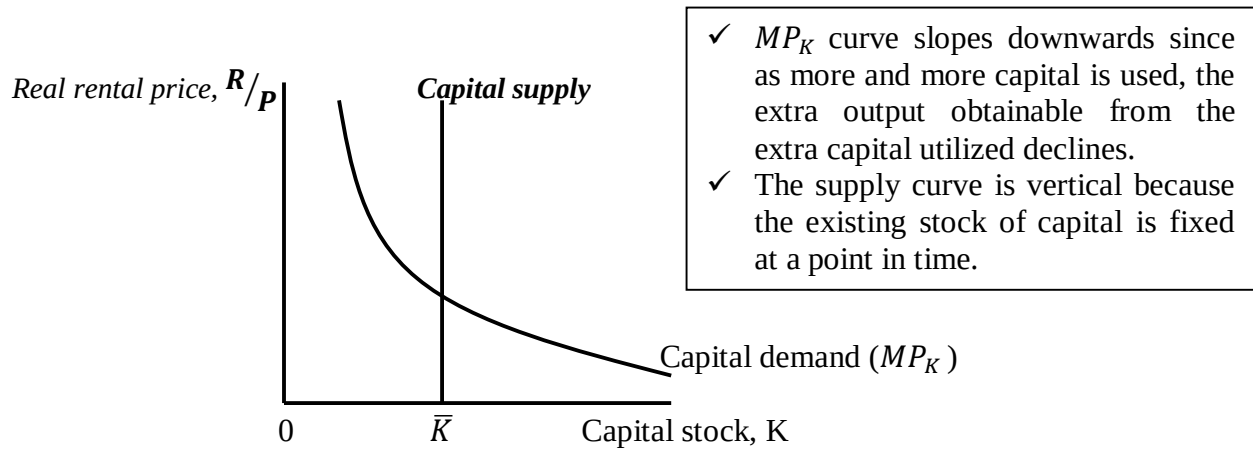


Fig:2.1: The Capital Rental Market and the decision of a production firm

What determines the real rental price of capital, R/P ?

🚦 To see what variables influence the equilibrium rental price, R/P , let's consider the Cobb-Douglas production function as a good approximation of how the actual economy turns capital and labor into goods and services.

- The Cobb-Douglas production function is: $Y = AK^\alpha L^{1-\alpha}$, where Y is output, K capital, L labor, and A a parameter measuring the level of technology, and α a parameter between 0 and 1 that measures capital's share of output.

- ✓ The marginal product of capital for the Cobb-Douglas production function is

$$MP_K = \alpha A(L/K)^{1-\alpha}.$$

Because the real rental price equals the marginal product of capital in equilibrium, we

$$\text{can write } R/P = \alpha A(L/K)^{1-\alpha}.$$

This expression identifies the **variables that determine the real rental price**. It shows the following:

- The **lower** the stock of capital, K , the **higher** the real rental price of capital
- The **greater** the labor employed, the **higher** the real rental price of capitals
- The **better** the technology, the **higher** the real rental price of capital.

Events that **reduce** the capital stock (eg: earthquake & war), or raise employment (eg: expansion in aggregate demand), or improve the technology (a scientific discovery), **raise the equilibrium real rental price of capital**.

B. Rental Firms' Investment Decision

Rental firms invest in new capital when the benefit exceeds the cost of doing so.

So, let's consider the benefit and cost of owning capital.

- ✓ R/P - is the benefit of capital rental firms earn from renting a unit of capital to production firms.
- ✓ For each period that a firm rents out a unit of capital, the rental firm bears **three costs**:
 - i. The interest cost**: when a rental firm borrows to buy a unit of capital to rent out, it must pay interest on loan, which equals the purchase price of a unit of capital P_K times the nominal interest rate, i , so iP_K . Notice that this interest cost would be the same even if the rental firm buys capital using cash on hand because it loses the interest it could have earned by depositing this cash in bank.
 - ii. Depreciation δ** defined as the fraction of value lost per period because of the wear and tear while the capital is rented out, so the dollar cost of depreciation is δP_K .
 - iii. Capital Loss**: While the rental firm is renting out the capital, the price of capital may change. If the price of capital rises, the firm gains, because the firm's asset has risen in value, and hence this reduces the firm's cost and vice versa. The cost of the loss or gain on the price of capital is denoted as $-\Delta P_K$.

Therefore, the **total nominal cost of renting out a unit of capital for one period is**: $= iP_K + \delta P_K - \Delta P_K = P_K(i + \delta - \Delta P_K/P_K)$

For example, consider the cost of capital to a car-rental company. The company buys cars for \$10,000 each and rents them out to other businesses. The company faces an interest rate i of 10 percent per year. Moreover, car prices are rising at 6 percent per year and cars depreciate at 20 percent per year. **Calculate the company's nominal cost of capital?**

$$\begin{aligned} \text{Nominal Cost of capital} &= 0.1(\$10,000) + 0.2(10,000) - 0.06(10,000) \\ &= \$1000 + \$2000 - \$600 = \$2400 \end{aligned}$$

To make the expression simple, assume that the price of capital rises with the price of other goods. Thus, $\Delta P_K / P_K$ equals the overall rate of inflation, π . Because $i - \pi = r$, we can write the **cost of capital** as:

$$\text{Nominal Cost of capital} = P_K(r + \delta)$$

Finally, we want the **real cost of capital**- the cost of capital relative to the price of other goods in the economy.

✚ **The Real Cost of Capital** = $P_K / P (r + \delta)$.

Where r is the real interest rate and P_K / P equals the relative price of capital.

- This equation states that the real cost of capital positively depends on the relative price of capital, the real interest rate and the depreciation rate.

Now consider a RENTAL FIRM'S DECISION about whether to increase/ decrease its capital stock.

For each unit of capital, the firm **earns real revenue** R/P & bears the **real cost** $P_K / P (r + \delta)$.

Therefore, the real profit per unit of capital is:

Profit rate = Revenue - Cost

$$= R/P - (P_K / P)(r + \delta) \text{ or}$$

$$\text{Profit rate} = MP_K - P_K / P (r + \delta), \text{ B/c } R/P = MP_K.$$

Thus, the **change in the capital stock**, called **net investment on new capital** depends on the difference between the MP_K and the **real cost of capital or the profit rate**.

- If the $MP_K > \text{the real cost of capital}$, the firm is making profits on the last unit of capital it is using and it could increase profits by investing more on new capita. In other words, if the profit rate is positive then increasing capital is profitable.
- If the $MP_K < \text{real cost of capital}$, firms let their capital stock shrink.

❖ Net Investment & Gross Investment

Now, we can derive the investment function in the Neoclassical model of investment.

$$\text{Net investment} = \Delta K = I_n \left[MP_K - \frac{P_K}{P} (r + \delta) \right],$$

where $I_n(.)$ is the function showing how **net investment** responds to the **incentive to invest**.

Total spending on business fixed investment (i.e., gross investment) is the sum of net investment and the replacement of depreciated capital. The gross investment function, I , therefore, is:

$$I = \Delta K + \delta K$$

$$\therefore I = I_n \left[MP_K - \left(\frac{P_K}{P} \right) (r + \delta) \right] + \delta K$$

- **Business fixed investment depends on the marginal product of capital, the cost of capital and the amount of depreciation.**

This model shows how investment depends on the interest rate. An increase in the real interest rate raises the cost of capital therefore reduces the amount of profit from owning capital and reduces the incentive to accumulate more capital, the converse holds true.

- The investment function is down ward sloping to the interest rate. And an increase in the marginal productivity of capital shifts the investment function outward at any given interest rate.

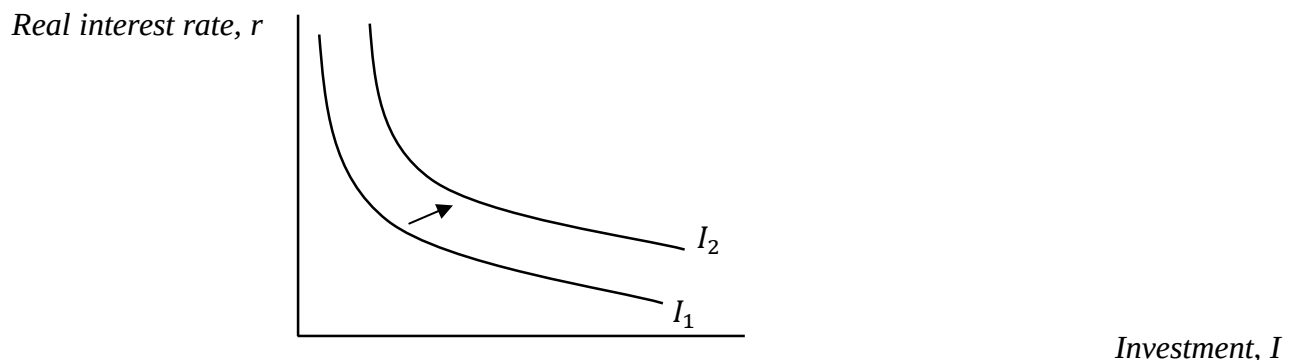


Fig: 2.2: The investment function and marginal productivity of capital

The above figure presents the investment function. Any event that raises the marginal product of capital / Eg: technological innovation that increases the production

function parameter A increases the profitability of investment and causes the investment function to shift outward for any given interest rate /i.e., the amount of capital that rental firms wish to buy increases.

C. Taxes and Investment

Tax laws influence firms' incentives to accumulate capital in many ways. We can amend the profit rate relationship we derived above to include the effects of government taxes. Let τ be the tax rate on firm revenues. Then, **the after tax profit rate of capital** is;

$$\text{Profit rate} = (1 - \tau) MP_K - \left(\frac{P_K}{P} \right) (r + \delta)$$

So we can see that the tax reduces the profits earned from capital by reducing the Marginal benefit of using and owning capital.

🚦 Here we can consider three of the most important tax policies affecting investment:

i. Corporate Income Tax

The Corporate income tax is a tax on corporate profits. An increase in the *Corporate income tax* will effectively increase τ and hence decreases profit and thereby investment. *This sort of policy will discourage the accumulation of capital.*

ii. Investment Tax Credits

This is a *tax provision* policy that reduces a firm's taxes by a certain amount for each dollar spent on capital goods. This effectively lowers τ , reducing the cost of capital and causing an increase in investment. *This sort of policy will encourage the accumulation of capital.*

iii. Tax Treatment of Depreciation

Corporate profits are taxed and profits are just revenue minus costs. Part of the costs of production firms is depreciation on capital so that firms are allowed to include depreciation of their capital as a cost when calculating the amount of profit to be taxed. *But depreciation is calculated using historical cost, not replacement cost which tends to understate the true cost of depreciation in periods of high inflation, which*

overstates the profits being made. This means a tax is levied even if the economic profit is zero, which effectively increases τ , reducing investment, and discouraging the accumulation of capital.

➤ **Summary of Business Fixed Investment**

- i. Higher interest rate increases the cost of capital and reduces business fixed investment.
- ii. Improvements in technology and tax policies, such as the corporate income tax and investment tax credit, shift the business fixed investment function.
- iii. During booms higher employment increases the MP_K and therefore, increases business fixed investment.

2.3.1.2. The Accelerator Theory Of Business Fixed Investment

The other theory for explaining the level of business fixed investment and why it behaves as it does is called the **accelerator theory of investment**.

This theory argues that firms produce output and they need capital to produce this output. It is likely that a firm has some desired stock of capital in mind when producing any given amount of output. If the desired capital stock differs from what the firm actually has, then it must change its capital stock and how much it changes it is the amount of investment.

The accelerator model begins with an assumption that firms' desired capital-output ratio is roughly constant. If Y_t output is required to be produced and α is capital-output ratio (which is equal to K/Y), the required amount of capital to produce Y_t output will be given by the following equation:

$$K_t = \alpha Y_t \dots \dots \dots (1)$$

Under the assumption of constant capital-output ratio, changes in output are made possible by changes in the stock of capital. Thus, when income is Y_t as above, then required stock of capital is $K_t = \alpha Y_t$. When output or income is equal to Y_{t-1} then required stock of capital will be

$$K_{t-1} = \alpha Y_{t-1} \dots \dots \dots (2)$$

Hence, **the increase in the stock of capital** in period t is given by the following equation: $K_t - K_{t-1} = \alpha Y_t - \alpha Y_{t-1} \dots \dots \dots (3)$

Since increase in the stock of capital in a year ($K_t - K_{t-1}$) represents investment in that year (I_t), the above equation can be written as:

$$K_t - K_{t-1} = \alpha(Y_t - Y_{t-1}) ;$$

Where, $\Delta K_t = K_t - K_{t-1} = I_t$ & $\Delta Y_t = Y_t - Y_{t-1}$

Therefore, $I_t = \alpha \Delta Y_t$ (4)

Equation 4 above reveals that *as a result of increase in income in any year t from a previous year $t - 1$* , increase in the investment will be α times more than the increase in income. Hence, it is α i.e., capital-output ratio which represents the magnitude of the accelerator.

Thus, the simplest accelerator model predicts that investment is proportional to the increase in output.

2.3.1.3. Tobin's Q Theory Of Investment

Many economists see a link b/n *fluctuations in investment and fluctuations in the stock market*. The term **stock** refers to the shares in the ownership of corporations, and the **stock market** is the market in which these shares are traded. Stock prices tend to be high when firms have many opportunities for profitable investment, because these profit opportunities mean higher future income for the shareholders. *Thus, stock prices reflect the incentive to invest.*

*The Nobel-Prize-winning economist James Tobin proposed that firms base their investment decisions on the following ratio, which is now called **Tobin's q**:*

$$q = \frac{\text{Market Value of Installed Capital}}{\text{Replacement Cost of Installed Capital}}$$

- ✓ Numerator: the stock market value of the economy's capital stock (i.e., it is the number of shares outstanding times their market price).
- ✓ Denominator: the actual cost to replace the capital goods that were purchased when the stock was issued.
- ✚ Tobin reasoned that net investment should depend on whether **q** is greater or less than one.
 - If $q > 1$, investors buy more capital to raise the market value of their firms.
 - If $q < 1$, the stock market values capital at less than its replacement cost and thus, firms will not replace their capital stock as it wears out.

As such, the q equation provides a correct indication of the current worth and likely prospects for the business. If a firm faces a $q > 1$, then this is a signal that it should buy additional capital because the present value of the future earnings from such capital will be greater than its cost.

2.3.1.4. The Keynesian Investment Explanation

In the simplest Keynesian model, aggregate investment in physical capital is a function of the aggregate level of income Y_t and on the interest rate r_t such that $I_t(Y_t, r_t)$ where $\frac{\partial(I_t(Y_t, r_t))}{\partial Y_t} > 0$ and $\frac{\partial(I_t(Y_t, r_t))}{\partial r_t} < 0$. Thus, the Keynesians predict that investment should increase with output. This prediction follows from an assumption of a long-run relationship between the capital stock K_t and Y_t . On the other hand, higher interest rates r_t means that it is more expensive for producers to take fund from financial institutions, which should decrease investment.

In addition to the above, for Keynesians: ***“Investment is driven by perceptions of future profitability”***

- ✚ Mr. J.M. Keynes emphasizes the role of expectations in the determination of investment.
- ✚ *Keynes argued that in addition to the interest rate, profit expectations, and the degree of confidence or weight that managers place in their profit forecasts, determines investment.*
- ✚ For Keynes, investments are made until the present value of expected future revenues-i.e., the expected return on investment; (marginal efficiency of capital in Keynes terminology), is equal to the opportunity cost of capital (*the discount rate determined by central banks*).
- ✚ The computation of the expected return on investment (i.e., marginal efficiency of capital) involves calculation of benefits and costs of the investment project under consideration which basically depends on expectations (*where expectations depend on available information*).
- ✚ Keynes believed that the degree of confidence manager's place in their forecasts will also influence investment expenditures. If forecasts are deemed reliable by the

decision-maker, then they will be more inclined to base investment decisions on their forecasts. The problem facing researchers has been how to measure this subjective cognitive process. Lacking any direct measure of profit expectations, researchers have been inherently unable to test Keynes' proposition that the **profit expectations** and the **weight assigned to profit expectations** (i.e., the degree of confidence decision-makers hold regarding their forecasts) are key determinants of investment.

- *In general, despite direct measure is lacking, since investment is founded on expectations of future profitability for Keynes, business cycles are largely the outcomes of fluctuations in these expectations.*

2.3.2. RESIDENTIAL CONSTRUCTION INVESTMENT

Now, let's consider the determinants of residential investment by looking at a simple neoclassical model of the housing market. Residential investment includes the purchase of new housing both by people who plan to live in it themselves and by landlords who plan to rent it to others.

2.3.2.1. Neoclassical Model of Residential Investment

The neoclassical model of residential investment directly uses an important concept when thinking about capital: the existing capital is a stock variable whereas investment is a flow variable. This is true of any capital and investment but is made explicit in the model we will learn about.

*The idea is that there are **two parts to housing**:*

- ✓ *The market for the existing stock of houses which determines the equilibrium housing prices, and*
- ✓ *The supply of new housing which determines the flow of residential investment.*

A. The Stock Equilibrium and the Flow Supply

Note the following variables:

P_H = the going price of housing in an economy

P = the price level in an economy

K_H = stock of residential capital or housing, I_H = investment in housing.

i. The Market for Existing Houses -Determining the Equilibrium Price

P_H/P

- ✓ The relative price of housing, P_H/P is determined jointly by demand & of supply for housing.

- ✓ The stock of houses is fixed in the short-term, and so supply curve of existing houses is vertical.
- ✓ The demand for housing is of the normal downward sloping form in the relative price of housing.

ii. The Supply of New Housing -Determining Investment

We assume the law of supply that supplying extra quantities of new housing is more expensive because of diminishing returns to the factors of production used by the firms. That is, the existing resources become more expensive as firms bid up their prices to get them, additional factors require higher payments to induce them to supply their services, or the additional factors are just not as productive as existing resources used. The Marginal benefit for firms or developers from building new housing is the relative price of housing. To induce construction firms to build new houses the Marginal benefit, or relative price, of building new housing has to be higher than the marginal cost. This tells us that supply of new housing curve is upward sloping in the relative price of new housing. Look the following figure.

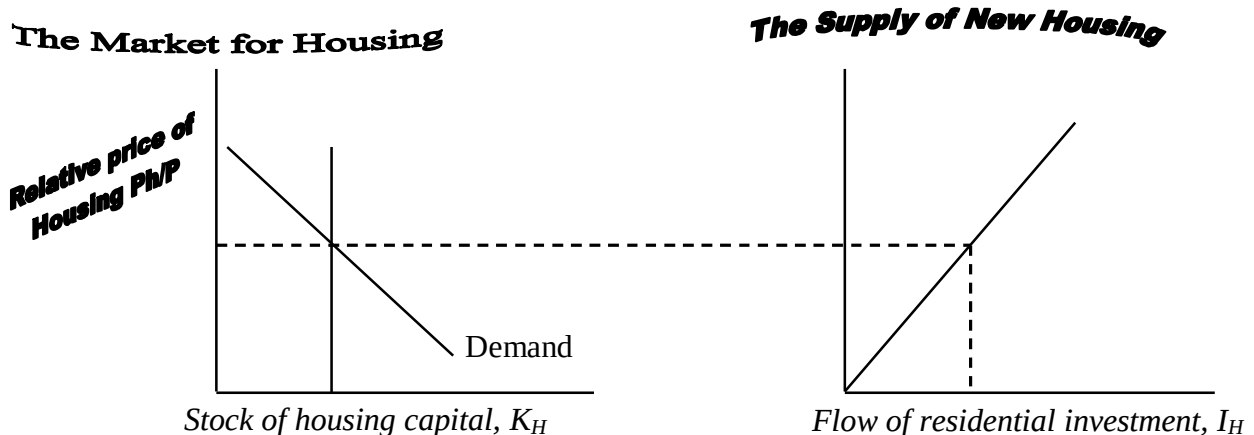


Fig: 2.3: The Housing Market

B. Changes in the Demand for and Supply of Housing

- ✚ Any change in the relative price of housing just involves movements along the existing curves, as per normal supply and demand curves.
- ✚ Any other change involves a shift of a curve which will have other flow on effects. For instance, a country's *population increases* either because of an *increase in the reproduction rate* or because of *higher net immigration*. This causes the following sequence of things to happen:

- i. Demand for housing curve shifts outwards
- ii. P_H/P increases to clear the housing market
- iii. Supply of new housing increases, increasing the future stock of housing (this will then cause a fall in P_H/P although it will still be higher than the original level).

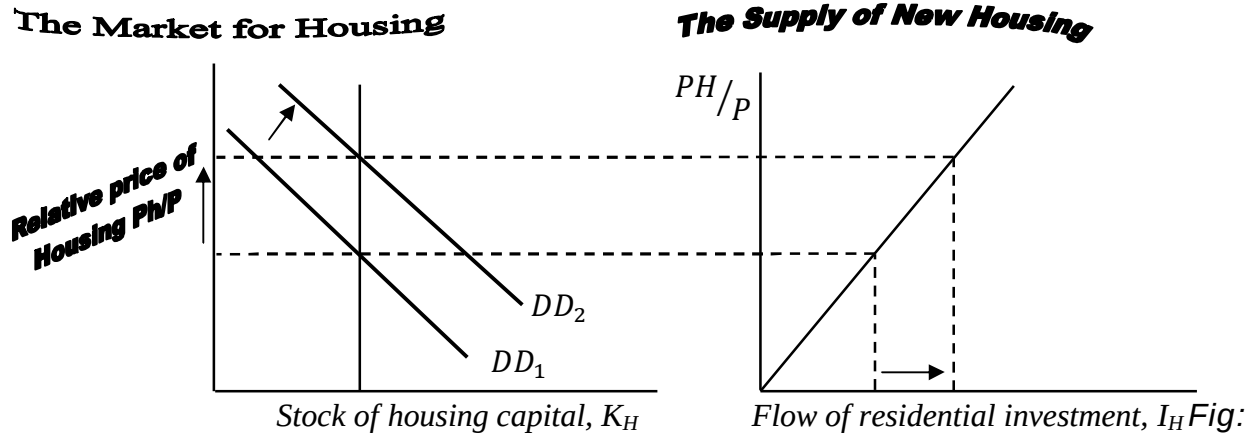


Figure 2.4: Housing Investment

There are many other things that can affect the demand for housing including: the *interest rate* (i.e. because it affects mortgage costs, or the *opportunity cost of holding wealth in housing*); better technology, especially in building materials and construction techniques (as it lowers the cost of supplying new houses and thus shifts the supply curve down); incomes of people (i.e. how much income people have affects their demand curves); and a host of other factors.

Summary of Residential Investment

- i. An increase in the interest rate increases the cost of borrowing for home buyers and reduces residential housing investment.
- ii. An increase in population and tax policies shift the residential housing-investment function.
- iii. In a boom, higher income raises the demand for housing and increases residential investment.

2.3.3. INVENTORY INVESTMENT

Inventory investment, the goods that businesses put aside in storage, is at the same time negligible and of great significance. It is one of the smallest components of spending—but its volatility makes it critical in the study of economic fluctuations.

2.3.3.1. Reasons for Holding Inventories

Before developing a theory of inventory investment let us see the reasons why inventories are held. Knowing the reasons for holding inventories will help us

understand better the factors which influence the level of inventory investment. There are four reasons for holding inventories:

i. Production Smoothing: Sales fluctuate, but many firms find it cheaper to produce at a steady rate. That means,

When sales < production, inventories rise.

When sales > production, inventories fall.

ii. Inventories as a Factor of Production

- ✓ Firms carry inventories of spare parts in case a machine breaks down, so that the entire assembly line does not come to a stop.
- ✓ Samples for retail sales purposes

iii. Stock-Out Avoidance

To avoid lost sales and profits when sales may be unexpectedly high e.g. a book retailer may carry multiple copies of a book if they are not sure how popular the book will be.

iv. Work in Process: Partly completed products are counted as inventories (i.e., Goods not yet completed are counted in inventory).

2.3.3.2. Accelerator Model of Inventory Investment

The **accelerator model** assumes that firms hold a stock of inventories that is proportional to the firm's level of output. Thus, if N is the economy's stock of inventories and Y is output, then

$$N = \beta Y$$

where β is a parameter reflecting how much inventory firms wish to hold as a proportion of output. Inventory investment I is the change in the stock of inventories, ΔN .

Therefore, $I = \Delta N = \beta \Delta Y$

The accelerator model predicts that inventory investment is proportional to the change in output.

- ✓ When output rises, firms want to hold a larger stock of inventory, so inventory investment is high.
- ✓ When output falls, firms want to hold a smaller stock of inventory, so they allow their inventory to run down, and inventory investment is negative.

Inventories and the Real Interest Rate

- ✚ Like other components of investment, inventory investment depends on the real interest rate. When a firm holds a good in inventory and sells it tomorrow rather than selling it today, it gives up the interest it could have earned between today and tomorrow. Thus, the real interest rate measures the opportunity cost of holding inventories.

When the interest rate rises, holding inventories becomes more costly, so rational firms try to reduce their stock. Therefore, an increase in the real interest rate depresses inventory investment.

✚ **Summary of Inventory Investment**

- i. Higher interest rates increase the cost of holding inventories and decrease inventory investment.
- ii. According to the accelerator model, the change in output shifts the inventory investment function.
- iii. Higher output during a boom raises the stock of inventories firms wish to hold, increasing inventory investment.

END OF THE CHAPTER!

Have U Achieved the Chapter Objectives?

(Go back & Read, then check whether u achieved the objectives defined at the beginning of the chapter)- Do the ff questions to better refresh your understanding.

✚ **Review Exercises**

1. In the Neoclassical model of business fixed investment, under what conditions will firms find it profitable to add to their capital stock?
2. What is Tobin's q , and what does it has to do with investment?
3. Explain why an increase in the interest rate reduces the amount of residential investment?
4. Use the Neoclassical model of investment to explain the impact of each of the following events on the real rental price of capital, the cost of capital and investment:
 - A. Anti-inflationary monetary policy raises the real interest rate
 - B. An earthquake destroys some of the capital stock.
 - C. Immigration of foreign workers increases the size of the labor force.
5. Suppose that the firm faces a Cobb–Douglas production function with two inputs: K is capital, L is labor. The production function is given as: $Y = AK^{0.3}L^{0.7}$.
 - What happens to the real rental price of capital if?
 - A. Immigration raises the labor force by 10 percent.

- B. A gift of capital/Machines from abroad raises the capital stock by 10 percent. (Use graph)
- C. Technological advancement raises the value of the parameter A by 10 percent.
6. Addis Ababa, the capital of Ethiopia experienced a large increase in population owing to high birth rate and rural-urban migration in recent times. Use the Neoclassical model of residential investment to predict the impact of this event on housing prices and residential investment?
7. Our theories of investment suggest that investment depends on national income. That is, higher income might induce firms to invest more. Explain why investment might depend on national income?
8. Suppose that the firm faces a Cobb–Douglas production function with two inputs: K is capital (the number of machines = 15), L is labor (the number of workers = 60). The production function is $Y = 100K^{0.3}L^{0.7}$. Suppose further the cost of capital to a Machine-rental company. The company buys Machines for \$6,000 each and rents them out to other businesses, and the economy's price level, P is \$25. The company faces a real interest rate r of 10 percent per year and the machines depreciate at 20 percent per year.
- A. Compute the marginal product of machines and the real cost of machines, and interpret the result (should the rental firm add to its capital stock -invest in new machines)?
- B. What happen to the firms' decision if the relative price of machines increases by 10% and 15% in order?
- C. What happen to the real rental price of capital/Machine if:
- Immigration raises the labor force by 10 percent.
 - An earthquake reduces the capital stock by 10 percent. (Support your explanation using graph).
 - Technological improvement raises the value of the parameter A by 10 percent.
9. If a firm invests out of retained profits rather than borrowed funds, will its investment decision still be affected by the changes in the real interest rate? Explain.
10. The theories of investment suggest that investment is related to the business cycle. Explain why investment rises during expansion and falls during recessions by referring to the theories?

CHAPTER THREE

MONEY DEMAND AND MONEY SUPPLY

Chapter Objectives:

At the end of this chapter, you will be able to:

- ✓ Explain who affects the money supply/ how the amount of money in an economy is determined?
- ✓ Discuss how central bank policies influence the money supply
- ✓ Explain the motives why people hold money and/or discuss what factors affect the demand for money.

3.1. INTRODUCTION: What is Money?

What economists mean by money?

Economists tend to use the term “money” differently than most people. While people may say a rich person “*has a lot of money*”, economists use the term “money” to mean stock of assets that can be readily used /the most liquid part of a person's assets/ to make transactions.

➤ Why do we need to have money? There are three basic purposes/functions that money serves:

📊 **Money serves as a store of value:** - money is a way to transfer purchasing power from the present to the future. If I work today and earn \$100, I can hold the money and spend it tomorrow, next week, or next month. Of course, *money is an imperfect store of value*: if prices are rising, the amount you can buy with any given quantity of money is falling. Even so, people hold money because they can trade the money for goods and services at some time in the future.

📊 **Money serves as a unit of account:** - money provides the terms in which prices are quoted and debts are recorded. Microeconomics teaches us that resources are allocated according to relative prices —the prices of goods relative to other goods —yet stores post their prices in dollars and cents. A car dealer tells you that a car costs \$20,000, not 400 shirts (even though it may amount to the same thing). Similarly, most debts require the debtor to deliver a specified

number of dollars in the future, not a specified amount of some commodity. *Money is the yardstick with which we measure economic transactions /i.e., it is an agreed measure for stating prices/.*

 **Money serves as a medium of exchange:** - money is what we use to buy goods and services. When we walk into stores, we are confident that the shopkeepers will accept our money in exchange for the items they are selling. The ease with which money is converted into goods and services—is sometimes called **money's liquidity**. This is a fundamental reason why we hold it, and is a definite advantage over a barter economy. Without money, in order to carry out transactions we would have to wait for a **“double coincidence of wants”** - that is, we would have to wait until we found a person who would give us something we wanted in return for something we have.

3.2. The Supply of Money

The quantity of money available in an economy is called the **money supply**. Because money is the stock of assets generally accepted for transactions, the quantity of money is the quantity of those assets. In a simple economy that uses only gold as a medium of transaction, the quantity of money is the quantity of gold in that economy. But how can we measure the quantity of money in more complex economies? The answer is not obvious, because no single asset is used for all transactions. People can use various assets, such as cash or checks, to make transactions, although some assets are more convenient than others. This ambiguity leads to numerous measures of the quantity of money.

The most obvious asset to include in the quantity of money is **currency**, the sum of outstanding paper money and coins. Most day-to-day transactions use currency as the medium of exchange. A second type of asset used for transactions is **demand deposits**, the funds people hold in their checking accounts. If most sellers accept personal checks, assets in a checking account are almost as convenient as currency. In both cases, the assets are in a form ready to facilitate a transaction. Demand deposits are, therefore, added to currency when measuring the quantity of money.

Many other assets can also become candidates for inclusion. Funds in savings accounts, for example, can be easily transferred into checking accounts; these assets are almost as convenient for transactions. Money market mutual funds allow investors to write checks against their accounts, although restrictions sometimes apply with regard to the size of the check or the number of checks written. Because it is hard to judge which assets should be included in the money stock, various measures are available. Table 2.1 presents four measures of the money stock.

Table 3.1: Four alternative measures of the money stock

<i>Symbol</i>	<i>Assets Included</i>
C	Currency
M_1	Currency plus demand deposits (i.e., checking account deposits)
M_2	M_1 plus saving deposits, small time deposits, etc
M_3	M_2 plus large time deposits, repurchase agreements, etc.

N.B: Liquidity declines as we move from currency (the most liquid asset) to M_3 .

- ✓ The most common measures for studying the effects of money on the economy are M_1 and M_2 . There is no consensus, however, about which measure of the money stock is best.

In this chapter, we focus on M_1 , that is, the *Money supply includes the currency (the sum of outstanding paper money and coins) in the hands of the public, and demand deposits (i.e., the funds people hold in their checking accounts) at banks that households can use on demand for transactions, both of which can be used directly to pay for goods and services.*

$$\text{Money supply} = \text{Currency} + \text{Demand deposits}$$

$$\therefore M = C + D$$

Where, M- the money supply, C currency/cash and D demand deposits /checking account deposits/

The Money Supply Process

- *Now, to better understand the money supply process, we must understand the interaction between currency and demand deposits and how the Central bank policy influences these two components of the money supply.*
- **There are three Players in the Money Supply Process**
 - ✓ Central banks (National Bank of Ethiopia in our case)- in charge of money supply management
 - ✓ Banks (depository institutions; financial intermediaries)
 - ✓ Depositors (individuals and institutions)
- *The interaction between these three sets of people determines the money supply.*

100 Percent- Reserve Banking

- *Imagine a world without banks, where all money takes the form of currency and the quantity of money is simply the amount of currency that the public holds (Say there is \$1000 of currency in the economy).*
- *Now introduce banks, and at first assume that banks accept deposits but do not make loans. The only purpose of the banks is to provide a safe place for deposits to keep their money. The deposits that banks have received but have not lent out are called **reserves**. Some reserves are held at the Central bank. If all deposits are held as reserves, such a system is called '**100 Percent Reserve Banking**'.*
- *Before the creation of the first Bank, the money supply was \$1000 of currency and after the creation of the bank the money supply is \$1000 of demand deposits; if the public is depositing the entire money in the bank. A dollar deposited in a bank reduces currency by one dollar and raises deposits by one dollar, so the money supply remains the same. If banks hold 100 percent of deposits in reserve, the banking system does not affect the supply of money.*

Fractional-Reserve Banking

- *Now imagine that banks start to use some of their deposits to make loans – for example, to families who are buying houses or firms that are investing in new plants and equipment. The banks earn interest from loans. The banks must keep some reserves on hand so that reserves are available whenever depositors want to make withdrawals. This kind of system is called '**fractional reserve banking**,' a system under which banks keep only a fraction of their deposits in reserve.*

- Assume the reserve deposit ratio-the fraction of deposits kept in reserve-is 20%. The first bank increases the supply of money by \$800 when it makes the loan. Before the loan is made the money supply was \$1000, equal with the deposits in the first bank. After the loan is made, the money supply is \$1800; the depositors still have a demand deposit of \$1000, but now borrower holds \$800 in currency. *Thus, in a system of fractional reserve banking, banks create money.*

The creation of money does not stop with the first bank. If the borrower deposits \$800 in another bank (or if the borrower uses the \$800 to pay someone who then deposits it), the process of money creation continues despite it does not create an infinite amount of money.

Letting rr denote the *reserve-deposit ratio*, the amount of money that the original \$1000 creates is as shown in Table-2.2.

Table-2.2: The Money Supply Process

<i>Description</i>	<i>Quantity (\$)</i>
1. Original deposit	1000
2. First bank lending	$(1 - rr) \times 1000$
3. Second bank lending	$(1 - rr)^2 \times 1000$
4. Third bank lending	$(1 - rr)^3 \times 1000$
Total money supply	$[1 + (1 - rr) + (1 - rr)^2 + (1 - rr)^3$ $+ \dots \dots] \times 1000$ $= (1/rr) \times 1000$

- Each \$1 of reserve generates $\$(1/rr)$ of money. If in our example above $rr = 0.2$, so the original \$1000 generates \$5000 of money (i.e., $(1/0.2) \times \$1000 = \5000). *The banking system ability to create money is the primary difference between banks and other financial institutions.* Financial markets have the important function of transferring the economy's resources from those households and firms that wish to borrow to buy investment goods to be used in future production. This process is called '*financial intermediation*'. Yet of these financial institutions, only banks have the legal authority to create assets *that* are part of the money supply,

such as *checking accounts*. Therefore, banks are the only financial institutions that directly influence the money supply.

Note that although the system of fractional–reserve banking creates money, it does not create wealth. When a bank loans out some of its reserves, it gives borrowers the ability to make transactions and therefore increases the supply of money. The borrowers are also undertaking a debt obligation to the bank, however, so the loan does not make them wealthier. *In other words, the creation of money by the banking system increases the economy's liquidity, not its wealth.*

3.2.1. A Model of the Money Supply- What determines the money supply?

This model of the money supply is based on fractional-reserve banking. The model has three exogenous variables.

- i. **Monetary Base (B):** It is the total number of dollars held by the public as currency (C) and by the banks as reserves (R). It is directly controlled by the Central banks.
 - ii. **Reserve-Deposit Ratio (rr):** It is the fraction of deposits that banks hold in reserve. It is determined by the business policies of banks, and the laws regulating banks.
 - iii. **Currency-Deposit Ratio (cr):** It expresses the preferences of the public about how much money to hold in the form of currency (C) and how much to hold in the form of demand deposits (D).
- ✓ *This model shows how the money supply depends on the monetary base, the reserve-deposit ratio, and the currency-deposit ratio.* In other words, it examines how Central bank policies and the choices of banks and households influence the money supply.

The model begins with the definition of money supply and monetary base.

$$M = C + D \text{ — — — — — (1)}$$

$$B = C + R \text{ — — — — — (2)}$$

- ✓ *The first equation states that the money supply is the sum of currency and demand deposits.*
 - ✓ *The second equation states that the monetary base is the sum of currency and bank reserves.*
- If we divide the first equation by the second, we get:

$$\frac{M}{B} = \frac{C + D}{C + R} = \dots \dots \dots (3)$$

If we divide both the top and bottom of the expression on the right by D:

$$\frac{M}{B} = \frac{C/D + 1}{C/D + R/D} \dots \dots \dots (4)$$

Note that C/D is the currency-deposit ratio (cr), and that R/D is the reserve-deposit ratio (rr). Making these substitutions, and bringing B from the left to the right side of the equation, we obtain:

$$M = \frac{cr + 1}{cr + rr} \times B \dots \dots \dots (5)$$

This equation shows how the money supply depends on the three exogenous variables. Money supply is proportional to the monetary base. The factor of proportionality $[(cr + 1)/(cr + rr)]$ is denoted by m and is called the ‘money multiplier’.

That is, $M = m \times B \dots \dots \dots (6)$

Each dollar of the monetary base produces m dollars of money. Since the monetary base has a multiplier effect on the money supply, the monetary base is sometimes called ‘**High-Powered Money**.’

Numerical Example:

If the monetary base is \$300 billion, the reserve deposit ratio is 0.1 and currency-deposit ratio is 0.4; calculate the money multiplier and the money supply.

Answer: $m = \frac{0.4+1}{0.4+0.1} = 2.8$

Money supply (M) = $2.8 \times 300 = 840$

➤ Each dollar of the monetary base generates 2.8 dollars of money, so the total money supply is \$840 billion.

The influence of the three exogenous variables (B , rr , and cr) on the money supply can be summarized as follows.

- i. The money supply is proportional to the monetary base. Thus, an increase in the monetary base increases the money supply by the same percentage.

- ii. The lower the reserve-deposit ratio, the more loan banks make, and the more money banks create from every dollar of reserves. Thus, a decrease in the reserve-deposit ratio raises the money multiplier and the money supply.
- iii. The lower the currency-deposit ratio, the fewer dollars of the monetary base the public holds as currency, the more base dollars banks hold as reserves, and the more money banks can create. Thus, a decrease in the currency-deposit ratio raises the money multiplier and the money supply.

3.1.2. Instruments of Monetary Policy

The Central bank controls the money supply indirectly by altering either the monetary base or the reserve-deposit ratio. To do this, Central Banks have three instruments of monetary policy: open market operations, reserve requirements, and the discount rate.

- i. **Open Market Operations (OMO):** (Policy instruments that the Central bank uses most often). These are the purchases and sales of government bonds by the Central bank.
 - *When the Central bank buys bonds from the public, the dollars it pays for the bonds increase the monetary base and thereby increase the money supply.*
 - *When the Central bank sells bonds to the public, the dollars it receives reduce the monetary base and thus decrease the money supply.*
 - ii. **Reserve Requirements:** Ratio of cash reserves to deposits that banks are required to maintain. These are Central bank regulations that impose on banks a minimum reserve-deposit ratio. An increase in reserve requirements raises the reserve-deposit ratio and thus lowers the money multiplier and the money supply.
 - iii. **Discount Rate:** It is the interest rate that the Central bank charges when it makes loan to banks. Banks borrow from the Central bank when they find themselves with too few reserves to meet reserve requirements. The lower the discount rate, the cheaper are borrowed reserves, and the more banks borrow at the Central bank's discount window. Hence, reductions in the discount rate raise the monetary base and the money supply.
- ✚ *Although these three instruments give central banks substantial power to influence the money supply, central banks cannot control the money supply perfectly. Banks discretion in conducting business can cause the money supply to change in ways the central banks did not anticipate. For instance, banks may choose to hold excess reserves. The higher the amount of excess reserves, the higher the reserve-deposit ratio and the lower the money supply.*

3.3. Demand for Money

We now turn to the other side of the money market, and examine what determines money demand. The demand for money arises from the important functions money performs, such as units of account, store of value and medium of exchange. Money as a unit of account does not by itself generate any demand for money, because one can quote prices in dollars without holding any. By contrast, money can serve its other two functions only if people hold it.

The purpose of studying money demand is to gain insight about the money demand functions. Just as studies of the consumption function rely on microeconomic models of the consumption decision, studies of the money demand function rely on microeconomic models of the money-demand decision.

Therefore, in this section, we will learn about the determinants of money demand by referring to the prominent microeconomic models of the money demand decision. *The theories of money demand emphasize the role of money either as a store of value or as a medium of exchange.*

3.3.1. Quantity Theory of Money Demand

- ✚ The concept of money holdings began to take a formal shape in the classical quantity theory. The American economist Irving Fisher (1911) examine the link b/n the total quantity of money M and the total amount of spending on final goods and services produced in the economy $P * Y$, where P is the price level and Y is real GDP, V is the velocity of money.
- ✚ Velocity, V , represents the number of times per year that a dollar is used in buying the total amount of goods and services produced in the economy.

$$V = \frac{P * Y}{M} \text{ — — — — — (1)}$$

Multiplying both sides of this equation by M , we obtain the *equation of exchange*, which relates nominal income to the quantity of money and velocity:

$$MV = PY \text{ — — — — — (2)}$$

The equation of exchange thus states that quantity of money multiplied by the number of times that this money is spent in a given year must be equal to nominal income. When, M , changes, nominal income, PY changes in the same direction. *To convert the equation of exchange into a*

theory of how nominal income is determined requires an understanding of the factors that determined velocity.

- Irving Fisher believed that velocity is determined by the institution in an economy that affects the way individuals conduct transactions. He thought the institutional and technological features of the economy would affect velocity only slowly over time, so **velocity would normally be constant in the short run**. Fisher's view that velocity is fairly constant in the short run transforms the equation of exchange into the quantity theory of money, which states that *nominal income is determined solely by movements in the quantity of money*. For classical economists, the quantity theory of money provided an explanation of movements in the price level (I.e., *Changes in M determines changes in P*).

The quantity theory as a theory of Money Demand

Dividing both sides of the exchange equation by V , thus rewriting it as

$$M = \frac{1}{V}PY \text{ — — — — — (3)}$$

When the money market is in equilibrium, the quantity of money M that people hold equals the quantity of money demand, M^d , so we can replace M in the equation by M^d , and using k to represent $1/V$, we can rewrite the equation as

$$M^d = kPY \equiv \left(\frac{M}{P}\right)^d = kY \text{ — — — — — (4)}$$

Because k is a constant, the level of transaction generated by a fixed level of nominal income PY determines the quantity of money M^d that people demand. **Therefore, Fisher's quantity theory of money suggests that the demand for money is purely a function of income, and interest rates have no effect on the demand for money.**

Fisher believed that people hold money only to conduct transactions and have no freedom of action in terms of the amount they want to hold, so he concludes that *the demand for money is determined by:*

- i. The level of transactions generated by the level of income, Y and
- ii. The institutions in the economy that affect the way people conduct transactions that determine velocity and hence k .

3.3.2. Keynes's Liquidity Preference Theory: why do people hold money?

In his famous 1936 book “*The General Theory of Employment, Interest, and Money*”, Keynes developed a theory of money demand which he called **liquidity preference theory**.

He postulated that there are three motives behind the demand for money: the transactions motive, the precautionary motive, and the speculative motive.

i. Transactions Motive: - Keynes emphasized that this component of the demand for money is determined primarily by the level of people's transactions. The transactions demand for money arises from the lack of synchronization of receipts and disbursements. In other words, people aren't likely to get paid at the exact instant you need to make a payment, so between paychecks people keep some money around in order to buy stuff. *Keynes believed that transactions demand for money to be proportional to income.*

ii. Precautionary Motive: - Keynes also recognized that people hold money not only to carry out current transactions, but also as cushion against an unexpected need. Because people are uncertain about the payments they might want, or have, to make. When people are holding precautionary money balances, you can take advantages of the sale. Keynes believed that the amount of precautionary money balances people want to hold is determined primarily by the level of transactions that they expected to make in the future and that these transactions are proportional to income. *So he considered the demand for precautionary money balances to be proportional to income.*

iii. Speculative Motive: - The transactions motive and the precautionary motive for money emphasized medium-of-exchange function of money, for each refers to the need to have money on hand to make payments. He considered that wealth is tied closely to income; the speculative component of money demand would be related to income. Keynes believed that interest rates have an important role to play in influencing the decisions regarding how much money to hold as a store of wealth.

Keynes divided the assets that can be used to store wealth into **two categories: money and bonds**, and examines why individuals hold their wealth in the form of money rather than bonds. Keynes assumed that the expected return on money was zero in his time, unlike today. ***For bonds, there are two components of the expected return. the interest***

payment and the expected rate of capital gains. As we know, when interest rates rise, the price of a bond falls. If you expect interest rates to rise, you expect the price of the bond to fall and suffer negative capital gains. In this case, people would want to store their wealth as money because its expected return is higher; its zero return exceeds the negative return on the bond.

Keynes assumed that when interest rates are low, people expect the interest rate on bonds to rise in the future and so expect to suffer capital loss. Therefore, people will be more likely to hold their wealth as money rather than bonds and the demand for money will be high and the converse holds true. Therefore, money demand is negatively related to the level of current interest rates.

- In general, Keynes developed the following demand for money equation, known as the **liquidity preference function**, which says that the demand for real money balances $(M/P)^d$ is a negative function of interest rate, i and a positive function of income, Y :

$$\frac{M^d}{P} = L(i, Y) \text{ — — — — — (5)}$$

- Because the transactions motive and precautionary motive demand for money is positively related to real income Y , speculative motive demand for money is negatively related to interest rate i , the demand for real money balances M^d/P can be rewritten as:

$$\frac{M^d}{P} = L_1(Y) + L_2(i) \text{ — — — — — (6)}$$

where L_1 means the transactions demand for money; L_2 means the speculative demand for money.

Implication for velocity

Keynes abandoned the classical view that velocity is constant, emphasized the importance of interest rates. By deriving the liquidity preference function for velocity PY/M , we can see that Keynes's theory of the demand for money implies that velocity is not constant but instead fluctuates with movements in interest rates. The liquidity preference equation can be rewritten as:

$$\frac{P}{M^d} = \frac{1}{L(i, Y)} \text{ — — — — — (7)}$$

Multiplying both sides of this equation by Y and recognizing that M^d can be replaced by M because they must be equal in money market equilibrium, we solve for velocity:

$$V = \frac{PY}{M} = \frac{Y}{L(i, Y)} \text{ --- (8)}$$

- Keynes liquidity preference theory of the demand for money indicates that velocity has substantial fluctuations as well. E.g: $\uparrow i \Rightarrow \downarrow L(i, Y) \Rightarrow \uparrow V$.

3.3.3. Portfolio Theories of Money Demand

✚ Theories of money demand that emphasizes the role of money as a store of value are called ‘**portfolio theories**’. These theories stress that people hold money as part of their portfolio of assets. The key insight is that money offers a different combination of risk and return than other assets. In particular, money offers a safe (nominal) return, whereas the prices of stocks and bonds may rise or fall. **Thus, some economists have suggested that households choose to hold money as part of their optimal portfolio.**

✚ **Portfolio theories** predict that the demand for money should depend on the risk and return offered by money and by the various assets households can hold instead of money. In addition, money demand should depend on total wealth, because wealth measures the size of the portfolio to be allocated among money and the alternative assets.

For instance, we might write the money demand function as:

$$(M/P)^d = L(r_s, r_b, \pi^e, W)$$

Where r_s , is the expected real return on stock, r_b , is the expected real return on bonds, π^e is the expected inflation rate, and W is real wealth.

- An increase in r_s , or r_b reduces money demand, because other assets become more attractive.
- An increase in π^e also reduces money demand, because money becomes less attractive.
- An increase in W raises money demand, because higher wealth implies a larger portfolio.

3.3.4. Transaction Theories of Money Demand

✚ Theories of money demand that emphasizes the role of money as a medium of exchange are called ‘**transactions theories**.’ These theories acknowledge that money, unlike other assets, is held to make purchases.

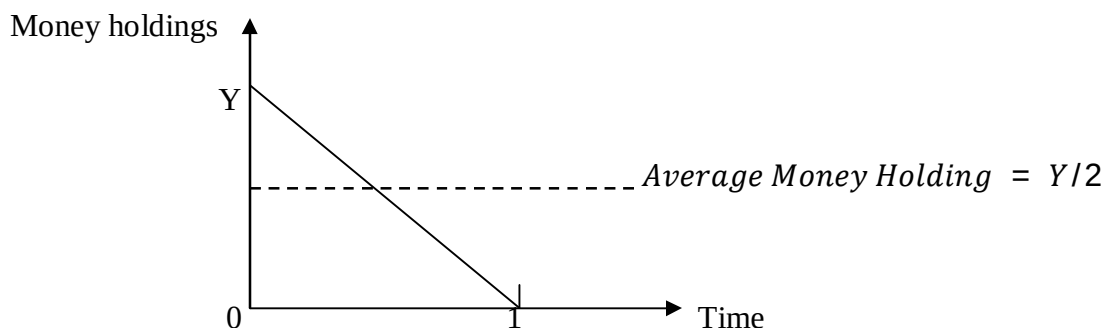
✚ Transactions theories of money demand take many forms, depending on how one model the process of obtaining money and making transactions.

- All these theories assume that money has the cost of earning a low rate of return and the benefit of making transactions more convenient.
 - People decide how much money to hold by trading off these costs and benefits.
- ✚ To see how transactions theories explain the money demand function, let's look at one prominent model of this type: the '**Baumol-Tobin Model**'. This model was developed in 1950s by economists *William Baumol* and *James Tobin*, and it remains a **leading theory of money demand**. *William Baumol and James Tobin independently developed similar demand for money models, which demonstrated that even money balances held for transactions purposes are sensitive to the level of interest rates.*

3.3.4.1. The Baumol-Tobin Model of Cash Management

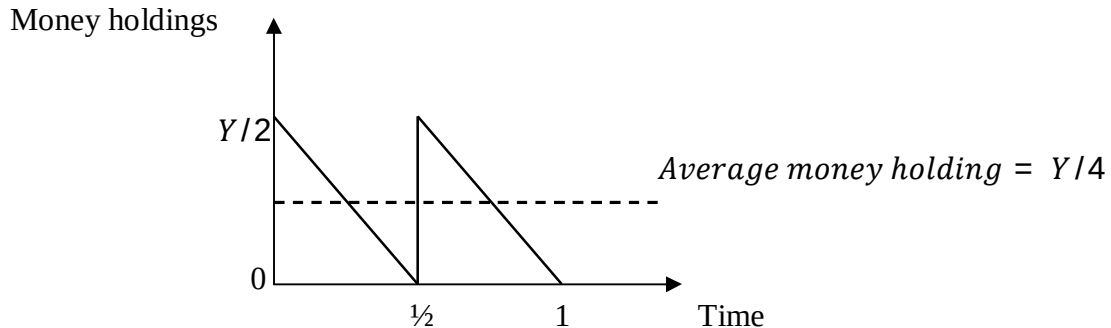
- ✚ The Baumol-Tobin model analyses the costs and benefits of holding money.
- **The benefit of holding money is convenience**; people hold money to avoid making a trip to the bank every time they wish to buy something.
 - **The cost of this convenience is the forgone interest** they would have received had they left the money deposited in a savings account that paid interest.
- ✚ To see how people trade off these costs and benefits, consider a person who plans to spend Y dollars gradually over the course of a year. (For simplicity, assume that the price level is constant, so real spending is constant over the year). How much money should he hold in the process of spending this amount? That is, what is the optimal size of average cash balances?
- ✚ Consider the different possibilities.
- The person could withdraw the Y dollars at the beginning of the year and gradually spend the money. Figure 3.1 shows the money holding over the course of the year under this plan. The money holding begin the year at Y and end the year at zero, averaging Y/2 over the year.

Figure-3.1: Money holdings with one trip to Bank



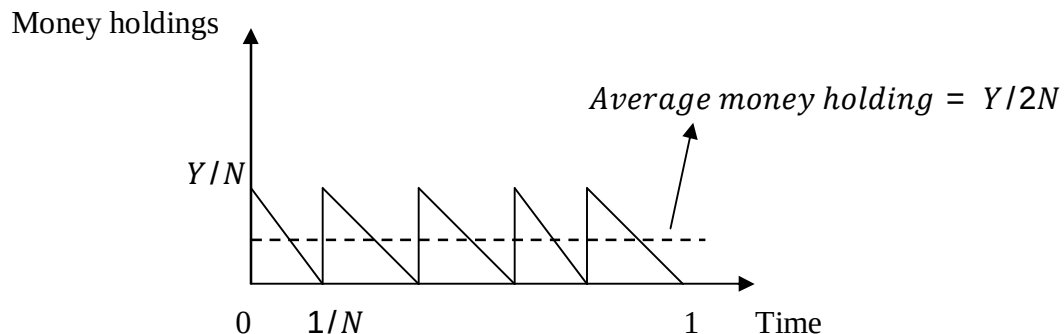
- A second possible plan is to make two trips to the bank. In this case, the person withdraws $Y/2$ dollars at the beginning of the year, gradually spends this amount over first half of the year, and then makes another trip to withdraw $Y/2$ for the second half of the year. Figure-3.2 shows that money holdings over the year vary between $Y/2$ and zero, averaging $Y/4$. This plan has the advantage that less money is held on average, so less interest is forgone, but it has the disadvantage of requiring two trips to the bank rather than one.

Figure: 3.2: Money holdings with two trips to bank ($N=2$)



- Generally, suppose the individual makes N trips to the bank over the course of the year. On each trip, he/she withdraws Y/N dollars; then spends the money gradually over the following $1/N$ th of the year. Figure-3.3 shows that money holdings vary between Y/N and zero, averaging $Y/2N$.

Figure: 3.3: Money Holdings with N trips to the Bank



- **The optimal choice of N is determined by different factors.** The greater N is, the less money is held on average and the less interest is forgone. But as N increases, so does the inconvenience of making frequent trips to the bank.
- ✚ Suppose that the cost of going to the bank is some fixed amount F . F represents the value of the time spent traveling to and from the bank and waiting in line to make the withdrawal.

(For example, if a trip to the bank takes 15 minutes and a person's wage is \$12 per hour, then F is \$3). Also let i denote the interest rate; because money does not bear interest, i measures the opportunity cost of holding money.

Now we can analyze the optimal choice of N , which determines money demand. For any N , the average amount of money held is $Y/(2N)$, so the forgone interest is $iY/(2N)$. Because F is the cost per trip to the bank, the total cost of making trips to the bank is FN .

➤ The total cost the individual bears is the sum of forgone interest and the cost of trips to the banks.

Total cost = Forgone interest + Cost of trips

$$C = iY/2N + FN$$

➤ The larger the number of trips, N , the smaller the forgone interest and the larger the cost of going to the bank.

Figure-3.4 shows how total cost depends on N . There is one value of N that minimizes total cost. **The optimal value of N denoted by N^* is:**

$$N^* = \sqrt{\frac{iY}{2F}}$$

(Note: To find the cost minimizing N , take the derivative of total cost with respect to N , set it equal to zero. I. e., $N^* = \frac{dC}{dN} = -\frac{iYN^{-2}}{2} + F = 0$)

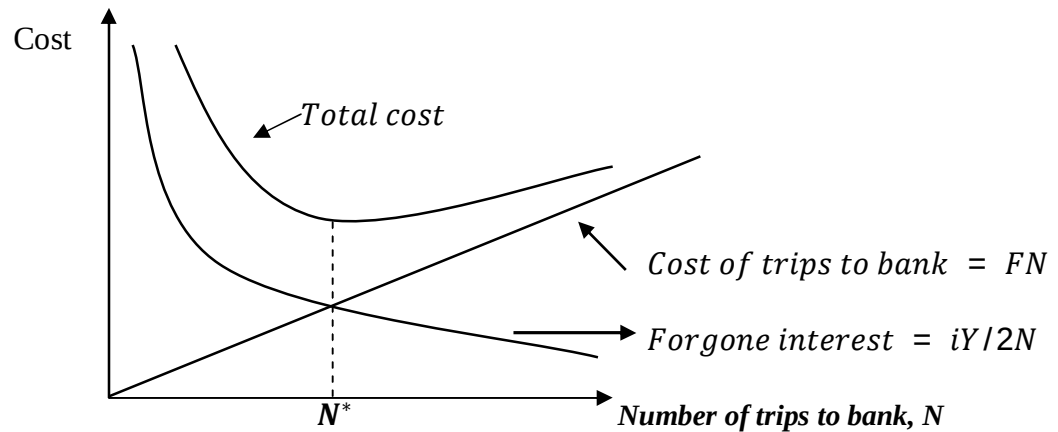
$$\text{Average Money Holding} = \frac{Y}{2N^*} = \sqrt{\frac{YF}{2i}}$$

$$\text{I.e., } \left(\frac{M}{P}\right)^d = \sqrt{\frac{YF}{2i}} = L(i, Y, F)$$

Therefore, the Baumol-Tobin money demand function shows that:

- ✓ The higher is i , the more income gain from holding interest bearing assets, the less likely to hold cash: Therefore $\uparrow i$, $\downarrow M^d$.
- ✓ The Higher income, Y , the higher is the demand for money, M^d
- ✓ The higher the cost of a trip to the bank, F , the higher is the demand for money, M^d

Figure: 3.4: The Cost of Money Holding



N^* is Number of trips that Minimize total cost.

Class Activity: Think of the impact of ATMs on money demand!

In recent years, Automatic Teller Machines are widely available in our country, Ethiopia.

- ✓ How do you think this affected N^* and money demand? Explain.
- ✓ Which categories of money demand is affected by introduction of ATMs: transactions or Portfolio demand for money or both? How?

Review Exercises

1. Explain how banks create money?
2. What are the three ways in which central banks can influence the money supply?
3. Briefly explain the notion of the quantity theory of money demand and Keynes liquidity preference theory.
4. Explain the difference between portfolio and transactions theories of money demand?
5. According to the Baumol-Tobin model, what determines how often people go to the bank? What does this decision have to do with money demand?
6. Suppose you spend birr 50,000 per year in cash/i.e., $Y = 50,000/$, and a trip to the bank takes 0.5 hour, and you earn birr 100 per hour. Assume further that the interest rate on your saving account is 5 percent. Calculate how often you should go to the bank to withdraw cash according to the Baumol-Tobin model?

CHAPTER FOUR

THE LABOUR MARKET

Chapter Objectives:

At the end of this chapter, you will be able to:

- ✓ *Elaborate how people make choices between leisure and hours of work;*
- ✓ *Identify the income and substitution effects of a wage increase on labour supply;*
- ✓ *Mention and discuss the determinants of labour demand;*
- ✓ *Show how the interaction b/n supply of and demand for labour determines wage rate and employment in the neoclassical model of labour market;*
- ✓ *Distinguish between the Neoclassical, Keynesian and New Keynesian view of the labour market and their implications;*
- ✓ *Mention and explain the causes of unemployment;*
- ✓ *Discuss how government policy interventions & firms decision affects employment.*

4.1. Introduction

 We have learnt about two major components of GDP, consumption and investment. We discussed the exports, and imports in the first half of the course. Another way of measuring GDP is through the *income approach*, which involves *factors of production*. We have already learnt about capital from the section on investment and so what we will now do is learn about the other important factor, labor.

 **Labor market** is where an important factor of production is traded with implications for the level of output produced. Knowing about labor as well as capital is very important to appreciate how the supply of aggregate output in the short-run, and then later on in the long-run is to be developed. We need to understand how labor markets work if we want to understand how economies work. However, understanding how the labor markets work is not easy as labor markets can exhibit substantial changes over time: and significant differences can exist in labor markets of different countries.

 In this section, therefore, we will attempt to learn how the labor market works by looking at neoclassical microeconomics model and the New Keynesian view of the labour market.

4.2. Neoclassical Microeconomics Model Of The Labour Market

In this section, we will develop the neoclassical model of labour supply and labour demand, and finally using the two components of the labour market, we will define the wage rate and amounts of labour traded/employed.

4.2.1. The Neoclassical Theory Of Labor Supply

To hold a paid job, you must first have decided to do so. This is the starting point of the so-called “Neoclassical” theory of labour supply. It posits that each individual disposes of a limited amount of time, which s/he chooses to allocate between paid work and leisure.

✓ *As such, the Neoclassical theory of labour supply is based on a model of a consumer making a choice between consuming goods and leisure.*

Therefore, in this section, we consider a simple model in which a person chooses the amount of consumption goods to consume and the amount of leisure to “consume”, and hence the amount of labor to supply, subject to a limited amount of resources. By doing so we will understand the factors influencing how much labor people want to supply (i.e. how much they want to work).

Choice Between Consumption and Leisure

It is assumed that consumer derives utility from two sources:

The quantity of consumer goods (aggregated to C with price normalized to unity) and the amount of leisure (time devoted to other activities than work, denoted by L)

Two key assumptions used by the model:

- i. People act to maximize their welfare, or utility (i.e. people do the best they can with what they have).
 - ii. People experience diminishing marginal benefit to consuming extra consumption goods or extra leisure (i.e. their welfare goes up but at a diminishing rate).
- ✓ The problem thus facing people is to decide how many goods and services to consume, how much leisure to consume and thus how much labor to supply so as to maximize their welfare subject to the resources they have.

We assume that a worker wants to achieve the mix of real income and leisure that is most satisfactory to him or her. To develop the labour supply model, we use the

microeconomics model of consumer behavior where the consumer maximizes utility subject to his/her budget constraint.

Notations:

R = The amount of non-labor income

C = the amount of consumer goods (aggregated to C with price normalized to unity)

T = the endowment of time a person has (say 24 hours in a day).

h = the amount of labour in hours supplied by the person.

L = the amount of leisure “consumed” by the person.

w = the going hourly real wage rate.

A. Consumer's Preferences

The trade-off b/n consumption and leisure can be depicted by the help of indifference curve analysis of a utility function, $U(C, L)$, where C and L designate, respectively the consumption of goods and the consumption of leisure. Assume the standard assumption that people prefer more consumption and more leisure to less of each of them. The following figure presents consumer's preference b/n the consumption of goods and the consumption of leisure.

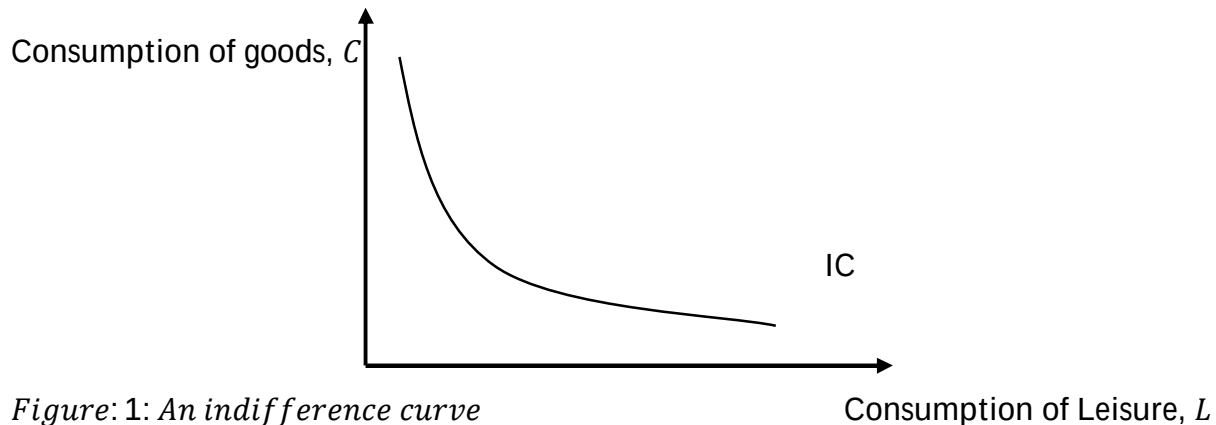


Figure: 1: An indifference curve

Note: The indifference curve exhibits the usual properties.

- The slope of the indifference curve at a given point defines the marginal rate of substitution b/n consumption and leisure (MRS_{CL}). It represents the hours of leisure which a consumer must give up in exchange for an hour of work or consumption goods, for his/her level of satisfaction to remain unchanged.

B. Consumer's Budget Constraint

Now let's consider all the resources available to a person. The consumer has non-labor income (R) and 24 hours a day which s/he can use to work and get income (where wT is the wage income from working all the possible hours available). People can then use this income to buy consumption goods (C) or leisure hours (wL). In doing this you should note that to buy one hours worth of leisure it costs the person one hour worth of working, which is just the wage rate. The wage rate is the opportunity cost of leisure. What we are doing here is conceptualizing labor income as though individuals work all the possible hours available to them and then use the proceeds to buy leisure time. Doing it this way makes it clear what the total endowment of individuals is, and the total possible income individuals can receive from their endowments. This gives us a budget constraint of the form:

$$C + wL = wT + R$$

This equation states that a person can use his/her income to either buy consumption goods, or to “buy” leisure time. Here the price of the leisure time we “buy” is the opportunity cost of that time, which simply equals the wage income we did not earn, but could have.

The following figure illustrates the consumer's budget constraint line

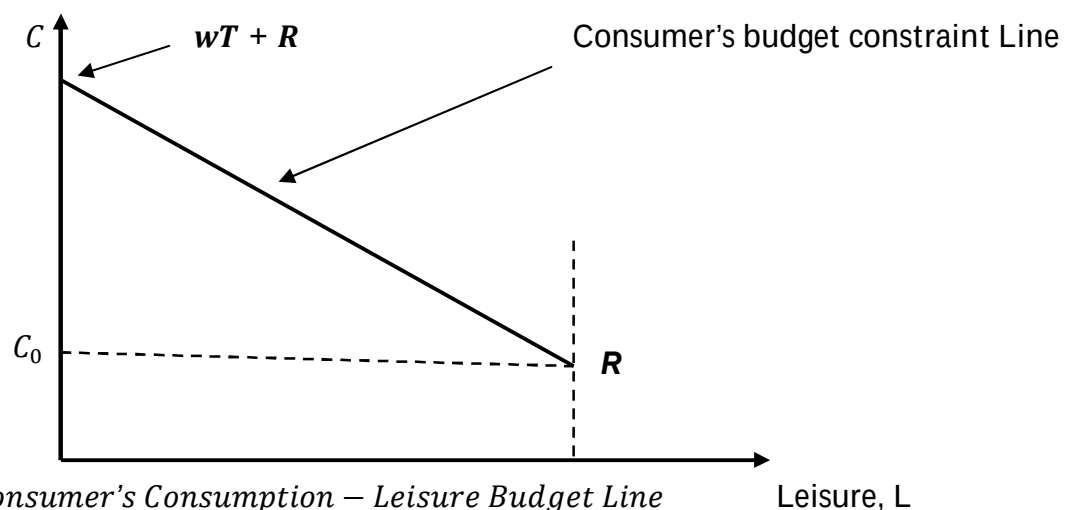


Figure: 2: Consumer's Consumption – Leisure Budget Line

The total amount of consumption that the person can have is amounted to, $wT + R$, if s/he allocates her/his time on working only.

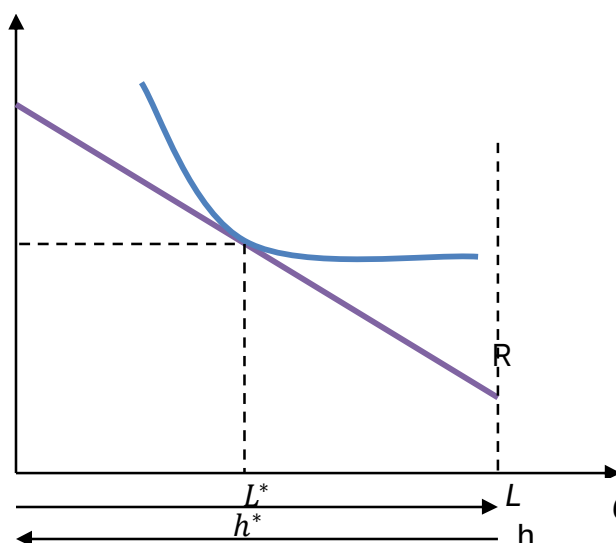
- If the persons enjoy all their time, they can still have positive amounts of consumption amounted to C_0 .
- The slope of the budget line can be worked out recalling that any changes in consumption or leisure expenditures must not violate the budget constraint, that is

$$\Delta C + w\Delta L = 0 \quad \text{i.e.,} \quad \frac{\Delta C}{\Delta L} = -w$$

C. Consumer's Choice-Optimal Allocation Of Time b/n Work and Leisure

- So far, we have developed the consumer's consumption-leisure preference curve and the Consumer's Consumption-Leisure Budget Line. Now, let's define consumer equilibrium.
- The optimal choice is where people are not wasting any of the resources they have at their disposal. That is, they are consuming all of the resources they have and the way they are consuming them cannot be rearranged to make them any better off. This situation is characterized by the budget line just touching the highest possible achievable indifference curve:

Consumption of goods, C



- The point at which the budget constraint line and the indifference curve meets gives us the *optimal consumption-leisure choice* (i.e., h^* –hours of work and L^* –hours of leisure).

Figure: 3: Optimal Consumption - Leisure Choice

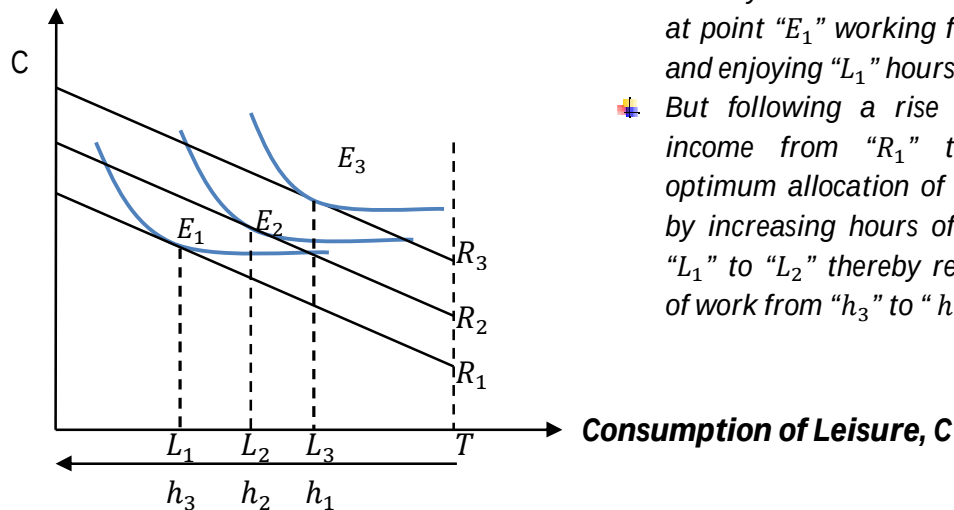
- ✓ For the individual to be at optimum, the slope of the indifference curve should be equal to the slope of the budget line. i.e., $MRS_{C,L} = w$

- At the optimal levels, the marginal rate of substitution (the rate at which a person is willing to give up leisure hours in exchange for additional consumption of goods) equals the wage rate (the rate at which the market allows the worker to substitute one hour of leisure time for consumption of goods).

D. The effect of Increase in Non-Labor Income on hours of work (Labour Supply)

- ✚ Income from non-labor sources is referred to as “unearned income”. Non-labor income may be received in the form of interest payments, rent, dividends, profits, transfer payments, lottery winnings, or any other income that does not vary with hours worked.
- ✚ Imagine an individual's non-labour income increases, what would happen to the amount of labor we would expect the individual to supply using our model? If we think of leisure as a normal good then we expect the consumption of it to increase as income increases. This would cause the amount of labor supplied to fall. The effect of a continual increase in non-labor income is shown in the following graph:

Consumption of Goods, C



- ✚ Initially the individual is at optimum at point “ E_1 ” working for “ h_3 ” hours and enjoying “ L_1 ” hours of Leisure.
- ✚ But following a rise in non-labor income from “ R_1 ” to “ R_2 ” the optimum allocation of time changes by increasing hours of leisure from “ L_1 ” to “ L_2 ” thereby reducing hours of work from “ h_3 ” to “ h_2 ”, etc.

Figure: 4: Effect of Non-Labor Income Increase on Labor Supply

- ✚ In general, increase in non-labor income reduces hours of work and hence increases hours of leisure.

E. Effect of Wage Increase on Labor Supply

Now consider what happens to labor supply when the wage rate rises. A change in Wage rate causes two effects:

1. **Substitution Effect**—as the price of leisure increases, the amount of leisure consumed goes down, and the amount of labor services supplied increases, holding utility constant.

2. **Income Effect**—as people's income increases, the amount of leisure consumed increases, and the amount of labor services supplied decreases. We can show these two effects graphically as follow:

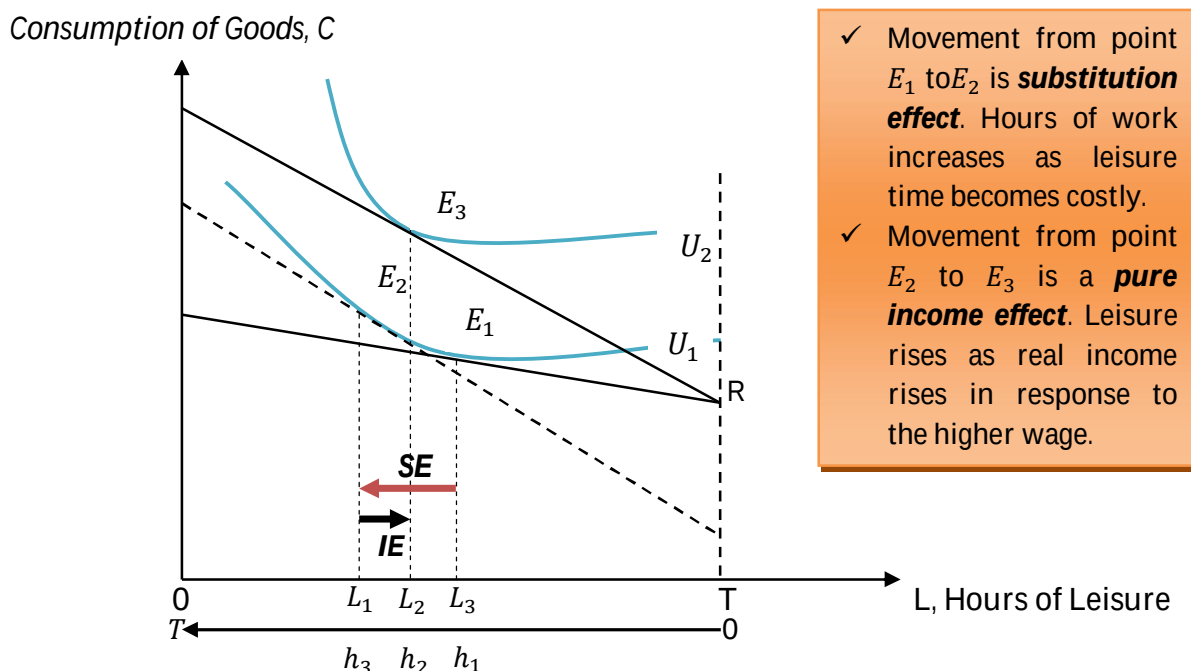


Figure: 5: The Effect of a Wage Increase

Summary: The net effects of a wage increase on labour supply:

- ✚ Higher wage will increase labour supply (decrease demand for leisure) if substitution effect is stronger than income effect:
- ✚ Higher wage will decrease labour supply (increase demand for leisure) if income effect is stronger than substitution effect:

o **Derivation of the Labor Supply Curve**

The labor supply curve traces out the relationship between the wage rate and hours of work. A person's labor supply curve can be derived from the utility-maximization problem that we solved earlier. Recall that an increase in the wage rate generates both an income and a substitution effect- where the income effect reduces hours of work; the substitution effect increases hours of work. The labor supply curve, therefore, is upward sloping if substitution effects dominate and downward sloping if the income effect dominates.

Economists believe that the substitution effect dominates for relatively moderate and low income levels (that is as the price of leisure increases the quantity demanded of it decreases as it is normal good). Contrary to this, at high income levels, the income effect dominates and hence we could get a backward bending labor supply curve.

As the figure below shows, at wages below the reservation wage (\$15)- the wage at which s/he is indifferent between working and not working, the person does not work. At wages higher than \$10, the person enters the labor market. But the hours of work varies with the wage rate. For example, as the figure below presents, s/he works 6 hours a day when the wage rate is \$25; 12 hours when the wage is \$50; and 15 hours a day when the wage is \$40.

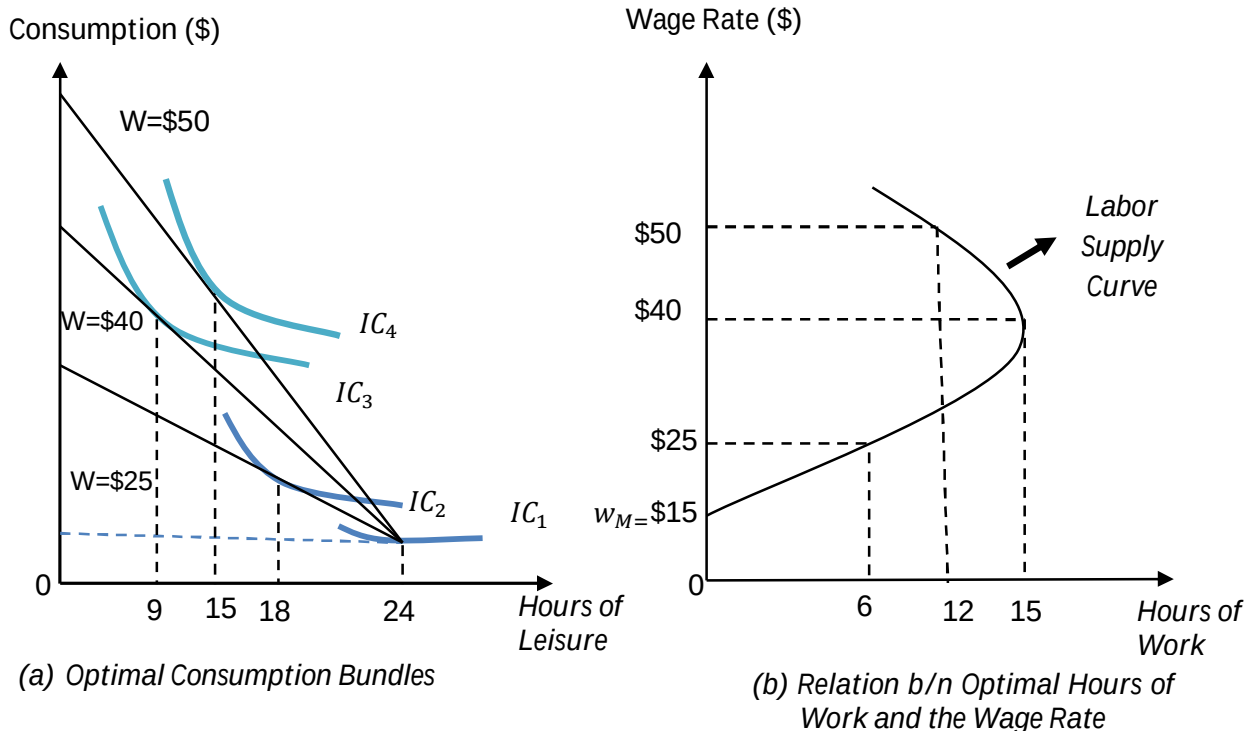


Figure: 5: Individual Labor Supply Curve

- ✓ The figure illustrates how an individual's labor supply curve can be derived from the utility-maximization problem that we solved earlier
- ✓ The individual labor supply curve which is a function of the market wage rate.
- ✓ Note that, as drawn, the figure implies that substitution effects dominate at lower wages and that income effects dominate at higher wages.

We can use the utility-maximization framework to derive a labor supply curve for every individual in the economy. The labor supply curve in the aggregate labor market is then

given by adding up the hours that all individuals in the economy are willing to work at a given wage.

- ✓ Normally, economists assume that most people are on the upward sloping portion of the labor supply curve. Therefore, the aggregate labor supply curve will take the following form.

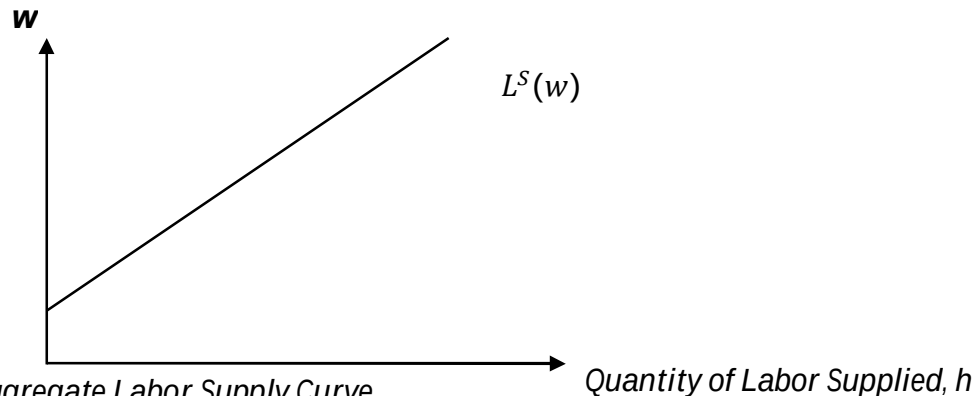


Figure: 6: Aggregate Labor Supply Curve

4.2.2. Labor Demand in the Neoclassical Model

So far, we have considered the supply side of the labor market in the neoclassical context. Now, let's consider the other part of the labour market-(I.e., Labor demand)- to add to our supply function of the labour market.

✚ Two key assumptions used by the model:

1. Firms act to maximise their profits.
 2. There exist diminishing marginal benefits to employing extra resources (i.e. output goes up but at a diminishing rate).
- ✓ The problem thus facing firms is to decide how much of each input to employ to maximize profits subject to the technologies they have and prices they face.

✚ The demand for labour is a demand for factors of production derived from demand for final goods. Firms in a perfectly competitive market use a short run production function (with predetermined capital stock and given technology) to maximize profits:

$$Y = AF(L, K)$$

Where Y –real output, K –given capital stock, L –amount of labour employed, A –technology.

Standard assumptions about the production function are adopted:

i.e, $F_L > 0$, $F_K > 0$, $F_{LL} < 0$, $F_{KK} < 0$, $F_{LK} = F_{KL} > 0$.

We know that for firms to maximise profits that the marginal benefit and marginal cost of the last unit of labour employed must be the same. The marginal cost of a unit of labour is the real wage ($\frac{W}{P} = w$) that has to be paid to the person who supplies it while the marginal benefit is the extra output that the extra unit of labour produces. Using the expressions for the MC and MB of employing labour we can specify the formal profit maximizing condition for employing labour, which is just

$$w = MP_L$$

Or equivalently,

Firms aim at maximizing short-run profits: $\pi = PY - WL - - - - - (1)$

$$Max \pi = PAF(L, \bar{K}) - WL.$$

$$\frac{d\pi}{dL} = 0 \Rightarrow AF_L(L^D, \bar{K}) = \frac{W}{P} - - - - - (2)$$

Demand for labour (L^D) is defined implicitly by (2). Let's take $w = W/P$ as a real wage.

➤ Then by totally differentiating (2) we get:

$$d(AF_L(L^D, \bar{K})) = dw$$

$$F_L dA + A(F_{LL} dL^D + F_{LK} d\bar{K}) = dw - - - - - (3)$$

By rearranging terms, we obtain:

$$dL^D = -\frac{F_L}{AF_{LL}} dA - \frac{F_{LK}}{F_{LL}} d\bar{K} + \frac{1}{AF_{LL}} dw - - - - - (4)$$

✓ This expression tells us everything we need to know about how the various factors affect the quantity of labour demanded by firms.

What Determines Labour Demand (L^D)?

From the above math expression we can claim that: $L^D(\uparrow) = f[w(\downarrow), A(\uparrow), K(\uparrow)]$.

i. An increase in real wage reduces demand for labour (Since $F_{LL} < 0$).

If the real wage rate increases then this means the MC to employing labour increases. If the firm was in equilibrium before the wage increase, it will now be making losses on the last units of labour it employs as their MB will be lower than their MC. The firm takes the real wage as given but it can influence the MB of employing labour. If it reduces the quantity of labour employed then the MB of the last unit it employs will increase and for a sufficiently large fall in labour employed it will no longer make any losses on any of the labour it employs. It will then be back maximizing profits. This tells us that there is a negative relationship b/n the real wage and the quantity of labour employed and can be characterized as the normal downward sloping labour demand curve, and in fact the labour demand curve is just the MP_L .

ii. An increase in capital stock increases demand for labour (Since $F_{LK} > 0$)

iii. An improvement in technology increases demand for labour (Since $F_L > 0$)

How changes in capital stock and technology affect demand for labour?

If k increases or production technology gets better then the MP_L increases, that is, each unit of labour employed becomes more productive. This is shown graphically by an outward shift of the MP_L schedule, (or equivalently, by outward shift of the labour demand curve). At each real wage a higher quantity of labour is demanded because the marginal benefit has increased making employing additional units more attractive compared to the initial situation. If k decreases or the production technology deteriorates then the opposite happens.

The following figure illustrates that labour demand is a decreasing function of the real wage rate (Meaning: It is by reducing the real wage rate that more workers can be employed.) and that the labour demand curve shifts out for increases in capital stock and technological improvement.

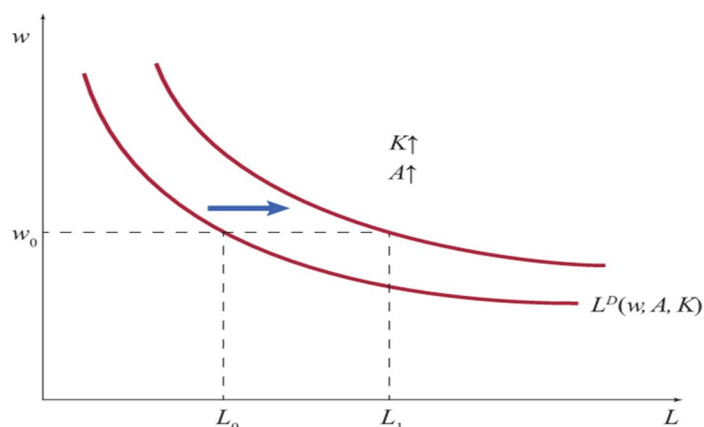


Figure: 7: Labor Demand in the Neoclassical Model

4.2.3. The Neoclassical Labor Market

We will now assume the equilibrium principle that forces exist to ensure that an unchanging quantity of labour is employed, to derive the market equilibrium. The forces here are that if an excess supply of labour exists pressure will be put on the real wage to fall and thus reduce the quantity of labour supplied and increase the quantity of labour demanded and make the two equal, and vice versa. This means we can now put the labour demand and labour supply curves together to get the traditional view of the labour market. In the labour market, the demand for labour and the supply of labour determine the level of output and employment.

✚ The demand for labour is a decreasing function of the real wage rate (indicating that it is by reducing the real wage rate that more workers can be employed). Contrary to this, the supply of labour is an increasing function of the real wage rate (showing that it is by increasing the real wage rate that more workers can be employed). When the L^D and L^S curves intersect at point E , the full employment level N_F is determined at the equilibrium real wage rate w^e .

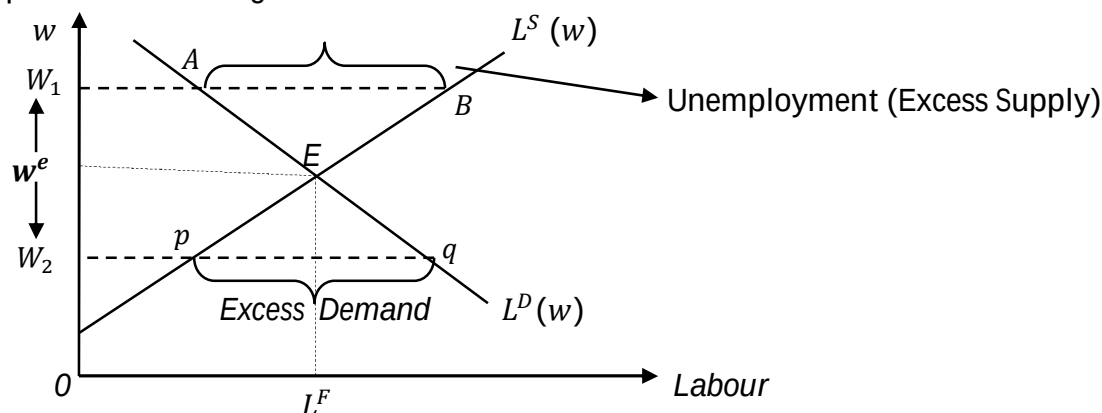


Figure: 8: Equilibrium in the Labour Market

- ✚ If the wage rate rises from w^e to w_1 the supply of labour will be more than its demand. Now at w_1 wage rate, **AB** workers will be involuntary unemployed b/c the demand for labour is less than the supply. With competition among workers for work, they will be willing to accept a lower wage rate. Consequently, the wage rate will fall from w_1 to w^e . Thus, the supply of labour will fall and the demand for labour will rise and the equilibrium point E will be restored and full employment level L^F is reached.
- ✚ On the contrary, if the wage rate falls from w^e to w_2 the demand for labour will be more than its supply. Competition by employers for workers will raise the wage rate from w_2 to w^e and the equilibrium point E will be restored and full employment level L^F is reached.
- ✚ In general, the forces in this market act to determine both the amount of labour traded and the level of the real wage. ***It is also worth noting that there is no involuntary unemployment in this model. People may be “unemployed”, but it is only because they choose not to work at the going real wage.*** According to this model, full employment is a normal situation and any deviation from this regarded as something abnormal, and the tendency of the economic system is to automatically provide full employment in the labor market.

o **Unemployment in the Neoclassical Model**

Recall that Neoclassical labour market is characterized by:

- ✓ Perfect competition and Perfect information
 - ✓ Market clearing, flexible wages
 - ✓ Homogenous workers
 - ✓ No involuntary unemployment

Is the Neoclassical model enough to explain the performance of the labor market today?

Possible Explanations for Unemployment

- O If wages don't clear the market, so that there exist unemployment, why don't they fall?
 - ✓ *Wage rigidity* (especially downward)—minimum wage legislation
 - ✓ *Efficiency wages*—it may be profitable for firms to pay higher wage than market-clearing wage.

- ✓ *Bargaining power of labor unions*
- ✓ *Heterogeneity of workers and jobs*—search for jobs and matching a worker with suitable job is costly and takes time.

Unemployment may result from the rigidity in the wage structure and interference in the working of free market system in the form of labor union, minimum wage legislation etc.

4.3. Keynesian and New Keynesian Views of the Labor Market

As we have noted before, *classical/Neoclassical economists presume that an excess supply of labor would exert a downward pressure on wages*. A reduction in wages would, in turn, reduce unemployment by raising the quantity of labor demanded. As a consequence, they argued that labor market is characterized by continual equilibrium-full employment, and deviation from full employment is a temporary phenomenon. However, in the Keynesian tradition, wage rates are sticky in the short-run. Hence, the Keynesians argued that full employment is a special situation, and the general situation is that economies encounter huge and prolonged unemployment like what happened during the Great depression.

The Keynesian tradition is basically concerned with the question of what causes surpluses and shortages in the labor market. Both in Keynesians and New Keynesians views persistent involuntary unemployment and economic fluctuations are central and continuing problems. However, referring to the role of nominal and/or real wages behavior in explaining unemployment, New Keynesian theories differ from the insight developed by Keynes in his *General Theory*.

4.3.1. Keynesian View Of the Labor Market

It was Keynes that first acknowledges the existence of disequilibria in the labor market. He emphasizes the possibility of downward nominal wage stickiness. However, concerning the capacity of nominal wages reductions in restoring full employment, Keynes did not believe that nominal wage rigidity was the main source of unemployment. As a consequence, he asserted that nominal wage cuts were not the proper cure for unemployment.

Keynes argued that greater wage flexibility might exacerbate economic downturns since it erodes business expectations. For him, nominal wage reduction leads to the expectation of a severe deflation which in turn adversely affect investment expenditure. That is, lower wages implies lower consumption and aggregate demand for goods and services thereby business expectation falls, and thus investment expenditures are discouraged. *(For this Keynes indicated that during the 1920s & 30s (in the Great Depression) nominal wages and prices were falling but this did not bring economic recovery rather it would lead to farther degeneration of economies.)* As a result, Keynes advocated the maintenance of a stable level of nominal wages and that authorities/governments would take positive action in order to restore full employment.

For Keynesians in general, output and employment would be driven by changes in effective demand. Thus, they argue that it is low effective demand which would be responsible for high unemployment and low output, and thus they propose governments to take expansionary policies (be it fiscal, monetary or both) rather than wage cuts to restore full employment.

The implication is that government through monetary and fiscal policies can alter aggregate demand, and thereby employment. Consider the IS-LM model of Keynes to understand how this works.

$$\textbf{IS (Investment-Saving): } Y = C(Y - T(\downarrow)) + I(r(\downarrow))(\uparrow) + G(\uparrow)$$

$$\textbf{LM (Liquidity Preferences-Money Supply): } \frac{M(\uparrow)}{P} = L(r(\downarrow), Y)$$

Note: Arrows indicate expansionary policies.

Expansionary policies shifts IS and LM curves to the right what provokes a shift in the same direction of the aggregate demand fixing a new and higher level of output and employment at a given price level. The following diagram presents how expansionary policies influences aggregate demand which then determines employment.

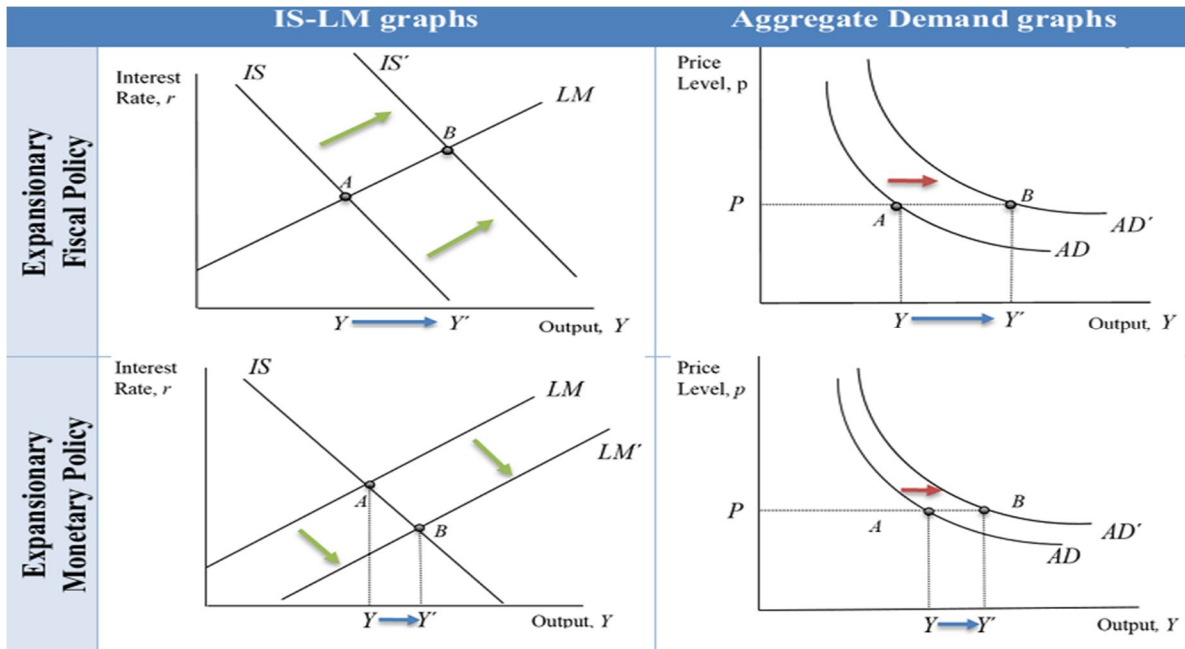


Figure: 9: Expansionary policies and Aggregate demand

4.3.2. New Keynesian View of the Labor Market

New Keynesian Economists (NKE) which obtains their inspiration from Keynes accepted the existence of wages and price rigidity. However, unlike Keynes, NKE argued that wage rigidity can be the source of unemployment. As a result, for NKE flexible wages would allow the economy to maintain full employment.

What was less clear in the traditional Keynesian view is: 'why labor markets do not clear at the microeconomic level if there is persistent unemployment' (regardless of the level of aggregate demand)? In a single labor market, if unemployed workers would offer to work for less pay and firms would be willing to hire them at that lower pay, real wages would be bid down and employment would increase. A contribution by New Keynesians is the provision of alternative explanations for such a phenomenon.

Thus, NKE attempts to provide micro foundations for the phenomena of wages and prices sluggish adjustments. For instance, NKE developed **efficiency wages** and **insider-outsider theories** to explain why firms do not cut wages when there are unemployed workers.

i. Efficiency wages: New Keynesian economists often turn to theories of what they call efficiency wages to explain why labour market-clearing mechanism may fail. In an asymmetric information framework, efficiency wage models describe several reasons for which cutting a wage adversely affect the quality or productivity of labor and increase at the end the firm's cost. These theories hold that high wages make workers more productive. *Hence, the influence of wages on worker efficiency may explain the failure of firms to cut wages despite an excess supply of labor.*

ii. Insider-Outsider Theory: in this theory, it is argued that the insiders (employed workers) have some power in determining firm's wage and employment decision due to the presence of turnover costs. Since for a firm it is costly to exchange insiders for outsiders (unemployed workers), the insiders can extract a share of the economic rent generated by such turnover costs. Furthermore, if insiders feel their position threatened by outsiders, they can refuse to cooperate with and train new workers coming from outside, as well as make their life at work unpleasant. This raises the disutility of work for outsiders and can contribute to explain also why they do not accept lower wages.

In general, New Keynesians theories of the labour market (e.g. efficiency wages and insider-outsider) emphasize the role of real wage rigidity in explaining involuntary unemployment.

O The differences b/n Keynesians and New Keynesian views of the labor market

First, the models above are fundamentally theories of real wage rigidity. They explain why real wage does not adjust in labor markets if there are involuntary unemployed workers. Thus, NKE concludes that real wage rigidity explains involuntary unemployment. On the contrary, Keynes focused on nominal wage rigidity while, in his view, the role of real wage is not so relevant in determining output and employment. As a result, while real wage rigidity models produce an unemployment equilibrium outcome which could be compatible with the Keynes's involuntary unemployment, this outcome is much less in harmony with the Keynes's involuntary unemployment theory since, theoretically

speaking, unemployment due to the presence of a downward rigid real wage rate which is too high as compared to the market-clearing value is more similar to the classical than Keynesian concept of unemployment.

Second, in these models the role of effective demand in restoring full employment is much less clear than in the Keynes's original insight. In fact, real wage rigidity alone is not enough to understand Keynesian unemployment because it leads only to 'classical' (but involuntary) unemployment and leaves in a large part unclear the role (if there is one) for aggregate demand.

4.4. Labour-Market Search and Unemployment

Empirical evidence has shown that the labour market in many countries is characterized by huge gross flows of workers leaving a job and entering unemployment and vice versa. This flow of workers determines the fraction of the labor force that is unemployed (the unemployment rate). *In this section, we develop a simple model of labor-force dynamics that shows what determines the natural rate of unemployment.*

We start with some notations: Let

L = the labor force (working-age population),

E = the number of employed workers (individuals who currently have Jobs),

U = the number of unemployed workers (individuals who do not currently have any job but who are actively looking for one).

Because every worker is either employed or unemployed, *the labor force is the sum of the employed and the unemployed:*

$$\text{i.e., } L = E + U$$

Here, the rate of unemployment is $\frac{U}{L}$ (i.e. the fraction of the labour force unemployed).

To see what determines the unemployment rate, we assume that the labor force, L is fixed and focus on the transition of individuals in the labor force between employment and unemployment. This is illustrated in Figure 10. Let s denote the rate of job separation, the fraction of employed individuals who lose their job each month. Let f denote the rate of job

finding, the fraction of unemployed individuals who find a job each month. Together, the rate of job separation s and the rate of job finding f determine the rate of unemployment.

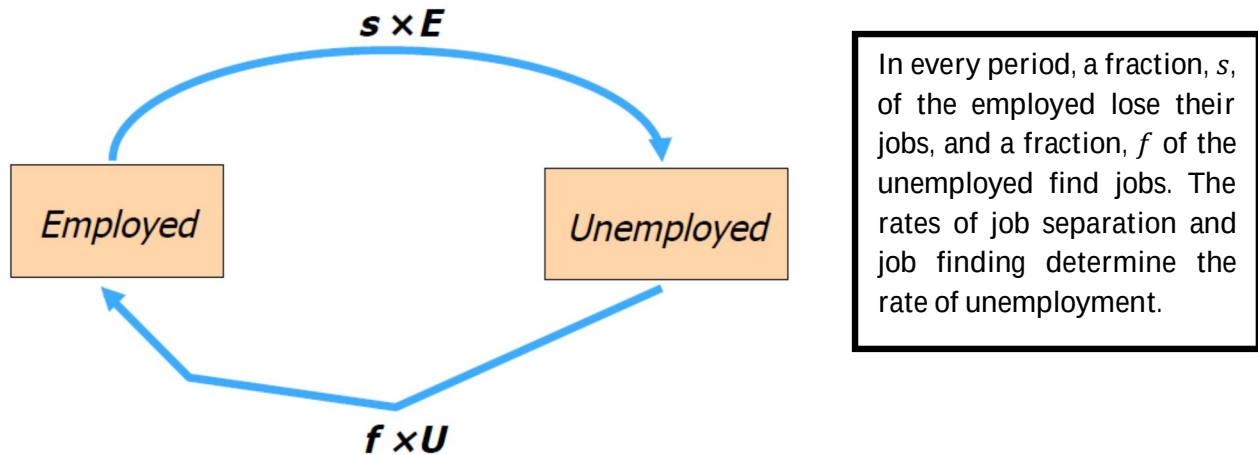


Figure: 10: The Transitions Between Employment and Unemployment

✚ If the unemployment rate is neither rising nor falling—that is, if the labor market is in a steady state—then the number of people finding jobs must equal the number of people losing jobs. The number of people finding jobs is fU and the number of people losing jobs is sE , so we can write the **steady-state condition** as:

$$fU = sE$$

We can use this equation to find the **steady-state unemployment rate**. From an earlier equation, we know that $E = L - U$. If we substitute $(L - U)$ for E in the steady-state condition, we find

$$fU = s(L - U).$$

Divide both sides of this equation by L to obtain:

$$f \frac{U}{L} = s \left(1 - \frac{U}{L} \right).$$

Now we can solve for U/L to find:

$$\frac{U}{L} = \frac{s}{s+f}$$

○ This equation shows that the steady-state rate of unemployment U/L depends on the rates of job separation s and job finding f . **The higher the rate of job separation, the higher will be the unemployment rate. The higher the rate of job finding, the lower will be the unemployment rate.**

- O This model of the rate of unemployment has important implication for public policy.
- ***Any policy that affects the rate of job separation or job finding also changes the rate of unemployment.***
 - ***That is, any policy aimed at lowering the natural rate of unemployment must either reduce the rate of job separation or increase the rate of job finding.***

4.5. Types and Causes of Unemployment

I. Job Search and Frictional Unemployment

One reason for unemployment is that it takes time to match workers and jobs. As a matter of fact, workers have different preferences and abilities, and jobs have different attributes. Furthermore, the flow of information about job candidates and job vacancies is imperfect, and the geographic mobility of workers is not instantaneous. For all these reasons, searching for an appropriate job takes time and effort, and this tends to reduce the rate of job finding. *The unemployment caused by the time it takes workers to search for a job is called **frictional unemployment**.*

Economists call a change in the composition of demand among industries or regions a **sectoral shift**. Because sectoral shifts are always occurring, and because it takes time for workers to change sectors, there is always **frictional unemployment**.

- ❖ In trying to reduce frictional unemployment, some policies unintentionally increase *frictional unemployment*. One such program is called **unemployment insurance**. In this program, workers can collect a fraction of their wages for a certain period after losing their jobs. However, it adversely affects workers labour supply decision and results in higher frictional unemployment.

II. Cyclical Unemployment

It is the unemployment that accompanies fluctuations in real GDP and sometimes called “**Demand deficient unemployment**”.

- ✓ Reflects business cycle conditions

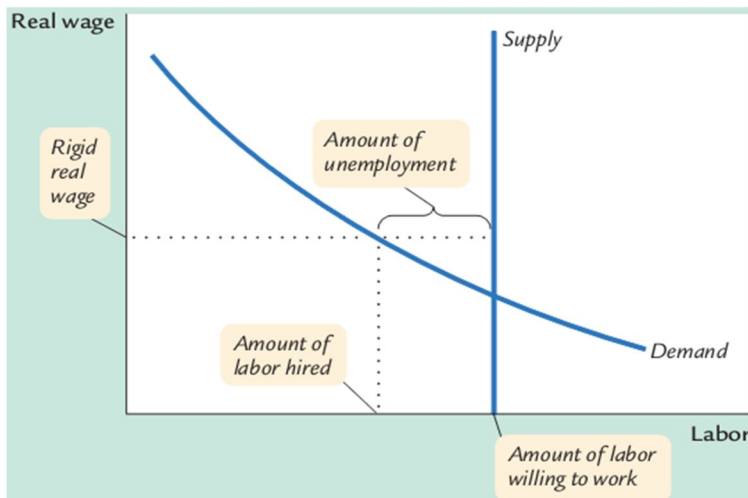
- ✓ When there is a general downturn in business activity, cyclical unemployment rises.

III. Seasonal Unemployment:

A product of regular, recurring changes in the hiring needs of certain industries on a monthly or seasonal basis.

IV. Real-Wage Rigidity and Structural Unemployment

Wage rigidity refers to the failure of wages to adjust until labor supply equals labor demand. The unemployment resulting from wage rigidity is called structural unemployment. Workers are unemployed not because they are actively searching for the jobs that best suit their individual skills but because, at the going wage, the supply of labor exceeds the demand. These workers are simply waiting for jobs to become available.



- ✓ If the real wage is stuck above the equilibrium level, then the supply of labor exceeds the demand. The result is unemployment.

Figure: 11: Real Wage Rigidity and Unemployment

To understand wage rigidity and structural unemployment, we must examine why the labor market does not clear. When the real wage exceeds the equilibrium level and the supply of workers exceeds the demand, we might expect firms to lower the wages they pay. Structural unemployment arises because firms fail to reduce wages despite an excess supply of labor. We now turn to three causes of this wage rigidity: minimum-wage laws, the monopoly power of unions, and efficiency wages.

A. Minimum-Wage Laws

The government causes wage rigidity when it prevents wages from falling to equilibrium levels. Minimum-wage laws set a legal minimum on the wages that firms pay their employees. For most workers, this minimum wage is not binding, because they earn well above the minimum. Yet for some workers, especially teenagers and unskilled workers, minimum-wage laws have significant unemployment impacts since it raises their wage above its equilibrium level, and thus reduces firm's demand for these labor types.

Economists believe that the minimum wage has its greatest impact on teenage unemployment. The equilibrium wages of teenagers tend to be low because teenagers are among the least skilled and least experienced members of the labor force; they tend to have low marginal productivity. For this reason, the wage at which the supply of teenage and/or unskilled workers equals the demand is low. The figure below shows how minimum wage laws hampers teenage/unskilled labor (where equilibrium wage rate is W_U).

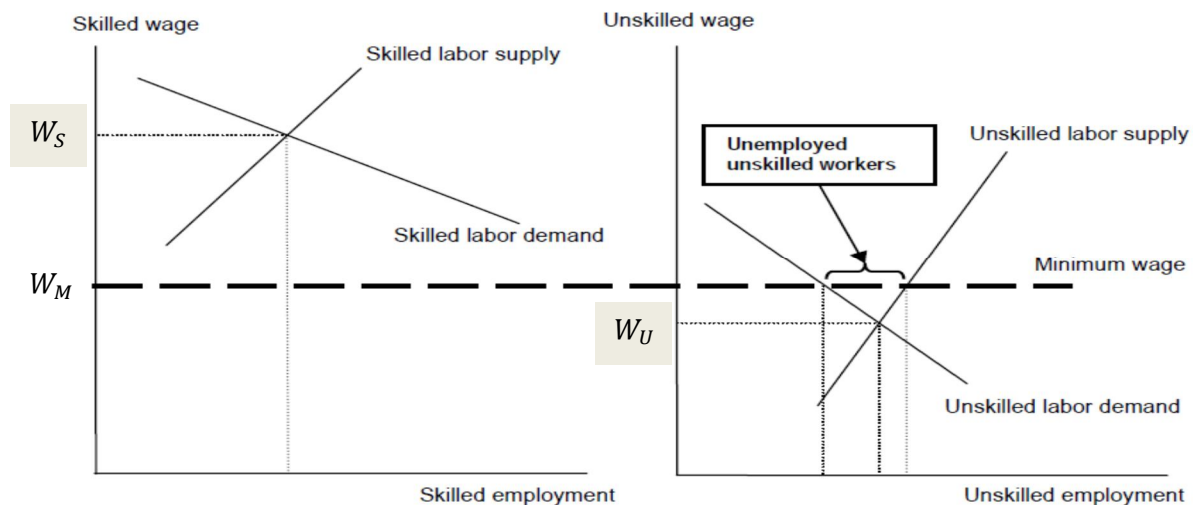


Figure: 12: The effect of minimum wage on employment

- ✓ As the above figure indicated, minimum wage laws do not affect employment of skilled labor as the wage rate of this type of labor (i.e., W_S) is well above the minimum wage, W_M .
- ✓ On the other hand, the minimum wage for unskilled workers which tends to be well above the market clearing wage rate (i.e., W_U) causes unemployment in this labor category.

❖ **Policy Options to Solve Unemployment caused by Minimum wage Laws**

i. Tax credits: Minimum wage increases are designed to help the working poor. However, it causes unemployment. Hence, for economists' provision of tax credits (and abolishing minimum wage laws) are better ways to increase the incomes of the working poor. The *earned income tax credit* is an amount that poor working families are allowed to subtract from the taxes they owe. Unlike the minimum wage, the *earned income tax credit* does not raise labor costs to firms and, therefore, does not reduce the quantity of labor that firms demand. It has the disadvantage, however, of reducing the government's tax revenue.

ii. Subsidizing unskilled labor: the government by subsidizing unskilled labor could push the labor demand up so that employment would be better.

iii. The government can directly employ some unskilled workers at the going minimum wage. Again, the demand for unskilled labor shifts to the right and unemployment is reduced.

➤ *N.B: For all the three options discussed above, there is a revenue requirement on the part of the government.*

iv. Provision of training to unskilled workers by the government: By making unskilled labor more productive, it is possible to stimulate the demand for those workers and employment. This can be taken as the most attractive option because the previously unemployed find work, they start to pay taxes, thus reducing the government's revenue requirement.

B. Unions and Collective Bargaining

Another cause of wage rigidity is the monopoly power of **unions**. In Sweden **84 percent** of workers, and in the United States, **only 18 percent** of workers belong to unions. Often, union contracts set wages above the equilibrium level and allow the firm to decide how many workers to employ. Consequently, the number of workers hired decreases, a lower rate of job finding, and unemployment increases.

The unemployment caused by unions and by the threat of unionization is an instance of conflict between different groups of workers—**insiders** and **outsiders**. Those workers already *employed* by a firm, the **insiders**, typically try to keep their firm's *wages high*. The *unemployed*, the **outsiders**, bear part of the cost of higher wages because at a lower wage they might be hired. These two groups inevitably have conflicting interests. The effect of any bargaining process on wages and employment depends crucially on the relative influence of each group.

C. Efficiency Wages

Efficiency-wage theories propose a third cause of wage rigidity in addition to minimum-wage laws and unionization. These theories hold that high wages make workers more productive. *The influence of wages on worker efficiency may explain the failure of firms to cut wages despite an excess supply of labor.* Even though a wage reduction would lower a firm's wage bill, it would also lower worker productivity and the firm's profits.

➤ Economists have proposed various reasons to explain how wages affect worker productivity.

1. **Wages influence nutrition** (*which is applied mostly to poorer countries*) __ Better-paid workers can afford a more nutritious diet, and healthier workers are more productive. *Hence, a firm may decide to pay a wage above the equilibrium level to maintain a healthy workforce.*
2. **High wages reduce labor turnover.** Workers quit jobs for many reasons—to accept better positions at other firms, to change careers, or to move to other parts of the country. The more a firm pays its workers, the greater their incentive to stay with the firm. By paying a high wage, a firm reduces the frequency of quits, thereby decreasing the time spent hiring and training new workers.
3. **The average quality of a firm's workforce depends on the wage it pays its employees.** If a firm reduces its wage, the best employees may take jobs elsewhere, leaving the firm with inferior employees.

4. High wage improves worker effort. This theory posits that firms cannot perfectly monitor their employees' work effort, and that employees must themselves decide how hard to work. Workers can choose to work hard, or they can choose to shirk. Economists recognize this possibility as an example of **moral hazard**—the tendency of people to behave inappropriately when their behavior is imperfectly monitored. The firm can reduce the problem of moral hazard by paying a high wage. By paying a higher wage, a firm induces more of its employees not to shirk and thus increases their productivity. *The result of this higher-than-equilibrium wage is a lower rate of job finding and greater unemployment.*

Review Exercises

1. Briefly explain the effect of non-labour income increase on labour supply in the neoclassical model?
2. Distinguish the income effect and the substitution effect of a wage increase on labor supply using the neoclassical model.
3. Distinguish between neoclassical, Keynesian and new Keynesian (un)employment views and their policy implications?
4. Give three explanations why the real wage may remain above the level that equilibrates labor supply and labor demand.
5. Consider an economy with the following Cobb –Douglas production function:

$$Y = K^{1/3}L^{2/3}$$

The economy has 1,000 units of capital and a labor force of 1,000 workers.

- A. Derive the equation describing labor demand in this economy as a function of the real wage and the capital stock.
- B. If the real wage can adjust to equilibrate labor supply and labor demand, what is the real wage and equilibrium employment (assume that labor & capital will work at any price)?
- C. Now suppose that the government, concerned about the welfare of the working class, passes a law requiring firms to pay workers a real wage of **1 unit of output**. How does this wage compared to the equilibrium wage? And, the government cannot dictate how many workers firms hire at the mandated wage. Given this fact, what are the effects of this law? Specifically, what happens to employment and output?

CHAPTER FIVE

MACROECONOMICS POLICY DEBATES- STABILIZATION POLICY

Chapter Objectives:

❖ *At the end of this chapter, you will be able to:*

- ✓ *Conceptualize some macroeconomics policy debates;*
- ✓ *Explain how policy makers might respond to macroeconomic shocks;*
- ✓ *Elaborate the way how policy responses might affect an economy;*
- ✓ *Analyze the inside lags and the outside lags of different policy responses: (Fiscal and Monetary);*
- ✓ *State the arguments for Active & Passive policy responses;*
- ✓ *Describe the arguments for policy by discretion and for policy by rule; and*
- ✓ *Mention the different monetary policy rules to be targeted.*

5.1. Introduction

- ✚ Some economists view the economy as *inherently unstable*. They argue that the economy experiences frequent shocks to aggregate demand and aggregate supply. Unless policymakers use monetary and fiscal policy to stabilize the economy, these shocks will lead to unnecessary and inefficient fluctuations in output, unemployment, and inflation. According to the popular saying, macroeconomic policy should “**lean against the wind**”, stimulating the economy when it is depressed and slowing the economy when it is overheated.
- ✚ Other economists, /such as Milton Friedman/, view the economy as *naturally stable*. They blame bad economic policies for the large and inefficient fluctuations we have sometimes experienced. They argue that economic policy should not try to “fine-tune” the economy. Instead, economic policymakers should admit their limited abilities and be satisfied if they do no harm. This debate has persisted for decades.
- ✚ The fundamental issue is how policymakers should use the theory of short-run economic fluctuations developed in the preceding chapters. In this chapter, we ask two questions that arise in this debate.

- ✓ *First, should monetary and fiscal policy take an active role in trying to stabilize the economy, or should policy remain passive?*
- ✓ *Second, should policymakers be free to use their discretion in responding to changing economic conditions, or should they be committed to following a fixed policy rule?*

5.2. Should Policy Be Active or Passive?

- + To many economists the case for active government policy is clear and simple. Recessions are periods of high unemployment, low incomes, and increased economic hardship. The model of aggregate demand and aggregate supply shows how shocks to the economy can cause recessions. It also shows how monetary and fiscal policy can prevent recessions by responding to these shocks. These economists consider it wasteful not to use these policy instruments to stabilize the economy.
- + Other economists are critical of the government's attempts to stabilize the economy. These critics argue that the government should take a hands-off approach to macroeconomic policy. At first, this view might seem surprising. If our model shows how to prevent or reduce the severity of recessions, why do these critics want the government to refrain from using monetary and fiscal policy for economic stabilization? *To find out, let's consider some of their arguments.*

1. Lags in the Implementation and Effects of Policies

- + Economic stabilization would be easy if the effects of policy were immediate. However, economic policymakers face the problem of long and variable lags that greatly complicate the conduct of monetary and fiscal policy. Besides, the lengths of the lags are hard to predict.
- + *Economists distinguish between two lags in the conduct of stabilization policy: the **inside lag** and the **outside lag**.*
 - o The **inside lag** is the time b/n a shock to the economy and the policy action responding to that shock. This lag arises b/c it takes time for policymakers *first* to recognize that a shock has occurred and then to put appropriate policies into effect.

- o The **outside lag** is the time b/n a policy action and its influence on the economy. This lag arises because policies do not immediately influence spending, income, and employment.
- + A long inside lag is a central problem with using fiscal policy for economic stabilization. But it depends on the legislative process.
 - ✓ If the structure of the government is presidential like the USA, where changes in spending or taxes require the approval of the president and both houses of Congress, policy action takes longer time.
 - ✓ If the structure of the government is parliamentary system, like Ethiopia and UK, the inside lag in using Fiscal policy is shorter because the party in power can often enact policy changes more rapidly.
- + Monetary policy has a much shorter inside lag than fiscal policy, because a central bank (i.e., National Bank of Ethiopia in our country case) can decide on and implement a policy change in less than a day, **but monetary policy has a substantial outside lag.** Monetary policy works by changing the money supply and thereby interest rates, which in turn influence investment. But many firms make investment plans far in advance. Therefore, a change in monetary policy is thought not to affect economic activity until about six months after it is made.
- + The long and variable lags associated with monetary and fiscal policy certainly make stabilizing the economy more difficult.
 - ✓ Advocates of passive policy argue that, b/c of these lags, successful stabilization policy is almost impossible. Indeed, attempts to stabilize the economy can be destabilizing. Suppose that the economy's condition changes between the beginning of a policy action and its impact on the economy. In this case, active policy may end up stimulating the economy when it is overheated or depressing the economy when it is cooling off.
 - ✓ Advocates of active policy admit that such lags do require policymakers to be cautious. But, they argue, these lags do not necessarily mean that policy

should be completely passive, especially in the face of a severe and extended economic downturn.

- o Some policies, called **automatic stabilizers**, are designed to reduce the lags associated with stabilization policy.

✚ **Automatic stabilizers** are policies that stimulate or depress the economy when necessary without any deliberate policy change. For example, *the system of income taxes automatically reduces taxes when the economy goes into a recession, without any change in the tax laws*, because individuals and corporations pay less tax when their incomes fall.

II. The Difficult Job of Economic Forecasting -Active Policy Requires Forecasts, Which Are Often Wrong

Because policy influences the economy only after a long lag, successful stabilization policy requires the ability to predict accurately future economic conditions. If we cannot predict whether the economy will be in a boom or a recession in six months or a year, we cannot evaluate whether monetary and fiscal policy should now be trying to expand or contract aggregate demand.

- ✓ One way forecasters try to look ahead is with **leading indicators**. A leading indicator is a data series that fluctuates in advance of the economy. A large fall in a leading indicator signals that a recession is more likely.
- ✓ Another way forecasters look ahead is with **macroeconomic models** developed for forecasting and policy analysis.

5.3. Ignorance, Expectations, and the Lucas Critique

✚ Nobel laureate Robert Lucas **has emphasized the issue of how people form expectations of the future. Expectations play a crucial role in the economy because they influence all sorts of economic behavior.**

✚ For instance, households decide how much to consume based on expectations of future income, and firms decide how much to invest based on expectations of future profitability. These expectations depend on many things, including the economic

policies being pursued by the government. Thus, when policymakers estimate the effect of any policy change, they need to know how people's expectations will respond to the policy change.

✚ Lucas has argued that traditional methods of policy evaluation—such as those that rely on standard **macroeconometrics models**—do not adequately take into account this impact of policy on expectations. This criticism of traditional policy evaluation is known as the **Lucas critiques**.

✚ **Lesson from Lucas critiques:**

- ✓ Economists evaluating alternative policies need to consider how policy affects expectations and, thereby, behavior.

5.4. Should policies be conducted by Rule or by Discretion?

- ✓ A second topic of debate among economists.

✚ **Policy is conducted by rule-** if policymakers announce in advance to the public and policy makers commit themselves to follow that particular policy.

✚ **Policy is conducted by discretion-** if policymakers are free to size up events as they occur and choose whatever policy seems appropriate at the time.

o The debate over rules versus discretion is distinct from the debate over passive versus active policy.

- ✓ Policy can be conducted by rule and yet be either passive or active. For example,
 - A *passive policy rule* might specify steady growth of money supply of 3% per year. Under this rule money supply grows at 3% per annum regardless of the state of the economy.
 - An *active policy rule* might specify that

$$\text{Money Growth} = 3\% + (\text{Unemployment Rate} - 6\%).$$

Under this rule, the money supply grows at 3% if the unemployment rate is 6%, but for every percentage point by which the unemployment rate exceeds 6%, money growth increases by an extra percentage point. This rule tries to stabilize the economy by raising money growth when the economy is in a recession.

- o In the following section, we will discuss *why policy might be improved by a commitment to a policy rule*. We then examine some possible policy rules.

i. Distrust of Policymakers and the Political Process

✚ Policy by discretion has its own advantage as it recognizes the existing macroeconomic situation. However, it has still its own limitation due to *incompetency* and *opportunistic* behavior of policy makers.

- *Incompetency in economic policy*- arises due to lack of sufficient knowledge of macroeconomics by politicians /policy makers/to make informed judgments.
- *Opportunism in economic policy* -arises when the interest of policymakers conflict with the interest/well-being of the public.

✚ *Politicians may use macroeconomic policy to further their own electoral ends*. If citizens vote on the basis of economic conditions prevailing at the time of the election, then politicians have an incentive to pursue policies that will make the economy look good during election years. A president might cause a recession soon after coming into office to lower inflation and then stimulate the economy as the next election approaches to lower unemployment; this would ensure that both inflation and unemployment are low on Election Day.

✚ *Manipulation of the economy for electoral gain*, called the **Political Business Cycle**. Distrust of the political process leads economists to propose constitutional amendments, like a **balanced-budget amendment (a rule preventing the government from spending more than it receives in revenues)**, that controls policy makers and insulate the economy from both *incompetence* and *opportunism*.

ii. The Time Inconsistency of Discretionary Policy

✚ *Discretionary policy is, by its nature, flexible.* If we can trust our policymakers, discretion at first glance appears superior to a fixed policy rule. Yet a case for rules over discretion arises from the problem of **time inconsistency** of policy.

- Policymakers may want to announce in advance the policy they will follow to influence the expectations of private decision makers.
- But later, after the private decision makers have acted on the basis of their expectations, these policymakers may be tempted to renege on their announcement. Understanding that policymakers may be inconsistent over time, private decision makers are led to distrust policy announcements.

✚ *The time inconsistency of policy arises in many contexts.* Here are some examples:

- ✓ To encourage investment, the government announces that it will not tax income from capital. But after factories have been built, the government is tempted to renege on its promise to raise more tax revenue from them.
- ✓ To encourage research, the government announces that it will give a temporary monopoly to companies that discover new drugs. But after a drug has been discovered, the government is tempted to revoke the patent.
- ✓ To encourage you to work hard, your Teacher announces that this course will end with an exam. But after you have studied and learned all the material, the Teacher is tempted to cancel the exam so that he or she won't have to grade it.

In each case, the solution is to take away the policymaker's discretion with a credible commitment to a fixed policy rule.

As a conclusion:

- ✓ *Advocates of discretionary policy argue that discretion gives more flexibility to policymakers in responding to various unforeseen macroeconomics situations.*
- ✓ *Advocates of policy rules argue that the political process cannot be trusted. They believe that politicians make frequent mistakes in conducting economic policy and sometimes use economic policy for their own political ends. In addition, advocates of*

policy rules argue that a commitment to a fixed policy rule is necessary to solve the problem of time inconsistency.

5.5. Rules for Monetary Policy

✚ Even if we are convinced that policy rules are superior to discretion, the debate over macroeconomic policy is not over. *If the Central Banks were to commit to a rule for monetary policy, what rule should they choose?*

✚ Let's discuss **three policy rules that various economists advocate.**

1. Monetarist's Policy Rule

✚ Economists, called **monetarists**, advocate that **Central Banks keep the money supply growing at a steady rate.**

- ✓ Monetarists believe that fluctuations in the money supply are responsible for most large fluctuations in the economy. They argue that slow and steady growth in the money supply would yield stable output, employment, and prices.
- ✓ *However, most economists believe that monetarist policy rule is not the best possible policy rule.* Steady growth in the money supply stabilizes aggregate demand only if the velocity of money is stable. But sometimes the economy experiences shocks, such as shifts in money demand that cause velocity to be unstable.
- ✓ Most economists believe that a policy rule needs to allow the money supply to adjust to various shocks to the economy.

2. Nominal GDP Targeting

- Under this rule, a central bank announces a planned path for nominal GDP.
 - ✓ If nominal GDP rises above the target, the central bank reduces money growth to dampen aggregate demand.
 - ✓ If GDP falls below the target, the central bank raises money growth to stimulate aggregate demand.
- Because a nominal GDP target allows monetary policy to adjust to changes in the velocity of money, most economists believe it would lead to greater stability in output and prices than a monetarist policy rule.

3. Inflation Targeting

- Under this rule, a central bank would announce a target for the inflation rate and then adjust the money supply when the actual inflation deviates from the target.
 - ✓ Like nominal GDP targeting, inflation targeting insulates the economy from changes in the velocity of money.

Review Exercises

(Read the Handout & Check Your Progress by Doing the Following Questions)

1. *What are the inside lag and the outside lag? Which has the longer inside lag- monetary or fiscal policy? Which has the longer outside lag? Why?*
2. *Describe the arguments for active policy and the arguments for passive policy.*
3. *Why would more accurate economic forecasting make it easier for policymakers to stabilize the economy? Describe two ways economists try to forecast developments in the economy.*
4. *Describe the Lucas critique.*
5. *Describe the arguments for policy by **discretion** and for **policy by rule**.*
6. *What is meant by the “time inconsistency” of economic policy? Why might policymakers be tempted to renege on an announcement they made earlier? In this situation, what is the advantage of a policy rule?*
7. *List three policy rules that National Bank of Ethiopia may choose from. Which of these would you advocate? Why?*

CHAPTER SIX

MODELS OF MACROECONOMICS GROWTH

Chapter Objectives:

❖ **At the end of this chapter, you will be able to:**

- ✓ Recognize the existence of huge difference in countries economic performance and societies living standards;
- ✓ Explain the growth implication of the Harrod-Domar Model;
- ✓ Appreciate the role of saving, investment, growth of labor force and technology in economic growth:
- ✓ Distinguish the differences b/n the Solow model and the endogenous model and analyze the respective growth implications;
- ✓ Demonstrate a high level of awareness about engines of economic growth;
- ✓ Analyze why societies in different countries experience differing levels of living standards.

6.1. Introduction: Modeling Economic Growth

Basic Questions:

1. Why some countries are richer than other countries?
2. What exactly is the engine of growth? Why some countries do exhibit sustained growth while others not?

To measure economic growth, economists use data on gross domestic product, which measures the total income of everyone in the economy. Over the past century, most countries in the world have enjoyed substantial economic growth. Real incomes have risen from generation to generation, and these higher incomes have allowed people to consume greater quantities of goods and services. Higher levels of consumption have led to a higher standard of living.

However, growth rates are not same across countries and over time. There are 215 countries in the world, whose GDPs are growing at different rates. In any given year, we can observe large differences in the standard of living among countries. Table 5.1 shows income per person in 2014 of some world countries. The United States tops the list with an income of \$54, 630 *per person*. Ethiopia has an income per person of only \$570—nearly 1 percent of the figure for the United States.

Table: 6.1: International Differences in the Standard of Living: 2014

Country	Income per Person (in U.S. dollars)	Country	Income per Person (in U.S. dollars)
United States	54,630	<i>India</i>	1,582
Canada	50,230	<i>Egypt</i>	3,365
Germany	47,774	<i>Nigeria</i>	3,203
Mexico	10,326	<i>Kenya</i>	1,358
Brazil	11,726	<i>Ethiopia</i>	570

❑ **Basic Facts of Economic Growth**

- a. *There is enormous variation in per capita income across economies.*
 - ✓ Countries at the top of the world income distribution are more than thirty times as rich as those at the bottom.
 - ✓ For instance, PCI (at PPP) in the US is more than 36 times higher than PCI in Ethiopia in 2014.
- b. *Rates of economic growth vary substantially across countries.*
 - ✓ Eg: GDP growth rate was, on average, 10% in Ethiopia but 5.5% for Singapore over the last decade.
- c. *Growth rates do vary over time.*
- d. *A countries relative position in the world distribution of per capita incomes is not immutable.*
 - ✓ Countries can move from being “poor” to being “rich” (Eg: East Asian countries) & vice versa (Eg: Argentina).

But why do growth rates and performances differ ever so much? Every country has a different starting point. The supply of labor, the stock of resources, physical capital and human capital, the level of infrastructure, the climate conditions, political and institutional environment, the volume of international trade, and the current GDP are some of the main aspects influencing the growth of countries.

Economists try to model the relationships and the development of these and other variables over time in trying to explain growth dynamics. They try to reveal how decisions made by economic agents, the value of the variables in the past, present and future, affect growth rates and achievements.

Our goal in this chapter is, therefore, to understand what causes these differences in income over time and across countries by examining different growth models.

6.2. Models of Economic Growth

6.2.1. The Harrod-Domar Growth Model

✚ The Harrod-Domar growth model is based on the work by these two authors in the 1930's. They developed their models independently, but they arrive at the same results. The model emphasizes the key role of investment in determining economic growth. Their argument can be stated as: Every economy must save a certain proportion of its national income, if only to replace worn-out or impaired capital goods. However, in order to grow, new investments representing net additions to the capital stock are necessary. If we assume that there is some direct economic relationship between the size of the total capital stock, K , and total GDP, Y —it follows that any net additions to the capital stock in the form of new investment will bring about corresponding increases in the flow of national output, GDP.

Notice that this relationship, known in economics as the capital-output ratio. If **we define the capital-output ratio as θ and assume further that** the national net savings ratio, s , is a fixed proportion of national output and that total new investment is determined by the level of total savings, we can construct a simple model of economic growth:

Model Assumptions

1. Net saving (S) is some proportion, s , of national income (Y) such that we have the simple equation: $S_t = sY_t$ ————— (1)

2. Capital stock (K) is proportional to output (Y): i.e.,

$$K_t = \theta Y_t \text{ ————— (2)}$$

3. We assume a closed economy (no trade or capital flows). Then savings S must be used for investments I , and investments can only come from savings. Hence they must be equal:

$$S_t = I_t \text{ ————— (3)}$$

4. In year t , the stock of capital is K_t . The year after, year $t+1$, the capital stock is K_{t+1} . Changes in the capital stock come from investments and the depreciation (wearing out) of the capital stock. Depreciation occurs at a constant rate δ , so an amount δK_t of capital disappears every year. Hence, the **capital accumulation equation** is:

$$K_{t+1} = K_t + I_t - \delta K_t$$

$$\Delta K = K_{t+1} - K_t = I_t - \delta K_t \text{ --- --- --- --- --- (4)}$$

Now, having the above assumptions, we can solve for the growth rate of capital and/or output. Therefore, the growth rate of capital is defined as

$$g_k = \frac{K_{t+1} - K_t}{K_t} = \frac{\Delta K}{K}$$

$$g_k = \frac{I_t - \delta K_t}{K_t} = \frac{I_t}{K_t} - \delta$$

And we know that $I_t = sY_t$, and $K_t = \theta Y_t$, so substitution yields:

$$g_k = \frac{sY_t}{\theta Y_t} - \delta$$

$$g_k = \frac{s}{\theta} - \delta \text{ --- --- --- --- --- (5)}$$

As we are ultimately interested in the rate of growth of GDP, we use that $Y_t = \frac{1}{\theta} K_t$. Then we get that growth in GDP is

$$g_y = \frac{Y_{t+1} - Y_t}{Y_t} = \frac{\frac{1}{\theta} K_{t+1} - \frac{1}{\theta} K_t}{\frac{1}{\theta} K_t} = \frac{K_{t+1} - K_t}{K_t}$$

This indicates that GDP and capital grow at the same rate. Hence, growth in GDP is

$$g_k = \frac{s}{\theta} - \delta \text{ --- --- --- --- --- (6)}$$

This equation states that, to grow, economies must save and invest a certain proportion of their GDP. Countries with lower savings and hence investment will end up at lower level of GDP. Thus, the more countries can save and invest, the faster they can grow.

□ **Evidence in support of the H-D Model**

As we know Chinese growth has been impressive registering a growth rate of 10% per year for over 30 years beginning in the early 1980s. Many international observers view that high savings as successful experience supporting China's rapid growth. The average rate of the world saving is 21%, less than half of that in China, which is averaged 47.4% in the last decade. In particular, china's savings to GDP ratio was 30.5%, 34.7%, 40.9%, and 47.4% in the 1970s, 1980s, 1990s, and 2000s, respectively. Researchers indicated that this high level of saving over the years was one major source of capital accumulation and rapid long-term growth in china.

6.2.2. The Neo-Classical model: The Solow Growth model

Robert Solow (1956), with the publication of his work “**A Contribution to the Theory of Economic Growth**”, provides a good growth model. His Model is designed to show how growth in the capital stock, growth in the labor force, and advances in technology interact in an economy, and how they affect a nation’s total output of goods and services.

✚ **The model is built around two equations: a production function and a capital accumulation equation.** As our first step in building the model, we examine how the supply and demand for goods determine the **accumulation of capital**. To do this, we **hold the labor force and technology fixed**. Later, we relax these assumptions, first by introducing changes in the labor force and then by introducing changes in technology.

6.2.2.1. Accumulation of Capital in the Solow Model

Let’s analyze the supply and demand for goods, and see how much output is produced at any given time and how this output is allocated among alternative uses.

1. The supply of goods and the production function.

✚ The supply of goods in the Solow model is based on the following production function: Which states output depends on the capital stock and the labor force.

$$Y = F(K, L) \text{ — — — — — (1)}$$

✚ The Solow growth model assumes that the production function has constant returns to scale. Thus, recall that a production function has constant returns to scale if

$$zY = F(zK, zL) \quad \text{for positive value, } z$$

That is, if we multiply both capital and labor by z , we also multiply the amount of output by z .

✚ **Production functions with constant returns to scale allow us to analyze all quantities in the economy relative to the size of the labor force. Besides, Per capita output is claimed to be the best measure of welfare and hence we are interested in explaining output per worker.** With this interest in mind, we can rewrite the production function in equation [1] in terms of **output per worker** and capital per worker.

✚ Divide both sides of equation [1], to obtain: $\frac{Y}{L} = F\left(\frac{K}{L}, 1\right)$

✓ This equation states that output per worker Y/L is a function of capital per worker K/L . **So, from now on, let's denote all quantities in per worker terms in lower case letters.** Thus, $y = Y/L$ is output per worker and $k = K/L$ is capital per worker.

✚ Equation [1] can, thus, be rewritten as:

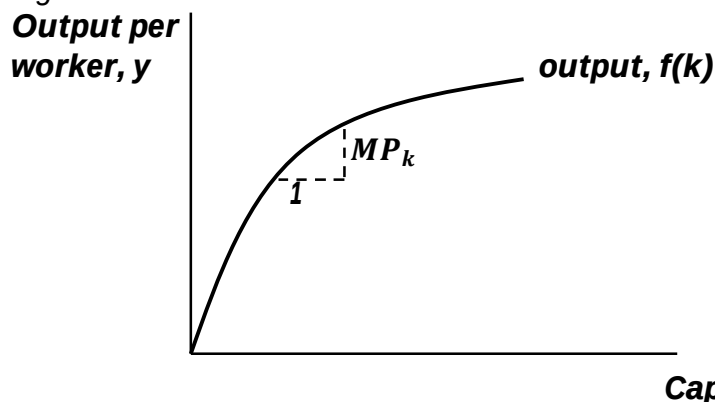
$$y = f(k) \text{ --- (2),}$$

Where $f(k) = F(k, 1)$.

This production function is graphed below. With more capital per worker, firms produce more output per worker. However, there are diminishing returns to capital per worker: each additional unit of capital we give to a single worker increases the output of that worker by less and less. This amount is the marginal product of capital MP_K . Mathematically, we write: $MP_K = f(k + 1) - f(k)$

✚ Note that in Figure 5-1, as the amount of capital increases, the production function becomes flatter, indicating that the **production function exhibits diminishing marginal product of capital**.

Figure: 6.1: The Production Function



The production function shows how the amount of capital per worker k determines the amount of output per worker $y = f(k)$.

➤ **Implication:** Countries with small capital stock (k) are more productive => grow faster

II. The Demand for Goods and the Consumption Function

✚ Solow assumes that the economy is closed, so that saving equals investment, and the only use of investment is to accumulate capital. The demand for goods in the Solow model comes from consumption and investment. In other words, output per worker y is divided b/n consumption per worker c and investment per worker i :

$$y = c + i \text{ ————— (3)}$$

This equation is the national income accounts identity for the economy. It omits government purchases (which for present purposes we can ignore) and because it expresses y , c , and i as quantities per worker.

✚ The Solow model assumes that the consumption function takes the simple form:

$$c = (1 - s)y \text{ ————— (4)}$$

Where, s , the saving rate.

This consumption function states that consumption is proportional to income. Each year a fraction $(1-s)$ of income is consumed, and a fraction s is saved.

To see what this consumption function implies for investment, substitute $(1 - s)y$ for c in the national income accounts identity:

$$y = (1 - s)y + i$$

Rearrange the terms to obtain

$$i = sy \text{ ————— (5)}$$

This equation states that **investment, like consumption, is proportional to income**. Since **investment equals saving**, the rate of saving s is also the fraction of output devoted to investment.

III. Growth in the Capital Stock and the Steady State

Having introduced the two main ingredients of the Solow model-the **production function** and the **consumption function**-we can now examine how increases in the capital stock over time result in economic growth.

✚ Two forces cause the capital stock to change:

- **Investment:** The capital stock rises as firms buy new plants and equipment.
- **Depreciation:** The capital stock falls as some of the old capital wears out.

Recall that investment per worker $i = sy$.

Let's substitute the production function for y from [2], and then we can express investment per worker as a function of the capital stock per worker:

$$i = sf(k), \text{ ————— (6)}$$

This equation relates the existing stock of capital k to the accumulation of new capital i . Figure 5-2 shows this relationship. *This figure illustrates how, for any value of k , the amount of output is determined by the production function $f(k)$, and the allocation of that output b/n consumption and saving is determined by the saving rate s .*

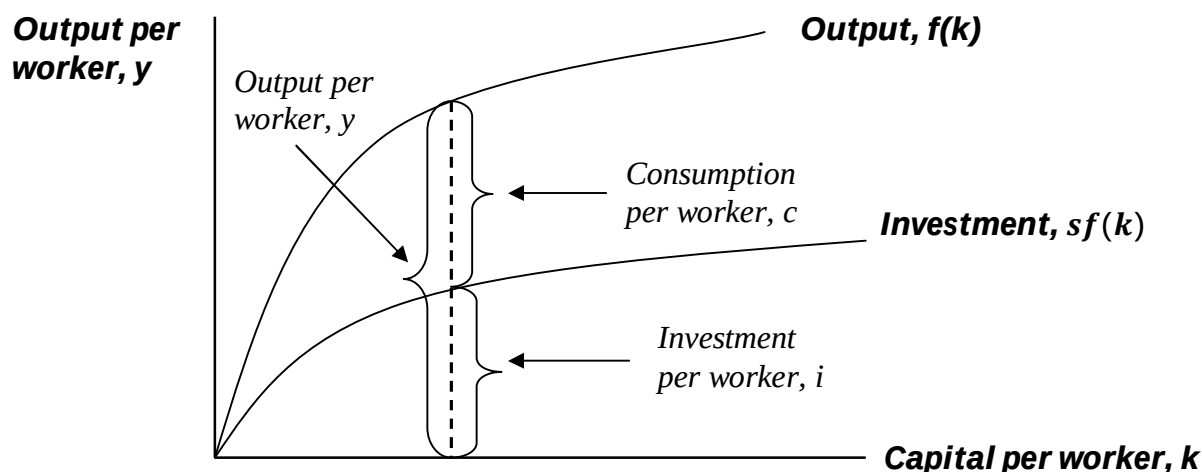


Figure 5.2: Output, Consumption, and Investment

Note: The saving rate s determines the allocation of output b/n consumption and investment.
At any level of capital k , output is $f(k)$, investment is $sf(k)$, and consumption is $f(k) - sf(k)$.

To incorporate depreciation into the model, we assume that a certain fraction δ of the capital stock wears out each year. We call δ the depreciation rate. So, depreciation is proportional to the capital stock.

Now, we can express the impact of investment and depreciation on the capital stock in the following equation:

$$\text{Change in Capital Stock} = \text{Investment} - \text{Depreciation}$$

$$\text{i. e., } \Delta k = i - \delta k \text{ --- (7)}$$

where Δk is the change in the capital stock between one year and the next. Because investment i equals $sf(k)$, we can write this as:

$$\Delta k = sf(k) - \delta k \text{ --- (8)}$$

Equation [8] is termed as the **capital accumulation equation**.

Figure 5-4 graphs the terms of this equation—investment and depreciation—for different levels of the capital stock k .

- ✓ **The higher the capital stock, the greater will be the amounts of investment and output. Yet the higher the capital stock, the greater will also be the amount of depreciation.**

Figure 5.3 shows that there is a single capital stock at which the amount of investment equals the amount of depreciation. If the economy has this capital stock, the capital stock will not change over time because the two forces acting to change it – investment and depreciation – just balance. That is, at this level of the capital stock, $\Delta k = 0$. **We call this the steady-state level of capital (long-run equilibrium of the economy) and designate it as k^* .**

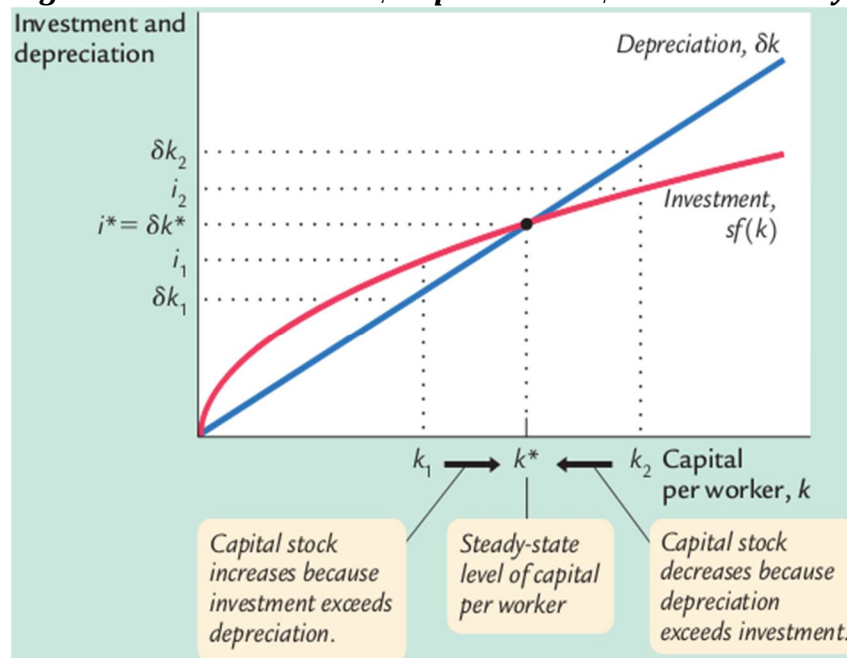
❖ **Implication: Approaching the Steady State**

- ✓ An economy at the steady state will stay there.
- ✓ An Economy which is not in the steady state will go there => convergence to the constant level of output per worker over time. But an economy not at the steady state will go there. That is, regardless of the level of capital with which the economy begins, it ends up with the steady-state level of capital. In this sense, the steady state represents the long-run equilibrium of the economy.

✚ To see why an economy always ends up at the steady state, suppose that the economy starts with less than the steady-state level of capital, such as level k_1 in Figure 5-3. In this case, the level of investment exceeds the amount of depreciation. Over time, the capital stock will rise and will continue to rise —along with output $f(k)$ —until it approaches the steady state k^* .

✚ Similarly, suppose that the economy starts with more than the steady-state level of capital, such as level k_2 . In this case, investment is less than depreciation: capital is wearing out faster than it is being replaced. The capital stock will fall, again approaching the steady-state level. Once the capital stock reaches the steady state, investment equals depreciation, and there is no pressure for the capital stock to either increase or decrease.

Figure: 6. 3: Investment, Depreciation, and the Steady State



✚ The steady-state level of capital k^* is the level at which investment equals depreciation- indicating that the amount of capital will not change over time.

- ✓ Below k^* , investment exceeds depreciation, so the capital stock grows.
- ✓ Above k^* , investment is less than depreciation, so the capital stock shrinks.

Numerical Example: Approaching the steady state

- Let's use a numerical example to see how the Solow model works and how the economy approaches the steady state. For this example, we assume that the production function is: $Y = K^{\frac{1}{2}}L^{\frac{1}{2}}$
- To derive the per worker production function $f(k)$, divide both sides of the production function by L : $\frac{Y}{L} = \frac{K^{\frac{1}{2}}L^{\frac{1}{2}}}{L}$

- Substitute y for Y/L , and rearrange to obtain: $y = \left(\frac{K}{L}\right)^{1/2}$

Since $k = K/L$, this becomes $y = k^{1/2}$ or $y = \sqrt{k}$
Output per worker is the square root of capital per worker.

- To complete the example, we assume that 30% of output is saved ($s = 0.3$), that 10% of the capital stock depreciates every year ($\delta = 0.1$), and that the economy starts with 4 units of capital per worker ($k = 4$). We can now examine what happens to this economy over time.

- Recall that: $\Delta k = sf(k) - \delta k$

Since $\Delta k = 0$ in the steady state, we know that

$$0 = sf(k^*) - \delta k^*$$

$$\text{or, equivalently, } \frac{k^*}{f(k^*)} = \frac{s}{\delta}$$

Substituting the values above, we obtain

$$\frac{k^*}{\sqrt{k^*}} = \frac{0.3}{0.1} = 3$$

Square both sides to find $k^* = 9$.

- ✓ The steady-state capital stock is 9 units per worker.

How Saving Affects Growth

Evidences tell us that the international difference in economic performance is partly due to differences in saving rates among countries.

- Hence, to understand the effect of savings, let's consider what happens to an economy when the saving rate increases in the Solow model. Fig. 5.4 shows such a change. We assume that the economy begins in a steady state with saving rate s_1 and capital stock k_1^* . The saving rate then increases from s_1 to s_2 , causing an upward shift in the $sf(k)$ curve. At the initial saving rate s_1 and the initial capital stock, k_1^* , the amount of investment just offset the amount of depreciation. The moment after the saving rate

risks, investment is higher, but the capital stock and depreciation are unchanged. Therefore, investment exceeds depreciation. The capital stock will gradually rise until the economy reaches the new steady state, k_2^* , with a higher capital stock and a higher level of output.

The Solow model shows that the *saving rate is a key determinant of the steady-state capital stock*. If the saving rate is high, the economy will have a large capital stock and a high level of output. If the saving rate is low, the economy will have a small capital stock and a low level of output.

According to the Solow model, higher saving leads to faster growth, but only in the short run. An increase in the rate of saving raises growth until the economy reaches the new steady state. If the economy maintains a high saving rate, it will also maintain a large capital stock and high level of output, but will not maintain a high rate of growth forever.

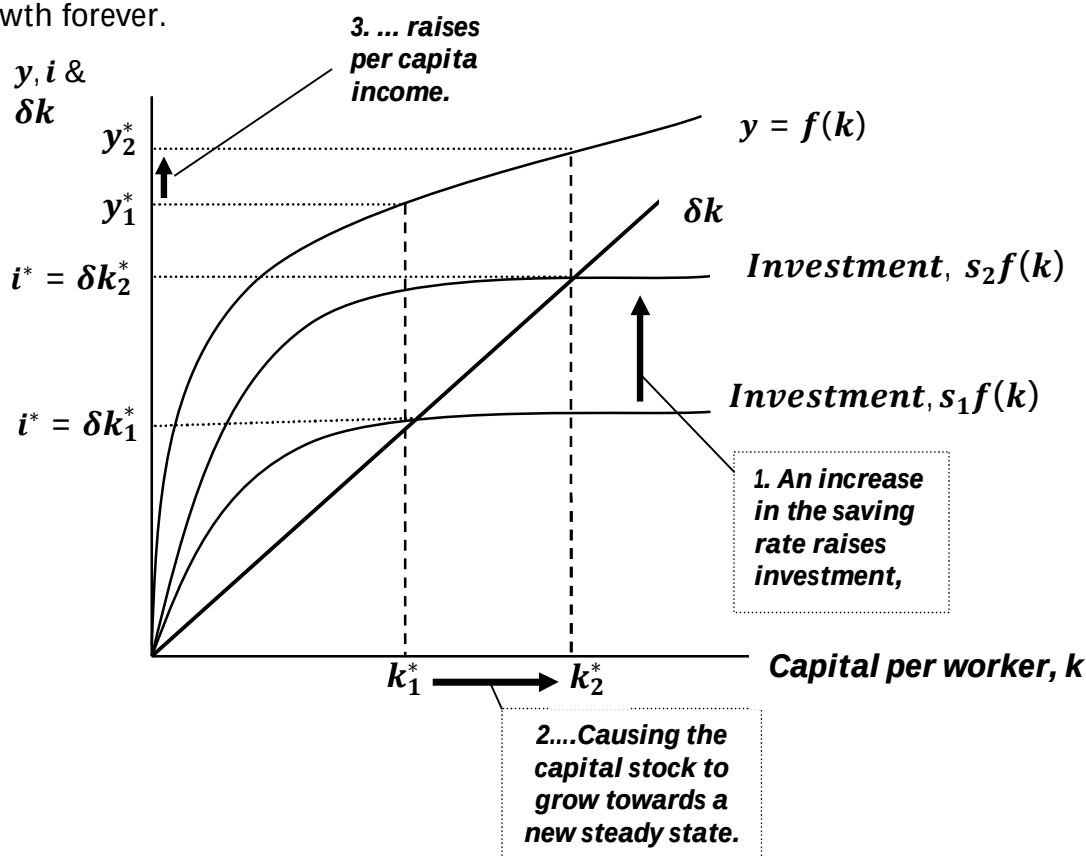


Figure: 6.4: Saving, Capital Stock & Economic Growth

Note: An increase in the saving rate, s implies that the amount of investment for any given capital stock is higher. It therefore shifts the saving function upward. At the initial steady state k_1^* , investment now exceeds depreciation. The capital stock rises until the economy reaches a new steady state k_2^* with more capital and output.

✚ The Golden Rule Level of Capital

As mentioned above, the Solow growth model allows us a dynamic view of how savings affects the economy over time. We also learned about the steady state level of capital. Now, we assume policy makers can set the savings rate to determine a steady state level of capital that maximizes consumption per worker. Note that: when choosing a steady state, the policymaker's goal is to maximize the wellbeing of the society (where wellbeing is measured by consumption level).

❖ **The steady-state value of k that maximizes consumption is called the Golden Rule Level of Capital and is denoted by k^*_{gold} .**

✚ To find steady-state consumption per worker, we begin with the national income accounts identity: $y = c + i$

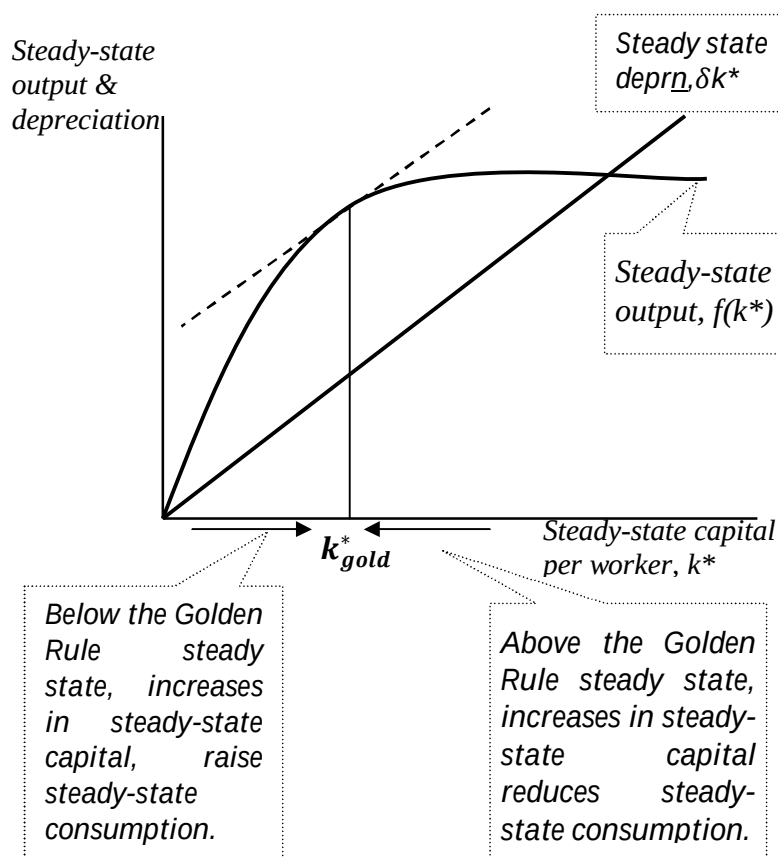
And rearrange it as: $c = y - i$

✚ Since, we want **to find steady-state consumption**; we substitute steady-state values for **output and investment**. From [2] **Steady-state output per worker is $f(k^*)$, where k^* is the steady-state capital stock per worker**. Furthermore, since investment is equal to depreciation, δk at the steady state. Substituting $f(k^*)$, for y and δk^* for i , we can write **steady state consumption per worker** as:

$$c^* = f(k^*) - \delta k^* \text{ — — — — — (9)}$$

- ✓ This **equation states that steady-state consumption is the difference between steady-state output and steady-state depreciation.**
- ✓ **Equivalently, because, consumption per worker is the difference between output per worker and investment per worker we want to choose k^* so that this distance is maximized. This is the golden rule level of capital k^*_{gold}**
- ✓ *The slope of the production function is the marginal product of capital MP_K . The slope of the δk^* line is δ . Because these two slopes are equal at k^*_{gold} , a **condition that characterizes the golden rule level of capital is $MP_K = \delta$.***
 - *At the Golden Rule level of capital, the marginal product of capital equals the depreciation rate.*
 - *If $MP_K - \delta > 0$, then increases in capital increase consumption, so k^* must be below the Golden Rule level.*
 - *If $MP_K - \delta < 0$, then increases in capital decrease consumption, so k^* must be above the Golden Rule level.*

Figure 5.5 graphs steady-state output and steady-state depreciation as a function of the steady-state capital stock. Steady-state consumption is the gap b/n output and depreciation. **This figure shows that there is one level of the capital stock—the Golden Rule level, k^*_{gold} —that maximizes consumption.**



The economy's output is used for consumption or investment.

- ✓ In the steady state, investment equals depreciation. Therefore, steady-state consumption is the difference between output $f(k^*)$ and depreciation δk^* .
- ✓ Steady-state consumption is maximized at the Golden Rule steady state. The Golden Rule capital stock is denoted k^*_{gold} , & the Golden Rule consumption is denoted c^*_{gold} .

NB: At the Golden Rule level of capital, the production function and the δk^* line have the same slope, and consumption is at its greatest level.

Figure: 6.5: Steady State Consumption

What is the golden rule level of savings?

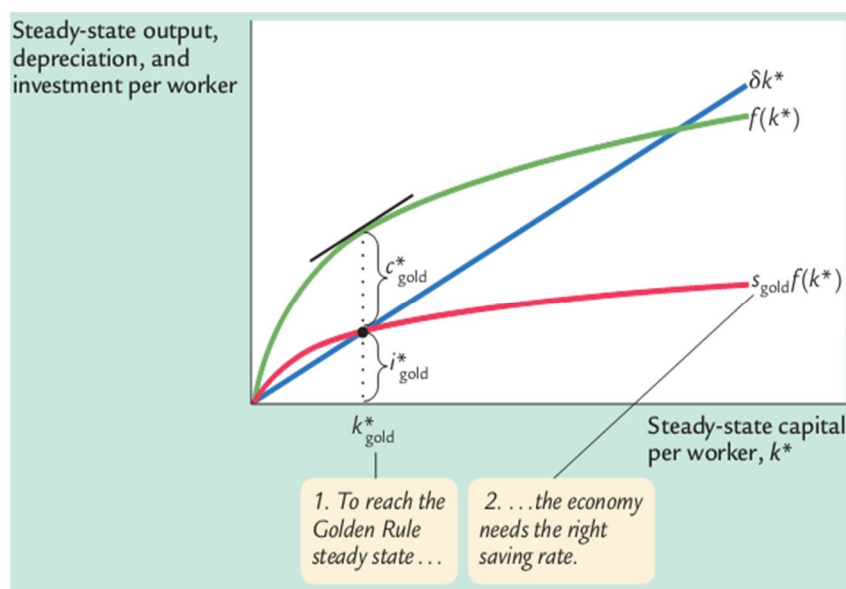


Figure: 5.6: The Saving Rate & the Golden Rule:

- ✓ There is only one saving rate that produces the Golden Rule level of capital k^*_{gold} . Any change in the saving rate would shift the $s f(k)$ curve & would move the economy to a steady state with a lower level of consumption.

Keep in mind that the economy does not automatically gravitate toward the Golden Rule steady state. If we want any particular steady-state capital stock, such as the Golden Rule,

we need a particular saving rate to support it. The figure below shows the steady state if the saving rate is set to produce the Golden Rule level of capital. If the saving rate is higher than the one used in this figure, the steady-state capital stock will be too high. If the saving rate is lower, the steady-state capital stock will be too low. In either case, steady-state consumption will be lower than it is at the Golden Rule steady state.

Golden Rule Steady State: A Numerical Example

Consider the production function in our earlier example; $y = \sqrt{k}$, and assume that **10%** of the capital stock depreciates each year. Then, calculate the golden rule level of steady state capital?

Solution: At the golden rule level, the slope of the production function in per capita terms equals with the rate of depreciation. That is, $MP_k = \delta$

Therefore, with $y = \sqrt{k}$, $MP_k = \frac{1}{2\sqrt{k}}$. Then, $\frac{1}{2\sqrt{k}} = 0.1$

Solve for k and we obtain: $k_{gold}^* = 25 \text{ units}$

6.2.2.2. Population Growth in the Solow Model

The basic Solow model shows that capital accumulation, by itself, cannot explain sustained economic growth: high rates of saving lead to high growth temporarily, but the economy eventually approaches a steady state in which capital and output are constant. To explain the sustained economic growth that we observe in most parts of the world, we must expand the Solow model to incorporate the other two sources of economic growth —population growth and technological progress. In this section we add population growth to the model.

The Steady State with Population Growth

How does population growth affect the steady state?

To answer this question, we must discuss how population growth, along with investment and depreciation, influences the accumulation of capital per worker over time. As we have noted before, investment raises the capital stock, and depreciation reduces it. But now there is a third force acting to change the amount of capital per worker: ***the growth in the number of workers causes capital per worker to fall.***

We continue to let lowercase letters stand for quantities per worker. Thus, $k = K/L$ is capital per worker, and $y = Y/L$ is output per worker. Keep in mind, however, that the number of workers is growing over time.

The change in the capital stock per worker (i.e., our capital accumulation equation) now is:

$$\Delta k = sf(k) - (\delta + n)k \longrightarrow \text{Discrete time version}$$

or $\dot{k} = sf(k) - (\delta + n)k \longrightarrow \text{Continuous time version}$

Proof:

We use the usual production function: $y = f(k)$ — — — — — (1)

The other key equation of the Solow model is an equation that describes how capital accumulates. Thus, the aggregate capital accumulation equation is given by:

$$\dot{K} = sY - \delta K \text{ — — — — — (2)}$$

Note that notationally, $\dot{K} = \frac{dK}{dt}$

- ✓ According to this equation, the change in the capital stock, $\dot{K}$, is equal to the amount of gross investment, sY , less the amount of depreciation that occurs during the production process, δK .

NB: The term on the left-hand side of equation [2] is the continuous time version of $K_{t+1} - K_t$, that is, the change in the capital stock per "period". We use the 'dot' notation to denote a derivative with respect to time.

❖ **To study the evolution of output per person in this economy, we rewrite the capital accumulation equation in terms of capital per person.**

Recall that $k = \frac{K}{L}$, Take the log & differentiate with respect to time:

$$\begin{aligned} \Rightarrow \ln k &= \ln K - \ln L \\ \Rightarrow \frac{d \ln k}{dt} &= \frac{d \ln K}{dt} - \frac{d \ln L}{dt} \\ \Rightarrow \frac{\dot{k}}{k} &= \frac{\dot{K}}{K} - \frac{\dot{L}}{L} \text{ — — — — — (3)} \end{aligned}$$

NB: the derivative of logarithm of a variable with respect to time gives its growth rate.

From the capital accumulation equation of [2], we have $\frac{\dot{K}}{K} = s \frac{Y}{K} - \delta$. Substituting this expression into [3],

$$\Rightarrow \frac{\dot{k}}{k} = s \frac{Y}{K} - \delta - n, \quad \text{where } n \text{ is the growth in labour force, } \frac{\dot{L}}{L}$$

$$\Rightarrow \frac{\dot{k}}{k} = s \frac{Y/L}{K/L} - \delta - n$$

$$\therefore \dot{k} = sy - (\delta + n)k \text{----- (4)}$$

- ✓ This equation shows how investment, depreciation, and population growth influence the per-worker capital stock. Investment increases k , whereas depreciation and population growth decrease k .

✚ We can think of the term $(\delta + n)k$ as defining **Break-Even Investment** —the **amount of investment necessary to keep the capital stock per worker constant**. Break-even investment includes the depreciation of existing capital, which equals δk . It also includes the amount of investment necessary to provide new workers with capital. The amount of investment necessary for this purpose is, because there are n new workers, and because k is the amount of capital for each worker. The equation shows that population growth reduces the accumulation of capital per worker much the way depreciation does. Depreciation reduces k by wearing out the capital stock, whereas population growth reduces k by spreading the capital stock more thinly among a larger population of workers.

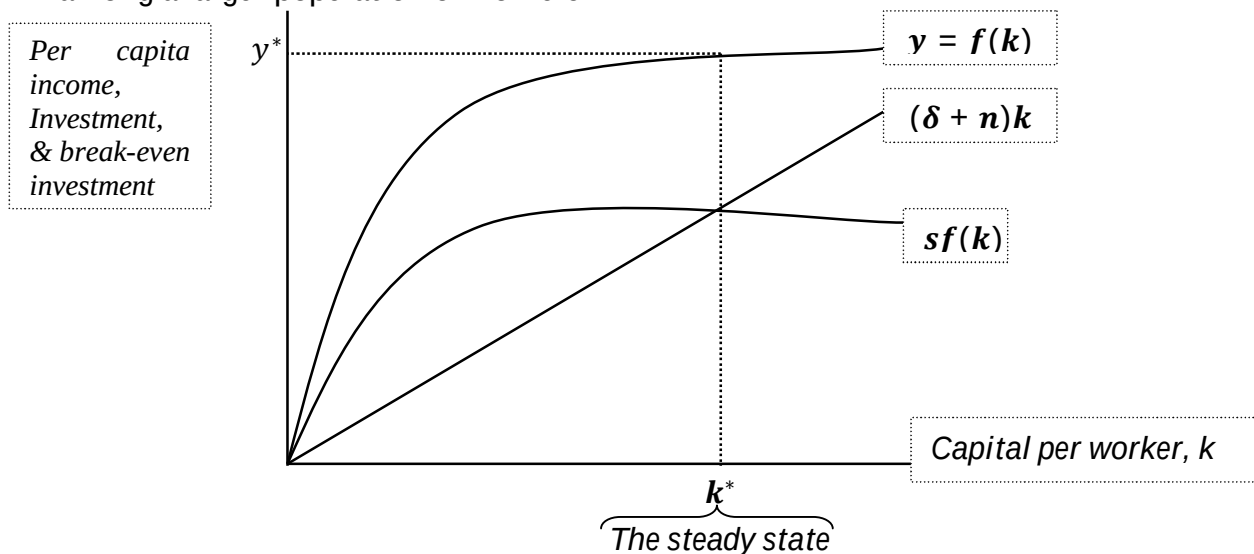


Figure: 6.7: Population Growth in the Solow Model

Note: For the economy to be in a steady state, investment $sf(k)$ must offset the effects of depreciation & population growth $(\delta + n)k$. This is represented by the crossing of the two curves.

- ✚ An economy is in a steady state if capital per worker k is unchanging. If k is less than k^* (a steady state level), investment is greater than break-even investment, so k rises. If k is greater than k^* , investment is less than break-even investment, so k falls. In the steady state, the positive effect of investment on the capital stock per worker exactly balances the negative effects of depreciation and population growth. That is, at k^* , $\Delta k = 0$ and $i^* = \delta k^* + nk^*$.

✚ The Effect of Population Growth in the Solow Model

- As the figure below presents, an increase in the rate of population growth from n_1 to n_2 reduces the steady-state level of capital per worker from k_1^* to k_2^* . Thus, the Solow model predicts that economies with higher rates of population growth will have lower levels of capital per worker and therefore lower per capita incomes.
- In the steady state, the positive effect of investment on the capital per worker just balances the negative effects of depreciation and population growth.
- Once the economy is in the steady state, investment has two purposes:
 - Some of it, (δk^*) , replaces the depreciated capital,
 - The rest, (nk^*) , provides new workers with the steady state amount of capital.

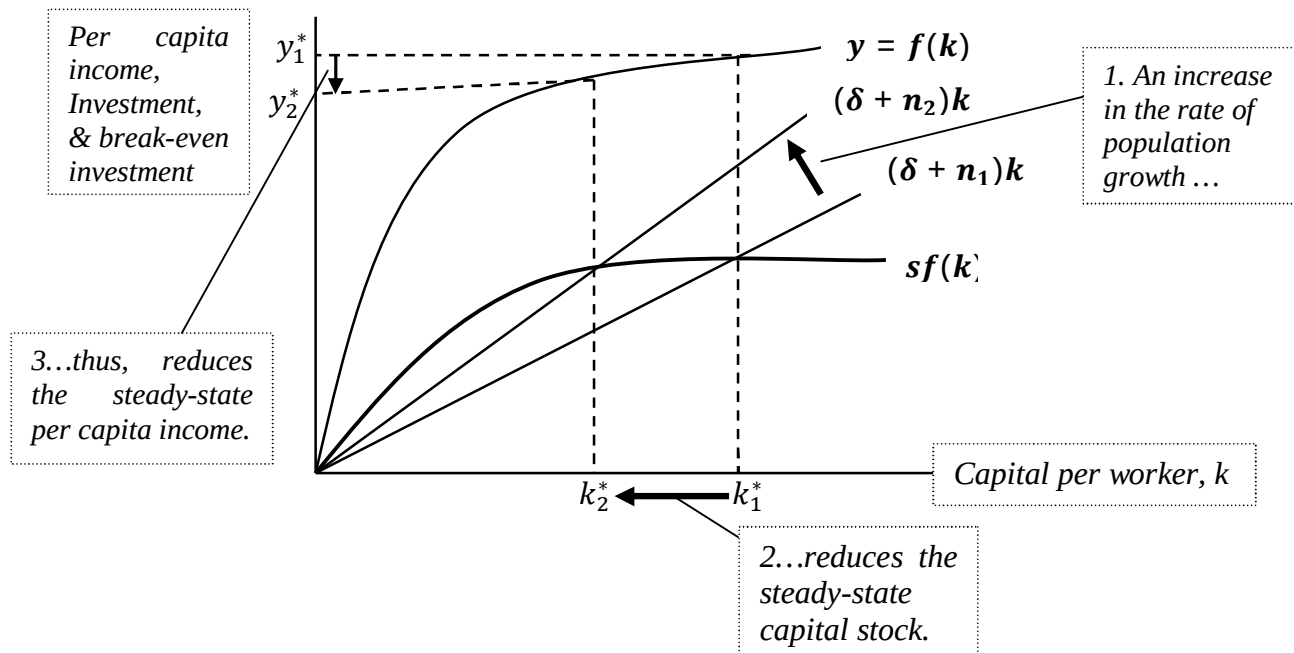


Figure: 6.8: The Impact of Population Growth in the Solow Model

Because k^* is lower, and because $y^* = f(k^*)$, the level of output per worker y^* is also lower. **Thus, the Solow model predicts that countries with higher population growth will have lower levels of GDP per person.**

Finally, population growth affects our criterion for determining the Golden Rule (consumption-maximizing) level of capital. To see how this criterion changes, note that consumption per worker is; $c = y - i$.

Because steady-state output is $f(k^*)$, and steady-state investment is $(\delta + n)k^*$, we can express steady-state consumption as

$$c^* = f(k^*) - (\delta + n)k^*.$$

The level of k^* that maximizes consumption is therefore, the one at which

$$MP_k = \delta + n,$$

or equivalently, $MP_k - \delta = n$

- ✓ That is, in the Golden Rule steady state, the marginal product of capital net of depreciation equals the rate of population growth.

Properties of the steady state

- ✓ An economy is in a steady state when per capita income and capital are constant.

Suppose the production function is given as:

$$Y = K^\alpha L^{1-\alpha} \text{-----(1)}$$

And we have the steady-state condition that $\dot{k} = 0$. so, we can solve for the steady state quantities of capital per worker and output per worker.

From our production function, we have

$$y = k^\alpha \text{-----(2)}$$

Therefore, by substitution we obtain:

$$\dot{k} = sk^\alpha - (n + \delta)k,$$

and setting this equation equal to zero yields:

$$\Rightarrow sk^\alpha = (n + \delta)k$$

$$\Rightarrow k = \frac{sk^\alpha}{(n+\delta)}$$

$$\Rightarrow k^* = \left(\frac{s}{n+\delta}\right)^{\frac{1}{1-\alpha}} \text{-----Steady state capital per worker}$$

Substituting this into the production function reveals the steady-state quantity of output per worker, y^* :

$$\Rightarrow y^* = \left(\frac{s}{n+\delta}\right)^{\frac{\alpha}{1-\alpha}} \text{-----Steady state output per worker}$$

✚ This equation reveals the Solow model's answer to the question "Why are we so poor and they so rich?" countries that have high saving/investment rates will tend to be richer, *ceteris paribus*. Such countries accumulate more capital per worker, and countries with more capital per worker have more output per worker. Countries that have high population growth rates, in contrast, will tend to be poorer, according to the Solow model.

✚ **Economic Growth in the Solow Model with population Growth**

What does economic growth look like in the steady state of this simple version of the Solow model? The answer is that there is **no per capita growth** in this version of the model! *Output per worker is constant in the steady state*. Output itself, Y , is growing, of course, but only at the rate of population growth. This can be shown as:

$$\Rightarrow y = \frac{Y}{L}, \text{ take log both sides, and then differentiate with respect to time}$$

$$\Rightarrow \ln y = \ln Y - \ln L$$

$$\Rightarrow \frac{d \ln y}{dt} = \frac{d \ln Y}{dt} - \frac{d \ln L}{dt}$$

Since the growth in per capita output is zero at the steady state

$$\Rightarrow 0 = \frac{d \ln Y}{dt} - \frac{d \ln L}{dt},$$

$$\Rightarrow \frac{d \ln Y}{dt} = \frac{d \ln L}{dt} \Rightarrow \frac{\dot{Y}}{Y} = \frac{\dot{L}}{L}$$

Thus, this model fails to predict a very important fact: **that economies exhibit sustained per capita income growth**. In this model, economies may grow for a while, but not forever. For example, an economy that begins with a stock of capital per worker below its steady-state value will experience growth in k and y along the *transitional path* to the steady state. Over time, however, growth slows down as the economy approaches its steady state, and eventually growth stops altogether.

To see that growth slows down along the transition path, *notice two things*.

1st, from the capital accumulation equation:

$$\dot{k} = sy - (\delta + n)k, \text{ and from equation (2) above we have, } y = k^\alpha, \text{ thus:}$$

$$\Rightarrow \frac{\dot{k}}{k} = s \frac{k^\alpha}{k} - (\delta + n)$$

$$\Rightarrow \frac{\dot{k}}{k} = \frac{s}{k^{1-\alpha}} - (\delta + n)$$

- Therefore, because α is less than one, as k rises, **the growth rate of k gradually declines**.

2nd, the growth rate of y is proportional to the growth rate of k , so that the same statement holds true for **output per worker**- i.e.

$$\begin{aligned} y &= k^\alpha, & \Rightarrow \ln y &= \alpha \ln k \\ & & \Rightarrow \frac{d \ln y}{dt} &= \alpha \frac{d \ln k}{dt} \\ & & \Rightarrow \frac{\dot{y}}{y} &= \alpha \frac{\dot{k}}{k} \end{aligned}$$

- ✓ This indicates that when the growth rate of k is zero at the steady state, output growth would also be zero. **Thus, the model doesn't explain growth at the steady state.**

As a Conclusion

The Solow model developed so far shows how saving and population growth determine the economy's steady-state capital stock and its steady-state level of income per person. As we have seen, it sheds light on why countries that save and invest a high fraction of their output are richer than countries that save and invest a smaller fraction, and why countries with high rates of population growth are poorer than countries with low rates of population growth.

What the model cannot do, however, is explain the persistent growth in living standards we observe in most countries. In the model we developed so far, when the economy reaches its steady state, output per worker stops growing. To explain persistent growth, we need to introduce technological progress into the model.

6.2.2.3. Technology and the Solow Model

To generate sustained growth in per capita income, we need to introduce technological progress to the model. This is accomplished by adding a technology variable, **A**, to the production function:

$$Y = F(K, AL) = K^\alpha (AL)^{1-\alpha} \text{ — — — — — (1)}$$

Entered this way, the technology variable A is said to be **"labour-augmenting" or "Harrod-neutral"**. Technological progress occurs when **A** increases over time – a unit of labour, for example, is more productive when the level of technology is higher.

An important assumption of the Solow model is that technological progress is **exogenous**: in a common phrase, technology is like **"manna from heaven"**, in that it descends upon the economy automatically and regardless of whatever else is going on in the economy. Instead of modelling carefully where technology comes from, we simply

recognize for that there is technological progress and make the assumption that A is growing at a constant rate, $\frac{\dot{A}}{A} = g_A = g$ where g is a parameter representing the growth rate of technology.

- Note that we use *lower case letters for per capita terms* and **lower case with a hat for per effective worker levels**.

✚ Per capita levels:

$$k_t = \frac{K_t}{L_t}, \quad y_t = \frac{Y_t}{L_t} \Rightarrow y_t = \frac{F(K, L)}{L} = f(k), \quad i_t = \frac{I_t}{L_t} \quad \& \quad c_t = \frac{C_t}{L_t}$$

✚ Per effective worker levels:

$$\hat{k}_t = \frac{K_t}{A_t L_t}, \quad \hat{y}_t = \frac{Y_t}{A_t L_t} \Rightarrow \hat{y}_t = \frac{F(K, AL)}{AL} = f(\hat{k}), \quad \hat{i}_t = \frac{I_t}{L_t} \quad \& \quad \hat{c}_t = \frac{C_t}{L_t}$$

The capital accumulation equation in this model is the same as before- i.e.

$$\dot{K} = sY - \delta K$$

$$\frac{\dot{K}}{K} = s \frac{Y}{K} - \delta \quad \text{--- (2)}$$

- ✓ Capital grows with investment ($I = sY$), and a part of it is destroyed each period with depreciation (δK).
- ✚ To see the growth implication of this model, first rewrite the production function [1] in terms of output per worker:

$$\begin{aligned} \Rightarrow Y &= F(K, AL) = K^\alpha (AL)^{1-\alpha} \\ \Rightarrow \frac{Y}{L} &= \frac{F(K, AL)}{L} = \frac{K^\alpha A^{1-\alpha} L^{1-\alpha}}{L} \\ &\Rightarrow y = \left(\frac{K}{L}\right)^\alpha A^{1-\alpha} \\ &\Rightarrow y = k^\alpha A^{1-\alpha} \end{aligned}$$

- ✓ Taking the logs and differentiating the above expression with respect to time:

$$\begin{aligned} \ln y &= \alpha \ln k + (1 - \alpha) \ln A \\ \frac{d \ln y}{dt} &= \alpha \frac{d \ln k}{dt} + (1 - \alpha) \frac{d \ln A}{dt} \\ \frac{\dot{y}}{y} &= \alpha \frac{\dot{k}}{k} + (1 - \alpha) \frac{\dot{A}}{A} \quad \text{--- (3)} \end{aligned}$$

Notice from the capital accumulation equation [2] that the growth rate of K will be constant if and only if Y/K is constant. Furthermore, if Y/K is constant, y/k is also constant (i.e., $\frac{Y}{K} = \frac{Y/L}{K/L} = \frac{y}{k}$) and most important, y and k will be growing at the same rate. A situation in which capital, output, and consumption are growing at constant rates is called a **Balanced Growth Path**.

Along the balanced growth path, $g_y = g_k$. Substituting this relationship into equation [3]:

$$g_y = \alpha g_k + (1 - \alpha)g$$

Since $g_y = g_k$ along the balanced growth path

$$g_y = \alpha g_y + (1 - \alpha)g$$

$$g_y(1 - \alpha) = (1 - \alpha)g$$

$$g_y = g_k = g$$

- That is, along the balanced growth path in the Solow model, output per worker and capital per worker both grow at the rate of exogenous technological change, g .

Notice that in the simple model (i.e., the model without technology), there was no technological progress, and therefore there was no long-run growth in output per worker or capital per worker; $g_y = g_k = 0$.

- ❖ **The Solow model with technology reveals that technological progress is the source of sustained per capita growth.**

✚ The Solow Diagram with Technology

In the case where we introduce technology in the Solow model, the only important difference is that the variable k is no longer constant in the long run. So we have to write our differential equation in terms of another variable. The new state variable will be $\hat{k} = K/AL$. Notice that this is equivalent to k/A and is obviously constant along the balanced growth path because $g_k = g_A = g$. The variable $\hat{k}$ therefore represents the ratio of capital per worker to technology (or capital per effective worker). We refer to this as the "capital-technology" ratio (b/c the numerator is capital per worker).

- ✚ **Rewriting the production function, $Y = K^\alpha (AL)^{1-\alpha}$ in terms of per effective worker, we obtain:**

$$\hat{y} = Y/AL = \left(\frac{K}{AL}\right)^\alpha$$

$$\hat{y} = \hat{k}^\alpha \text{-----[4]}$$

Where, $\hat{y} = Y/AL = y/A$ and $\hat{k} = K/AL = k/A$.

Note: $\hat{y}$ is called the "output-technology ratio" (output per effective labour).

Equation [4] implies that output per effective worker depends on capital per effective worker.

Recall that: $\hat{k} = K/AL$. If we log linearize this, and differentiate with respect to time (to get growth rate of $\hat{k}$):

$$\begin{aligned} \Rightarrow \ln \hat{k} &= \ln K - (\ln A + \ln L) \\ \Rightarrow \frac{d \ln \hat{k}}{dt} &= \frac{d \ln K}{dt} - \left(\frac{d \ln A}{dt} + \frac{d \ln L}{dt} \right) \\ \Rightarrow \frac{\dot{\hat{k}}}{\hat{k}} &= \frac{\dot{K}}{K} - \frac{\dot{A}}{A} - \frac{\dot{L}}{L} \end{aligned} \quad (5)$$

From the aggregate capital accumulation equation [2], we know that $\frac{\dot{K}}{K} = sY/K - \delta$

Substituting this into the above expression, we get:

$$\begin{aligned} \Rightarrow \frac{\dot{\hat{k}}}{\hat{k}} &= sY/K - \delta - g - n \\ \Rightarrow \frac{\dot{\hat{k}}}{\hat{k}} &= s \frac{Y/AL}{K/AL} - \delta - g - n \\ \Rightarrow \frac{\dot{\hat{k}}}{\hat{k}} &= s \frac{\hat{y}}{\hat{k}} - \delta - g - n \\ \therefore \dot{\hat{k}} &= s\hat{y} - (n + g + \delta)\hat{k} \end{aligned}$$

Or, $\dot{\hat{k}} = s\hat{y} - (n + g + \delta)\hat{k}$ ----- [6]

(Fundamental Differential Equation for Capital Stock Accumulation in the Neoclassical Solow Model with Exogenous Technological Progress)

- ✓ As before, the change in the capital stock $\dot{\hat{k}}$ equals investment $s\hat{y}$ minus break-even investment $(n + g + \delta)\hat{k}$.
- ✓ Now, however, because $\hat{k} = K/AL$, **Break-Even Investment** (minimum amount of investment required to keep capital per-effective worker constant) includes three terms: to keep capital-per effective worker, $\hat{k}$ constant, each period, new investment amounted to $\delta\hat{k}$ is needed to replace depreciating capital, $n\hat{k}$ is needed to provide capital for new workers, and $g\hat{k}$ is needed to provide capital for the **new "effective workers"** created by technological progress.

Equation [6] would give the Solow diagram with technology as given below. There is one level of $\hat{k}$, denoted $\hat{k}^*$, at which capital per effective worker and output per effective worker are constant.

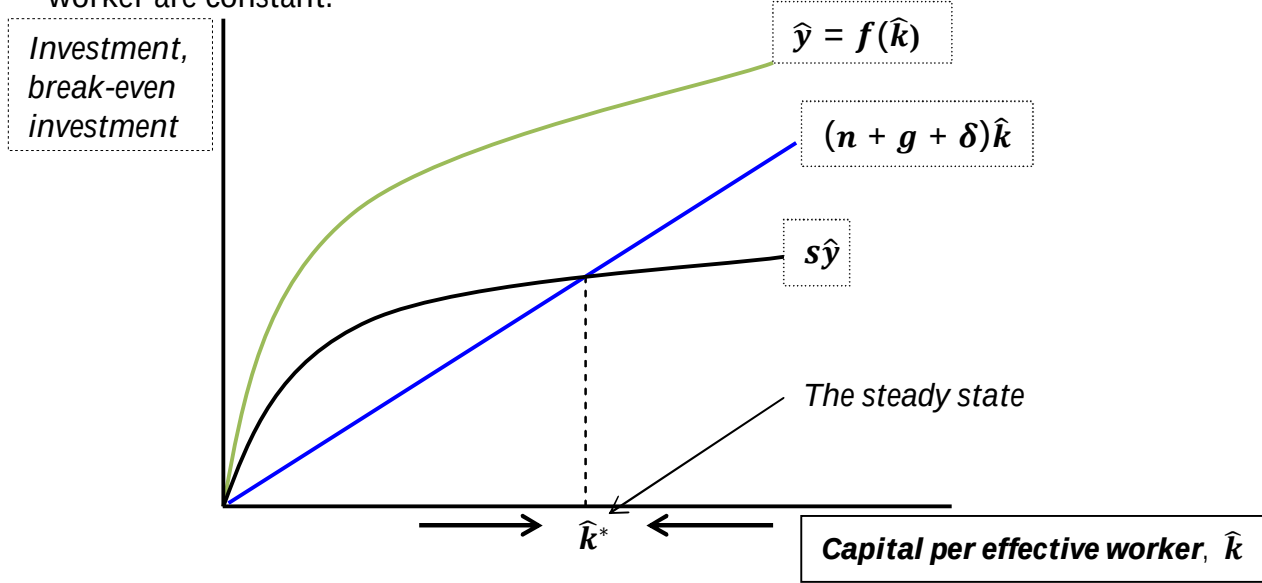


Figure: 6.9: Technological Progress and the Solow Growth Model

- Note that: In the steady state, investment, $s\hat{y}$ exactly offsets the reductions in $\hat{k}$ attributable to depreciation, population growth, and technological progress.
- ✚ If the economy begins with a Capital per effective worker that is below its steady-state level, the Capital per effective worker will rise gradually over time because the amount of investment being undertaken exceeds the amount needed to keep the Capital per effective worker constant.
- ✚ The introduction of technological progress modifies the criterion for the Golden Rule (consumption-maximizing) level of capital. The Golden Rule level of capital is now defined as the steady state that maximizes consumption per effective worker.

To see how this criterion changes, note that consumption per effective worker is;
 $\hat{c} = \hat{y} - \hat{i}$

Because steady-state output is $f(\hat{k}^*)$, and steady-state investment is $(\delta + g + n)\hat{k}^*$, we can express steady-state consumption as

$$\hat{c}^* = f(\hat{k}^*) - (\delta + g + n)\hat{k}^*.$$

The level of $\hat{k}^*$ that maximizes consumption is therefore, the one at which

$$MP_k = \delta + g + n,$$

or equivalently,
$$MP_k - \delta = n + g$$

- ✓ That is, in the Golden Rule steady state, the marginal product of capital net of depreciation equals the rate of population growth plus the rate of technology growth.

Steady State Solutions and Growth rates /Effects of Technological Progress/

The steady-state capital and output-per effective worker is determined by the production function and the steady-state condition – i.e. $\hat{k} = 0$. Setting $\hat{k} = 0$ we can solve for $\hat{k}^*$ as follows:

$$\Rightarrow 0 = s\hat{y} - (n + g + \delta)\hat{k}$$

$$\Rightarrow s\hat{y} = (n + g + \delta)\hat{k}$$

From equation [2], we have $\hat{y} = \hat{k}^\alpha$

$$\Rightarrow s\hat{k}^\alpha = (n + g + \delta)\hat{k}$$

$$\Rightarrow \frac{\hat{k}}{\hat{k}^\alpha} = \left(\frac{s}{n + g + \delta}\right)$$

$$\therefore \hat{k}^* = \left(\frac{s}{n + g + \delta}\right)^{1/1-\alpha}$$

← **Steady State Capital per effective worker**

To see what this expression implies about capital per worker, rewrite the equation as:

$$\hat{k} = K/AL = k/A$$

$$k = \hat{k} A$$

$$k^*(t) = \hat{k} A(t)$$

$$k^*(t) = A(t) \left(\frac{s}{n + g + \delta}\right)^{1/1-\alpha} \text{-----} [7]$$

Since s , n , g and α are all constants, capital per worker grows with growth in technology.

Finding the growth rate (e.g., if $Y = X.Z$, thus, $g_Y = g_X + g_Z$)

Hence, $g_k = g_A + 0 = g$

➤ **Thus, the growth rate of capital per worker is the growth rate of labour augmented technology (at a rate, g)**

- ✓ Substituting the steady state $\hat{k}^*$ in to the production function, $\hat{y} = \hat{k}^\alpha$, we get the **Steady state output per-effective worker.**

$$\therefore \hat{y}^* = \left(\frac{s}{n + g + \delta}\right)^{\alpha/1-\alpha}$$

← **Steady State Output per effective worker**

To see what this expression implies about output per worker, rewrite the equation as

$$\hat{y} = Y/AL = y/A$$

$$y = \hat{y} A$$

$$y^*(t) = \hat{y} A(t)$$

$$y^*(t) = A(t) \left(\frac{s}{n + g + \delta} \right)^{\alpha/1-\alpha} \text{-----} [8]$$

- ✓ Since s , n , g , δ and α are all constants, output per worker grows with growth in technology.

o Finding the growth rate, we get: $g_y = g_A + 0 = g$

- Thus, the growth rate of output per worker is the growth rate of labour augmented technology (at a rate, g).

An interesting result from equation [8] is that changes in the investment rate or the population growth rate affect the long-run level of output per worker but do not affect the long-run growth rate of output per worker.

Conclusion on the Effect of Technological progress on Economic Growth

With the addition of technological progress, the Solow model explains the sustained increases in standards of living. That is, we have shown that technological progress can lead to sustained growth in output per worker. By contrast, a high rate of saving leads to a high rate of growth only until the steady state is reached. Once the economy is in steady state, the rate of growth of output per worker depends only on the rate of technological progress.

- **According to the Solow model, only technological progress can explain persistently rising living standards.**

Summary: Steady-State Growth Rates in the Solow Model with Technological Progress

<i>Variable</i>	<i>Symbol</i>	<i>Steady State level formula</i>	<i>Steady State Growth Rates</i>
<i>Capital per effective worker</i>	$\hat{k}^* = \frac{K}{AL}$	$\hat{k}^* = \left(\frac{s}{n + g + \delta} \right)^{\frac{1}{1-\alpha}}$	$g_{\hat{k}^*} = 0$
<i>Output per effective worker</i>	$\hat{y}^* = \frac{Y}{AL}$	$\hat{y}^* = \left(\frac{s}{n + g + \delta} \right)^{\frac{\alpha}{1-\alpha}}$	$g_{\hat{y}^*} = 0$
<i>Capital per worker</i>	$k^* = \frac{K}{L} = \hat{k} \cdot A$	$k^*(t) = A(t) \left(\frac{s}{n + g + \delta} \right)^{\frac{1}{1-\alpha}}$	$g_{k^*} = g$
<i>Output per worker</i>	$y^* = \frac{Y}{L} = \hat{y} \cdot A$	$y^*(t) = A(t) \left(\frac{s}{n + g + \delta} \right)^{\frac{\alpha}{1-\alpha}}$	$g_{y^*} = g$
<i>Aggregate Capital, K_t</i>	$K_t = \hat{k}^* \cdot A \cdot L$	$K_t = A(t)L(t) \left(\frac{s}{n + g + \delta} \right)^{\frac{1}{1-\alpha}}$	$g_K = g + n$
<i>Aggregate Output, Y_t</i>	$Y_t = \hat{y}^* \cdot A \cdot L$	$Y_t = A(t)L(t) \left(\frac{s}{n + g + \delta} \right)^{\frac{\alpha}{1-\alpha}}$	$g_Y = g + n$

 The Solow Model: Evaluation

How does the Solow model answer the key questions of growth? First, the Solow model appeals to differences in investment rates and population growth rates and (perhaps) to exogenous differences in technology to explain differences in per capita incomes. Why some countries are so rich and others so poor? According to the Solow model, it is because the rich countries invest more and have lower population growth rates, both of which allow the rich countries to accumulate more capital per worker and thus increase labour productivity. This hypothesis is supported by data across countries of the world.

Second, why do economies exhibit sustained growth in the Solow model? The answer is technological progress. Without technological progress, per capita growth will eventually cease as diminishing returns to capital set in. Technological progress, however, can offset the tendency for the marginal product of capital to fall, and in the long run, countries exhibit per capita growth rate of technological progress.

How, then, does the Solow model account for differences in growth rates across countries? At this juncture, the Solow model appeals to the transitional dynamics. An economy with a capital-technology ratio below its long-run level will grow rapidly until

the capital-technology ratio reaches its steady-state level. This reasoning may help explain why some countries such as Japan and Germany, which had their capital stocks wiped out by World War II, have grown more rapidly than the United States over the last fifty years. Or it may explain why an economy that increases its investment rate will grow rapidly as it makes the transition to a higher output-technology ratio. This explanation may work well for countries such as South Korea, Singapore, and Taiwan which increased their investment rates dramatically since 1950.

6.2.3. Endogenous Growth Theory

From the Solow growth model we saw that for an economy that exhibits CRS that simply accumulating factors of production will not lead to permanently growing values of capital, output, and consumption per capita. We also saw that while there was some evidence supportive of the Solow growth model there was also quite a bit of evidence which suggested that quite a bit still needed explaining, especially why countries kept on growing and what was causing technological change to occur.

Endogenous Growth theory began with the thesis of Paul Romer at the University of Chicago in 1986, and has flourished ever since. His main insight was that technological change is caused by the deliberate actions of people and that it responds to economic incentives as does any economic activity. He and others after him also explored what impact this view of technological change would have for economic growth.

- ***Models that try to explain the source and progress of technological progress often go by the label endogenous growth theory, because they reject the Solow model's assumption of exogenous technological change.***
- o Growth in this model is driven by technological change that arises from intentional investment decisions on R&D by profit-maximizing agents.

Some Characteristics of Technology:

- ✓ ***It requires huge investment:*** E.g: the creation of Boeing 747 jet aircraft, cost was about \$1.2 billion.
- ✓ ***Free rider problem:*** ownership of R&D outcomes (technology) is difficult. There is certain degree of non-exclusiveness problem. This in turn discourages reinvestment on R&D.
- ✓ ***Capital Spillover Effects:*** Romer's central insight was that the benefits of some forms of capital not only accrued to the individual with the capital but *spilled over* to others. Now, let's look how human capital produces Spillovers on others as an

example. That is, why the accumulation of human capital may not only benefit the individual accumulating it, but others around him or her:

i. Education:

Education increases the skills and intellectual capabilities of people and it is possible that there are spillovers in knowledge generation between the more educated. Knowledge generation is probably much greater in specialist research labs, than if each person worked alone because they can discuss ideas with other skilled people in the area they are working on. It may also be easier to acquire human capital when the people around you have a lot of human capital themselves (i.e. they help a person learn much faster (that is at a lower cost) than if the person had to learn by themselves).

ii. Learning-By-Doing

As firms produce more of a good, people in the work-force learn how to make it more efficiently. People share their experiences (that is, their human capital) with each other, helping them avoid errors and helping them find better ways of producing the goods. Thus, the human capital acquired by a person not only is an input directly into producing output, but also acts to increase the level of technology of the economy due to the resulting capital spillovers.

Note that: Capital spillovers imply that the aggregate production function exhibits

Increasing Returns to Scale, contrary to the Solow model.

Implications for Government Policies

1. Government Actions May Affect Growth Rates

What happens if the government subsidizes R&D? This causes that the cost of innovation to decrease, so that innovative activity increases. This is likely to lead to a higher rate of technological progress, and, therefore, a higher growth rate in output per worker.

2. Importance of Property Rights: For people and firms to have an incentive to invest in R&D and create new techniques or products, they need to have some security over the ownership of the capital they sacrifice to create.

A Simple Endogenous Growth Model: The AK Model

The model's structure is the same as the Solow growth model with the exception of the production function (i.e., The production function is modified so that $\alpha = 1$). Thus, the aggregate production function is now:

$$Y = AK \text{ --- [1]}$$

where Y is output, K is the capital stock, and A is a constant measuring the amount of output produced for each unit of capital.

 Notice that this production function does not exhibit the property of diminishing returns to capital. One extra unit of capital produces A extra units of output, regardless of how much capital there is. **This absence of diminishing returns to capital is the key difference between this endogenous growth model and the Solow model.**

- Now let's see how this production function relates to economic growth. As before, we assume a fraction s of income is saved and invested. We therefore describe capital accumulation with an equation similar to that we used previously:

$$\dot{K} = sY - \delta K \text{ --- [2]}$$

- ✓ This equation states that Capital grows with investment ($I = sY$), and a part of it is destroyed each period with depreciation (δK).

Combining equation [2] with the $Y = AK$ production function, we obtain, after a bit of manipulation,

$$\dot{Y}/Y = \dot{K}/K = sA - \delta \text{ --- [3]}$$

What determines the growth rate of output, $\dot{Y}/Y$?

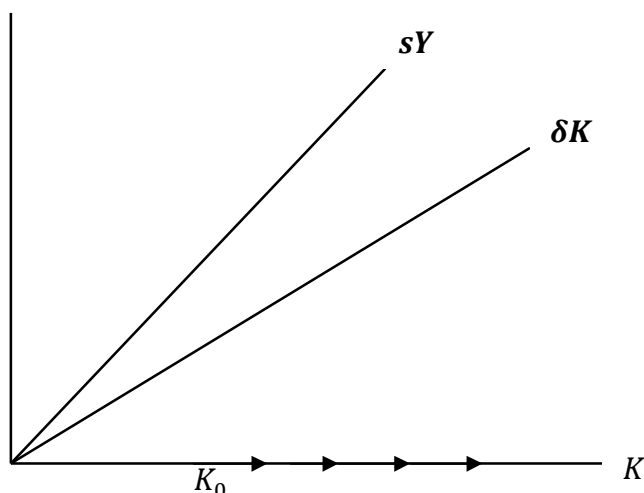


Diagram for the AK Model

- ✓ Notice from [3] that, as long as $sA > \delta$, the economy's income grows forever, even without the assumption of exogenous technological progress.
- ✚ Consider the above figure drawn for the AK model.
 - The δK line reflects the amount of investment that has to occur just to replace the depreciation of the existing capital stock.
 - The sY curve is total investment as a function of the capital stock. Notice that because Y is linear in K , this curve is actually a straight line, **a key property of the AK model**. We assume that total investment is larger than total depreciation, as drawn.
- ✚ Consider an economy that starts at point K_0 . In this economy, because total investment is larger than depreciation, the capital stock grows. Over time, this growth continues: at every point to the right of K_0 , total investment is larger than depreciation. **Therefore, the capital stock is always growing, and growth in the model never stops.**
 - The simple algebra $\dot{Y}/Y = \dot{K}/K = sA - \delta$ reveals a key result of the AK growth model: **the growth rate of the economy is an increasing function of the investment rate. Therefore, government policies that increase the investment rate of an economy permanently will increase the growth rate of the economy permanently.**
- ✚ This result can be interpreted in the context of the Solow model with $\alpha < 1$. Recall that in this case, the sY line is a curve, and the steady state occurs when $sY = \delta K$ (since we have assumed $n = 0$). The parameter α measures the “curvature” of the sY curve: if α is small, then the curvature is rapid, and sY intersects δK at a “low” value of K^* . On the other hand, the larger α is, the further away the steady-state value, K^* , is from K_0 . *This implies that the transition to steady state is longer.* The case of $\alpha = 1$ is the limiting case, in which the transition dynamics never end. **In this way, the AK model generates growth endogenously.** That is, we need not assume that anything in the model grows at some exogenous rate in order to generate per capita growth—certainly not technology.

Thus, a simple change in the production function can alter dramatically the predictions about economic growth.

- In the Solow model, saving leads to growth temporarily, but diminishing returns to capital eventually force the economy to approach a steady state in which growth depends only on exogenous technological progress.
- **By contrast, in this endogenous growth model, saving and investment can lead to persistent growth.**

✚ But is it reasonable to abandon the assumption of diminishing returns to capital? The answer depends on how we interpret the variable K in the production function, $Y = AK$. If we take the traditional view that K includes only the economy's stock of plants and equipment, then it is natural to assume diminishing returns. Giving 10 computers to each worker does not make the worker 10 times as productive as he or she is with one computer. *Advocates of endogenous growth theory, however, argue that the assumption of constant (rather than diminishing) returns to capital is more palatable if K is interpreted more broadly.*

✚ **Perhaps the best case for the endogenous growth model is to view knowledge as a type of capital.** Clearly, knowledge is an important input into the economy's production—both its production of goods and services and its production of new knowledge. Compared to other forms of capital, however, it is less natural to assume that knowledge exhibits the property of diminishing returns. (Indeed, the increasing pace of scientific and technological innovation over the past few centuries has led some economists to argue that there are increasing returns to knowledge.)

- If we accept the view that knowledge is a type of capital, then this endogenous growth model with its assumption of constant returns to capital becomes a more plausible description of long-run economic growth.

Conclusion

Endogenous growth theories attempt to explain the rate of technological progress by explaining the decisions that determine the creation of knowledge through research and development. By contrast, the Solow model simply took this rate as exogenous. In the Solow model, the saving rate affects growth temporarily, but diminishing returns to capital eventually force the economy to approach a steady state in which growth depends only on exogenous technological progress. By contrast, many endogenous growth models in essence assume that there are constant (rather than diminishing) returns to capital, interpreted to include knowledge. Hence, changes in the saving rate can lead to persistent growth.

✚ **Review Exercises**

1. Explain the growth implication of the Harrod-Domar growth model.
2. In the Solow model, how does the saving rate affect the steady-state level of income? How does it affect the steady-state rate of growth?
3. Why might an economic policymaker choose the Golden Rule level of capital?

4. In the Solow model, how does the rate of population growth affect the steady-state level of income? How does it affect the steady-state rate of growth?
5. In the Solow model with technology, what determines the steady-state rate of growth of income per worker?
6. How do you determine whether an economy has more or less capital than in the Golden Rule steady state in the Solow model with technology?
7. How does endogenous growth theory explain persistent growth without the assumption of exogenous technological progress? How does this differ from the Solow model?

II. Workout Questions

1. Country A and country B both have the production function, $Y = F(K, L) = K^{1/2}L^{1/2}$.
 - A. Does this production function have constant returns to scale? Explain.
 - B. What is the per-worker production function?
 - C. Assume that neither country experiences population growth nor technological progress and that 5 percent of capital depreciates each year. Assume further that country A saves 10 percent of output each year and country B saves 20 percent of output each year. Using your answer from part (b) and the steady-state condition that investment equals depreciation, find the steady-state level of capital per worker for each country. Then find the steady-state levels of income per worker and consumption per worker.
2. Consider an economy described by the production function $Y = F(K, L) = K^{0.3}L^{0.7}$
 - A. What is the per-worker production function?
 - B. Assuming no population growth or technological progress, find the steady-state capital stock per worker, output per worker, and consumption per worker as functions of the saving rate and the depreciation rate.
 - C. Suppose the saving rate, $s = 40$ percent, and depreciation rate, $\delta = 5$ percent.
 - i. Calculate the steady state level of capital per worker and output per worker.
 - ii. What are the golden rule levels of capital accumulation and consumption for this economy?
 - iii. Explain whether the economy's steady state capital is more or less than the Golden Rule steady state?
3. An economy described by the Solow growth model has the following production function: $\hat{y} = \sqrt{\hat{k}}$.
 - A. Solve for the steady-state value of $\hat{y}$ as a function of s, n, g , and δ .
 - B. A developed country has a saving rate of 28 percent and a population growth rate of 1 percent per year. A less-developed country has a saving rate of 10 percent

- and a population growth rate of 4 percent per year. In both countries, $g = 0.02$ and $\delta = 0.04$. Find the steady-state value of $\hat{y}$ for each country.
- C. What policies might the less-developed country pursue to raise its level of income?
4. Suppose the production function of two countries (developed and less developed) is given as; $Y = F(K, L) = K^{1/2}L^{1/2}$. The capital stock grows as the usual Neo-classical capital accumulation equation (i.e., $\frac{dK}{dt} = \dot{K} = sY - \delta K$), where δ is depreciation rate, s is saving rate, and t is time.
- A. Assuming that there is no technological progress, find the steady-state capital stock per worker, output per worker as functions of the saving rate, population growth rate and the depreciation rate **(2.5 points)**.
- B. A developed country has a saving rate (s) of 36 percent and a population growth rate (n) of 1 percent per year. A less-developed country has a saving rate of 18 percent and a population growth rate of 4 percent per year. In both countries 5 percent of the capital stock depreciates each year.
- Find the steady-state value of capital stock and output per worker and the corresponding steady state consumption per worker for each country **(3 points)**.
- C. In accordance with the Solow Growth Model, what are the implications of the results in **A** and **B** above? Explain what policies might the less-developed country pursue to raise its per capita income? **(2 points)**.
5. Suppose the production function of a hypothetical economy is given by $Y = K^\beta (AL)^{1-\beta}$; where Y is output, L is total labour force, K is total capital stock, and A is labour-augmenting technology. The capital stock of this hypothetical economy grows as the usual Neo-classical capital accumulation equation (i.e., $\frac{dK}{dt} = \dot{K} = sY - \delta K$), where δ is depreciation rate, s is saving rate, and t is time.
- A. Find the formula for the growth rate of output per worker, and capital per worker. **(3 points)**
- B. Suppose $\beta = 2/3$, $s = 25$ percent, $\delta = 3$ percent, the growth rate of the labour force is 1.5 percent, labour-augmenting technology is growing at 3 percent annually.
- What is the growth rate of capital and output per worker for this economy? **(2 points)**
- C. What are the golden rule level of capital stock and consumption for this economy? **(2 points)**.

CHAPTER SEVEN
MACROECONOMIC THEORIES AND AFRICAN ECONOMIES

Chapter Objectives:

- ❖ *At the end of this chapter, you will be able to:*
 - ✓ *Conceptualize and evaluate the debates on the applicability of macroeconomic theories to African Economies;*
 - ✓ *Mention and explain the basic distinguishing features of LDCs and African Economies in particular from the Industrial Economies ;*
 - ✓ *Evaluate the applicability of conventional theories to African Economies;*
 - ✓ *State the main objectives and Policy Measures of the structural Adjustment Program (SAP);*
 - ✓ *Justify why the SAP had failed to achieve the intended objectives?*
 - ✓ *Evaluate the Performance of African Economies after the reform program.*

7.1 . The Applicability of Conventional Theories to African Economies

7.1.1. Introduction

Over the past decades there has appeared a serious and growing concern for the problems and potentials of short-term macroeconomic policy in less developed countries (LDCs). There were also much more debate on the applicability of conventional macro model to African economies in particular and developing countries in general. But the major views can be categorized into two:

A) The Orthodox View

According to this view such macro models are applicable elsewhere be it developed or developing country. For this view, the major factors responsible for low economic development is misallocation of resources that emanates from government intervention in the form of price controls, rationing, giving priority to some sectors in the economy through subsidy. Generally, the intervention of government makes market weak in doing its job. The solution forwarded by this view is that to **let the market to allocate resources and limit government intervention**. The guiding principle of economic decision is “price and market system is right”.

B) The Structuralist View

The structuralist view can be classified as:

- i. The Early Structuralist View:** it argues that the *structure of developing countries is different from that of developed countries*. Developing countries have agricultural exports which are price and income inelastic in the international market while their imports are income and price elastic. As a result, the terms of trade continue to deteriorate and therefore to achieve long term growth government intervention is essential to help them developed their industries. This school is the proponent of infant industry argument which focuses on protecting the industries and on import substitution strategy.
- ii. The Recent Structuralist View:** *this view focuses on short term adjustments*. For this school monetary and fiscal policy prescriptions are not relevant for developing countries. The IMF and the World Bank's and multilateral institutions have undesirable result on these economies and hence they are against them. For instance, ***for short term adjustment problem (like inflation), to lower the money supply is one of the IMF prescription***. But for this school, this does not have any mechanism to solve such macroeconomic problem. For developing counties where production is predominantly agrarian, markets are fragmented and underdeveloped; labor productivity is extremely badly low, ***inflation is not a demand side phenomenon***. ***Removing structural rigidities, creating well functioning and integrated markets to eliminate supply side rigidities should therefore be in the priorities of development strategies***.

7.2 . Basic Features of African Economies

-  To see whether conventional theories are applicable for developing countries or not, it is better to review the main features of these countries. Therefore, this section is devoted to outline these special features so that it is possible to see the structural characteristics that differentiate a representative developing economy from the industrial-country model.

1. Openness to Trade in Commodities and Assets

- A. *African economies tend to be much more open to trade in goods and services than are the major industrial economies.*
- A standard measure of openness is the trade share, that is, the sum of the shares of exports and imports in GDP, $(\text{Export} + \text{import})/\text{GDP}$. Evidences show that developing nations tend to be substantially more open than the major industrial countries, with the mean value of the trade share amounting to 45 percent. Hence, the conventional closed economy model is not applicable.
- B. *African countries typically have little control over the prices of the goods they export and import; that is, they typically face exogenous terms of trade.*
- The **exogeniety of the terms of trade** for developing economies in general and African countries in particular is suggested both by **their small share in the world economy and by the composition of their exports**. They **export primary products** (the price of which is market determined – on the international market) and **import manufactured and semi-processed products** (the price of which is determined at the production center); **this implies that developing countries are price takers for both exports and imports**.
 - Furthermore, for various reasons the exports of the developing world are characterized by erratic prices; that is, these economies are badly affected by fluctuations in the prices of their exports in international markets. Coupled with the relatively large share of exports and imports in the domestic economic activity, such fluctuation in export prices represent substantial exogenous changes in national income from one year to the next, and constitutes an important **source of macroeconomic volatility** for such countries.
- C. *Developing countries tend to be capital importers, and the servicing of external debt is a central policy issue in many of them.*

2. Exchange Rate Management

D. *In contrast to the major industrial countries, the vast majority of African countries have neither adopted fully flexible exchange rates nor joined monetary union.*

- Industrial countries are typically modeled either as operating flexible exchange rates or as members of a currency union, whereas in developing countries officially determined rates, adjusted by a variety of alternative rules (loosely referred to as “managed” rates) predominate. There are many issues that can be raised with respect to this exchange rate system. For instance, exchange rate management is one of the most important policy areas for LDCs. On the other side, the fact that the exchange rate is not determined in a free market via the interaction of demand and supply (as in the flexible exchange rate regime) implies that there is a possibility for the emergence of free exchange markets (the parallel/illegal markets) in parallel with the official ones. Consequently, the analysis of the aggregate implications of such informal, dual-exchange-rate regimes remains a relevant topic for macroeconomists in LDCs.

3. Domestic Financial Markets

E. *Financial markets in developing nations have long been characterized by the prevalence of rudimentary financial institutions and “financial repressions”.*

- Several African countries didn’t have equity markets, as well as secondary markets for securities. Financial markets in these economies continue to be dominated by a single type of institution- **the commercial bank**. Thus, the menu of assets available to private savers is limited.
- ❖ The commercial banking sector in developing countries has traditionally been heavily regulated: it has often been subjected to high reserve and liquidity ratios as well as legal ceilings on interest rates together with sectoral credit allocation quotas. Credit rationing in the developing world has thus tended to be legally imposed rather than endogenously generated by information asymmetries, as is commonly taken to be the case in industrial countries. This set of policies toward the financial

sector is known as “**financial repression**”. The consequence of these restrictions was that the size of the commercial banking system was artificially curtailed.

- ❖ In response to financial repression, an informal financial sector has often arisen, in which loan interest rates have been market determined, again operating in parallel with the official sector, with market-determined loan interest rates. *The instruments of monetary policy and the nature of the monetary transmission mechanism in this setting tend to be quite different from their industrial-country counterparts.*

In large part, because of the nature of the financial system, but also because of some of the other features mentioned previously, the specialization of standard text book macroeconomic behavioural relationships (decision rules) may need to be modified in the developing country context.

4. The Government Budget

F. The composition of government budget differs markedly between industrial and developing countries.

- In many African nations, the state plays a pervasive role in the economy. This role is exercised through the activities of not just the non-financial public sector (consisting of the central government, local government, specialized agencies, and non-financial public enterprises), but also of financial institutions owned by the government.

5. The Labor Market

G. Although labour market institutions vary substantially across developing countries, the informal sector continues to play an important role in determination of wages and employment in many of them.

- The nature of short-run wage setting behaviour represents one of the key differences between the major schools of modern macroeconomics, but most participants in these disputes acknowledge that country specific institutional differences are important in determining the economy’s short-run supply behaviour. There is also increasing recognition of the macroeconomic role

played by the informal urban sector, which can account for a sizable proportion (up to 60 percent) of economic activity and total employment in developing countries. The consequence of formal-informal dualism is the segmentation of the urban labour market, which plays a crucial role in explanations of urban poverty, unemployment and underdevelopment in developing countries.

6. Stability of policy regimes and macroeconomic environment

H. A large part of the developing world is characterized by policy instability than in industrial countries.

- Political instability has characterized many developing countries and multiparty democracies with free elections have been rare. Changes in government frequently signal changes in ruling ideologies and correspondingly in economic policy regimes. Hence, developing countries have tended to exhibit higher fiscal deficits, substantially higher rates of inflation than the major industrial countries.

I. As a result of many of the phenomena described previously, the macroeconomic environment in developing countries is often much more volatile than in industrial countries.

- A critical feature of the macroeconomic environment in many developing countries is instability of macroeconomic outcomes. The roots of this macroeconomic instability are both external and internal. *Volatility in the terms of trade and in international financial conditions are directly transmitted to small developing countries that are price takers in international markets.* Coupled with the inflexibility of domestic macroeconomic instruments and political instability resulting in frequent and discrete changes in policy regimes in weak institutional settings, the macroeconomic experience of many developing countries has tended to be punctuated by a series of crises. Boom and bust phenomenon tend to be much more common in developing than in industrial countries; the higher degree of macroeconomic volatility that results from such phenomenon are correspondingly more costly in the developing countries.

7. Aggregate supply and Aggregate Demand functions

J. With regard to the aggregate supply function, developing countries have a slightly flat due to nominal wage rigidity accompanied by an excess supply of labor and idle capacity that suggests that there are no significant diminishing returns. But for developed countries it is close to full employment with less nominal wage rigidity.

- K. *The aggregate demand curve is downward sloping just like developed countries but much steeper.* The main reason for this is that the IS curve is relatively steep due to a high marginal propensity to import and low interest rate elasticity of investment demand.

7.3 . Some Comparative Statics

✚ Now in view of the above characteristics let us have a comparative static analysis on the new wave of short-run stabilization models of LDCs, particularly African countries forwarded by IMF which consist of devaluation and both monetary and fiscal restraint. In addition, we consider an income policy that alters the nominal wage rate. In the analysis, we contrast the standard developed countries and developing countries models; they differ not only because the slopes of the aggregate supply and demand curves are likely to be different, but also, and more importantly, because the supply and demand side of the economy interact.

Richard Porter and Susan Ranny (1982) in their seminal work attempt to develop a standard **least developed country(LDC)** model of aggregate demand and aggregate supply that can be readily compared with the standard **most developed country(MDC)** model and conducted a comparative static analysis on the conventional theories. Therefore, forthcoming discussion owes much to their work.

7.3.1. Devaluation and Developing Countries

The difference between the MDC and LDC models are perhaps most evident when considering devaluation. In the MDC, devaluation/depreciation improves the balance of trade shifting the IS and the aggregate demand curves (AD) to the right, resulting in an increase in both output and the price level; hence, restrictive monetary and fiscal policy are likely to be the appropriate stabilization policies to accompany devaluation.

In the LDC, the analysis is more complex. For a given price level, *devaluation shifts both the IS and LM curves to the left (to IS_1 and LM_1 , respectively as in the figure below), resulting in a decrease aggregate demand (to AD_1), but with an ambiguous impact on*

i [shown with no change in Figure 6.1(a)]. Even with no change in interest rate, the aggregate supply curve shifts up to AS_1 , since **devaluation directly increases variable costs. The combined impacts are a major reduction in output** [shown in Figure 6.1(b) as a fall from Y_0 to Y_2], perhaps with an increase in the price level from P_0 to P_1 . Devaluation has a direct impact of reducing the monetary base. Tax revenue from trade increase in terms of domestic currency, while that of BOP surplus is assumed to decline. **The net effect is ambiguous.**

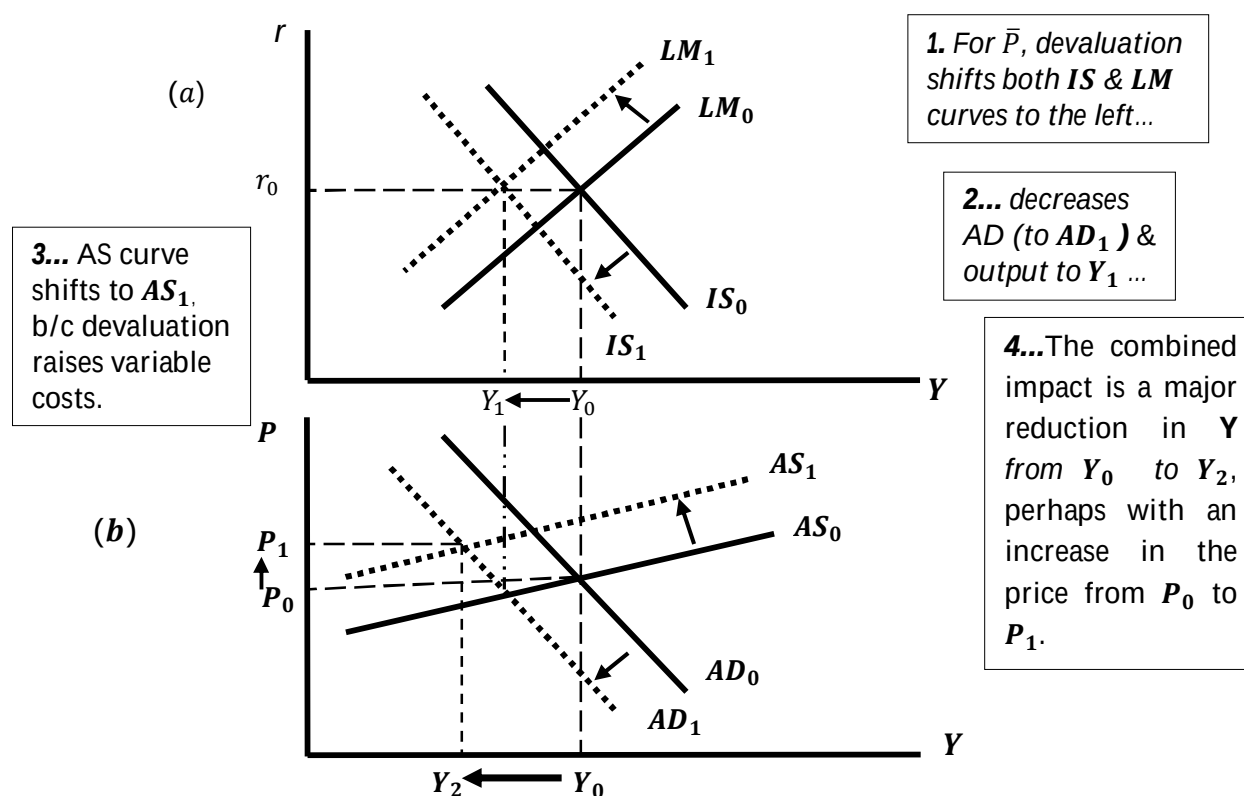


Figure: 7.1: Devaluation and African economies

7.3.2. Restrictive Monetary policy and Developing Countries

The impact of monetary contraction in the standard LDC model is shown in figure 6.2. An increase in the reserve ratio, reducing the money supply, has a large impact on the LM curve (for a given price level) because of the interest elasticity of money demand. The decrease in the availability of low interest commercial bank loans also reduces retained earning somewhat shifting the IS curve down. The interest rate rises from i_0

to i_1 and the fall in Y , from Y_0 to Y_1 , shows us the extent to which the **AD** curve shifts to the left and that the **AS** curve shifts upward (see figure 6.2 below)

- The impact of monetary contraction is thus a relatively small shift in the aggregate demand curve. And due to the flatness of the aggregate supply curve, any shift that does occur is relatively ineffective in reducing the price level. In LDC, this restrictive monetary policy also raises the aggregate supply curve through the increase in the interest cost on variable inputs. **The net impact of monetary contraction is, for MDC that the price level falls and output declines moderately; for the LDC output also falls, but the price level may increase.**

- The cumulating effects on the money supply are expansionary. Thus as time goes on, the initial impact of the increased reserve ratio, on output will be reversed and the LDC moves back toward its initial equilibrium level of income. Therefore, such policy is likely to increase both unemployment and inflation in the short run.

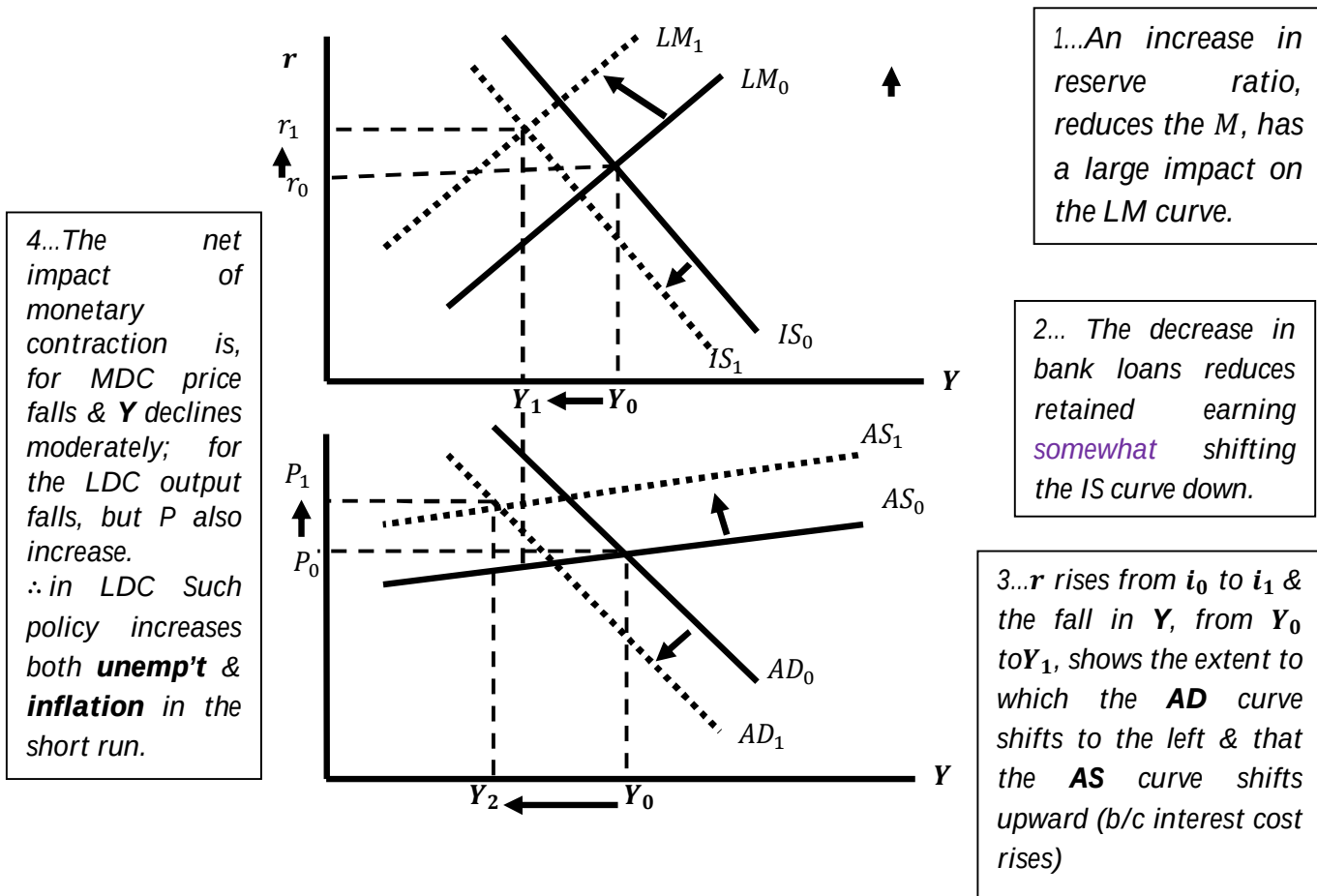


Figure: 7.2: Restrictive Monetary policy in LDCs

7.3.3. Restrictive Fiscal policy and Developing Countries

While restrictive monetary policy is likely to increase both unemployment and inflation in the LDC in the short run, restrictive fiscal policy is likely to be more successful in reducing the price level without the costs of a major recession. A decrease in government spending shifts down the IS curve, from IS_0 to IS_1 . This results in a small leftward shift in the aggregate demand curve from AD_0 to AD_1 at P_0 , but also a shift down in the aggregate supply because i has fallen from i_0 to i_1 . The net result is a decrease in the price level and possibly even an increase in output. In the standard MDC model of course, restrictive fiscal policy reduces both Y and P (See figure 6.2)

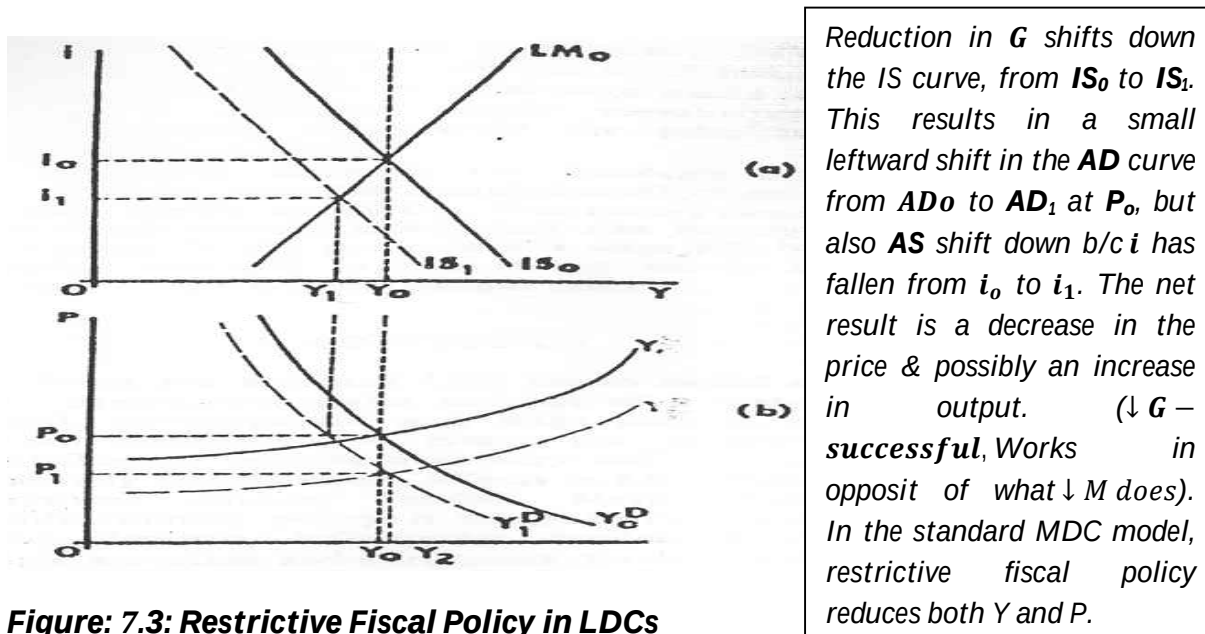


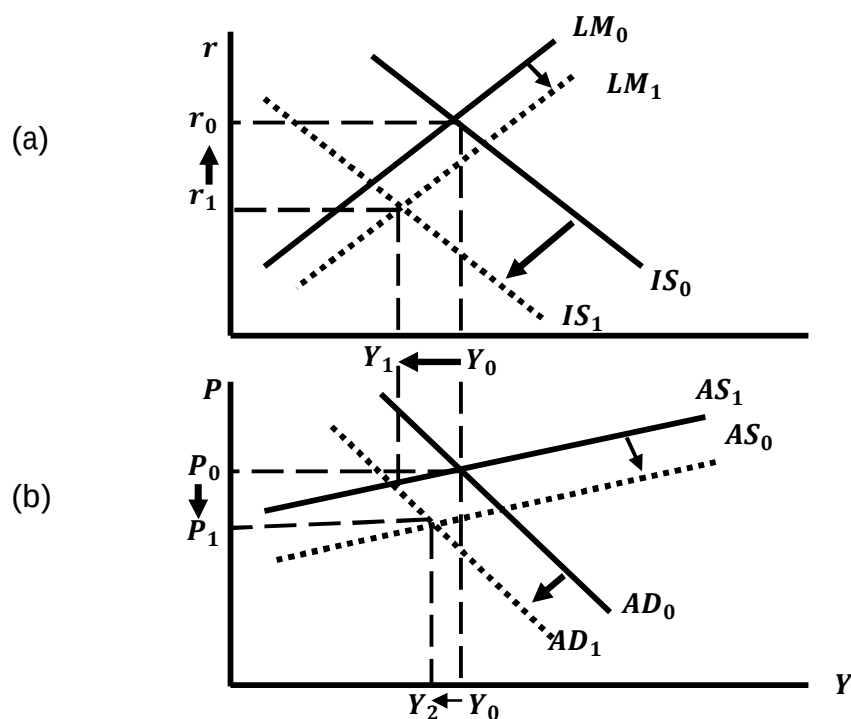
Figure 7.3: Restrictive Fiscal Policy in LDCs

7.3.4. Income policy and Developing Countries

An income policy that lowers the nominal wage affects the standard LDC economies in a number of ways.

- i. Shifts the IS curve to the left (since government spending is decreasing)
 - ii. The LM curve shifts to the right due to a decrease in money demand
 - iii. It decreases both wage and interest costs shifting aggregate supply down.
- Then, if the leftward shift in IS dominates the rightward shift in LM, then there is a decrease in aggregate demand. If the impact of a decrease in wage on government spending is large, or there is a substantial difference in the **MPC** for workers and

capitalists then the income policy will have the impact of decreasing both Price and output. Alternatively, if the shift in aggregate demand curve is small, then one may get an increase in output and employment. In the standard MDC model, on the other hand, a decrease in the rigid nominal wage would shift down aggregate supply curve; output unambiguously rises, while the price level unambiguously falls (see Figure 6.4).



An income policy that lowers the nominal wage affects LDC economies as follows:

- -Shifts the IS curve to the left (since government spending is decreasing),
- The LM curve shifts to the right due to a decrease in money demand.
- It decreases both wage and interest costs shifting **AS** down. **Then,**
- ✓ If the leftward shift in IS dominates the rightward shift in LM, then **AD** declines. If the impact of a decrease in wage on **G** is large, then the income policy will have the effect of decreasing both **P** & **Y**.
- ✓ Alternatively, if the shift in **AD** is small, then output & employment may increase. In the standard MDC model, on the other hand, a decrease in the rigid nominal wage would shift down **AS** curve; **output** unambiguously **rises**, while **Price** unambiguously **falls**.

Figure.7.4 Wage Restraint in LDCs

Conclusions

From the standard LDC macro model and the comparative static analyses of monetary, fiscal, exchange-rate and wage policies, one broad conclusion is clear: **the traditional stabilization policies have very different, and generally less desirable effects in LDCs than in advanced country models. Most notably, contractionary monetary policy is likely to fuel, rather than dampen, inflationary tendencies.** But there are other awkwardnesses, three of which merit attention.

- i. It is difficult to attack balance-of-payments crises, except through massive reduction of import demand.

- ii. All macro policies have effects on the government budget and the balance of payments, which affect the monetary base. This in turn severely limits policy-makers' ability to alter economic conditions in more than the very short run.
- iii. Most restrictive policy is likely to worsen the income distribution. These difficulties with short-run stabilization policy in LDCs indicate why policy-makers are- and indeed should be - so uncertain to undertake them. There are major conflicts between the means available to cure short-run crises and the long-run objectives of growth & equity.

7.4. Recent Macroeconomic Issues in Africa: The Structural Adjustment Program (SAP)

7.4.1. Introduction

- ✚ Structural adjustment program refers to the reorganization of institutions, economic activities, and the entire social system in line with the market-oriented economic system.
- ✚ This program focuses on two principal issues, namely **stabilization** and **adjustment**.
 - ✓ **Stabilization policies** emphasize on restoring macroeconomic balance, reducing government budget deficit, controlling the economy, and the like.
 - ✓ **Structural adjustment policies** focus on removing constraints on the supply side and pay a close attention to the production of export crops through a depreciation of the real exchange rate and other incentives.
- ❖ *The main objectives of SAP were revival of economic growth, reducing macroeconomic distortion, improving economic efficiency and resource allocation and expanding the productive capacity of the economy.*
- ✚ Structural adjustment program had three broad components, namely:
 - ✓ Expenditure-reducing policies,
 - ✓ Expenditure-switching policies and
 - ✓ Institutional policy reforms.

i. The Expenditure-Reducing Policies: also known as **demand management policies**.

- ✚ *They are expected to reduce spending levels in the economy.* Such policy measures include stringent monetary policy, i.e., tight money supply, credit control, and reduced public deficit. **These all aim to curtail domestic demand.**

ii. The Expenditure-Switching Policies, on the other hand, *aim to redirect productive resources from non-productive sectors to productive sectors or investments.* Such policies include exchange rate devaluation, trade interventions, taxes, tariff, and deregulation of prices.

iii. Institutional Policy Reforms: focus on *restructuring of institutions in order to enhance competitiveness and efficiency through market signals*. The required policy reforms entail trade liberalization, reduced role of the state in the economy, fiscal reform, privatization, financial market reform, and the like.

✚ Today, the aspects of underdevelopment in Africa look more serious than in any other regions of the world. The economic situation of Africa is quite dismal, and it looks especially miserable when compared to the economic achievements of the East Asian NICs. The average per capita GDP of East Asian NICs in 1995 exceeded \$10,000, while that of Sub-Saharan Africa (SSA) remained only around \$500. With these two groups at the opposite extreme, the differences in the Third World are now such that, as Barbara Stalling puts it, the composite term "Third World" has become an anachronism. Indeed, it might be said that Africa has now virtually become the poorest Fourth World.

Africa's economic performance has been very poor especially during the 1960's through 1990's. Moreover, many authors indicate that Africa is still seriously being marginalised from the world economy in the recent trends of globalisation. How can we account for this phenomenon? Over this question, the impacts of the Structural Adjustment Programmes (SAPs) implemented by most African states under the auspices of the IMF and the World Bank, are seriously analysed and discussed. It is because the SAPs have been the single most important economic policy during 1980's and 1990's that have determined the development course of SSA countries.

The purpose of this section is to look at the aspects of entrenched underdevelopment in SSA during recent decades, and to analyze the impacts of SAPs on these aspects. To do this, this essay will present some **general aspects of SAPs** first.

7.4.2. **What are the SAPs in Africa?**

✚ Structural adjustment in economic policies, by its own definition, means some measures to correct problematic sectors and redress the structure of a certain economy. However, after the 1980s, it has been used to refer to the specific sets of economic policies linked to the conditional loans of the IMF and the World Bank. So, Ankie Hoogvelt writes that "Structural adjustment is the generic term used to describe a package of measures which the IMF, the World Bank and individual Western aid

donors have persuaded many developing countries to adopt during the 1980s, ***in return for a new wave of loans***".

- ✚ Since the debt crisis broke in 1982, when Mexico first declared a moratorium on its international debt payments, the role of the IMF and the World Bank have become increasingly important in international financial markets, as private creditors hesitated to lend money to risky countries. This was especially the case with the Least Developed countries in Sub-Saharan Africa. *According to Hoogvelt, the 1980s saw 29 Sub-Saharan African countries accept the IMF/World Bank medicine.* The data of the IMF and the World Bank show that as of the end of 1992, 18 countries in SSA were still accepting the Structural Adjustment credits from the IMF, and 15 countries from the World Bank.
- ✚ The IMF and World Bank have linked severe conditionalities with their long-term, low-interest loans. They have demanded the borrowing countries to implement SAPs as a precondition for their lending. Most developing countries have accepted these conditionalities to get their badly needed financial resources. In this way, the IMF and the WB's intervention in the economic policies of the debtor countries became matched, perhaps even exceeded the direct administration of bygone colonial governments.

Consequently, the SAPs have affected the courses of development of the Third World greatly. And in recent years, many critics raise questions on the intention and results of the SAPs. The criticism especially gets harsh when it comes to the problems of SSA.

7.4.3. Main Tenets and Measures of the SAPs

The SAPs applied to various developing countries have had rather common features in their measures, under the apparent objectives in the context of international political economy. According to Adrian Leftwick, the aim of adjustment was to shatter the dominant postwar, state-led development paradigm and overcome the problems of developmental stagnation by promoting open and free competitive market economies, supervised by minimal states.

The IMF has focused on stabilization by austerity measures in fiscal and monetary policies, while the World Bank has emphasized structural adjustment by liberalization and privatization. Combined focuses of the two institutions have accompanied various measures, all of which have some common features despite the variety of economic situation of the debtor countries.

- *Michael Todaro summarizes the four basic components of the SAPs as follows:*
 1. *Abolition or liberalization of foreign exchange and import controls*

2. *Devaluation of the official exchange rate*
3. *A stringent domestic anti-inflation program consisting of control of bank credit, government deficit, wage increases, and dismantling of price control*
4. *Greater hospitality to foreign investment and international commerce.*

🌍 ***Underlying all these components were the neo-liberal tenets of the IMF and the World Bank that the "principle of free and competitive market" should work throughout all the developing countries.*** Considering the Cold War rivalry of the 1980s, it might be said that the SAPs had a clear message in the context of international political economy: that is, to incorporate all the developing economies into the system of Western capitalism.

The IMF and WB have explained their packages purely in terms of economic efficiency. However, most analysts could see the messages entailed in the packages from the very beginning. So, Samir Amin criticized that "the Berg Report", prepared by the World Bank in 1981 as a theoretical basis of the SAPs, meant nothing but a representation of the interests of Western capitalists. And now, aside from the political intention of the SAPs, their results of the past two decades especially in the Sub-Saharan Africa are severely criticised.

7.4.4. Economic Performance of Sub-Saharan Africa

Seeing from various statistical data, the situation of the Sub-Saharan economy after the implementation of the program was quite miserable. The World Bank data show that, Africa is the center of "***the poverty belt***" consisting of the poorest countries with less than 600 dollars of per capita GDP. Michael Chege writes that underdevelopment seems determined to make its last stand in Africa. According to Chege, the overall record of economic growth in SSA was very poor during the past decades. After registering average annual per capita growth rate of 2.9% in the immediate post independence period (1965-73), the rate has since experienced consistent decline, stagnating at 0.1% between 1973-80, and falling by 0.8% on average between 1980-92. At best, therefore, most Africans are as poor as they were at independence 30 years ago.

Chege shows the cases of 30 countries representing 80% of the Sub-Saharan population. Among them only 3 countries – Botswana, Lesotho, and Mauritius – managed a per capita growth record over 3% during the 1970-90 period. Some other countries such as Kenya, Ivory Coast, Malawi, Rwanda managed moderate growth records throughout the 1970s, but they have also succumbed to various negative factors over the recent past.

Ibrahim A. Elbadawi and Benno J. Ndulu show similar analysis. They show that, out of 32 SSA states considered in their joint study, only 6 managed to achieve consecutive positive rates of growth over the two decades of 1970-90. All other SSA countries have lost at least

one decade, and a significant number of these countries have had two consecutive, negative average, decadal growth rates. They show even more alarming fact that none of 29 non-SSA countries in their sample registered negative growth over the two consecutive decades.

The declining per capita income statistics from Africa are accompanied by a wide set of deteriorating sectoral indicators. *According to Michael Chege, the growth rates of African agriculture and industry also significantly slowed down during the recent two decades.* As a result Africa's world market share for its three main agricultural products – coffee, tea, and cotton – shrank by 13, 33, and 29% respectively between 1970 and 84. In the industrial sector, SSA's contribution to global manufacturing value added fell to 0.5% in 1988.

Social indicators also imply a depressing picture. Africa's infant mortality rates in 1992 were the highest of any region in the world, at 107 death per 1,000 live births, compared with 93 for South Asia, and 7 in Hong Kong. Even though some health standards such as rural child immunization improved under the auspices of UN Children's Fund (UNICEF), educational indicators generally aggravated during the 1980s.

Besides these, S.M. Wangwe and Flora Musonda show how much Africa is being marginalized from the world economy especially in the trends of globalization. They say that the weakness of Africa's global integration is evident in its falling share of world trade, investment and Information Technology. Africa's share of world trade varied from 4.1% to 4.9% during the 1960-70s and declined to 2-3% in 1990s. Its share of Foreign Direct Investment in the developing country-total fell from 11% during 1986-90 to 6% during 1991-93. Overall investment in technology is very low. With the exception of South Africa, the African countries' expenditure on telecommunications equipment in 1986 represented only 0.7% of world spending. Moreover, only a few countries were the main spenders.

7.4.5. Causes of the Economic Failure

- ✚ There are some arguments and disputes about the causes of the poor economic performance of Africa. Some scholars, especially from the West, indicate internal factors of Africa such as the poor governance and mismanagement as the most important causes. Michael Chege seems to be standing on this position when he says that political instability and mismanagement are the best predictors of economic regress in Africa.
- ✚ He argued that "For the purposes of economic development, what Sub-Saharan Africa really needs is the replacement of current administrative chaos, managerial incompetence, and corruption with efficient and reliable governance structures."

This position seems to reflect the reality somehow, but holds more importance as an official position of the IMF and the World Bank. The World Bank has claimed that the failure of the SAPs in Sub-Saharan Africa is attributed to the failure of the governance in the countries there. It claimed that the countries which implemented the policies only partially or not at all, or backslided, have paid the price with negligible or deteriorating growth. On the other hand, it claimed, six countries which have followed the guideline faithfully achieved a success by getting their macro-economic fundamentals right. Those six countries were Ghana, Tanzania, Gambia, Burkina Faso, Nigeria, and Zimbabwe.

In the 1990s, especially after the collapse of the Former USSR and socialist regimes in Eastern Europe, such position has had considerable influence on the new aid policies of the Western industrialized countries. So, France and Britain, the leading source of official development assistance to Africa, officially announced that their external aid would henceforth be restricted to countries that observe democratic practice and human rights.

Ankie Hoogvelt criticises this position of 'good governance' or 'democracy initiative' by raising questions as follows:

"How does this 'democracy is good, state is bad' agenda assist economic reform? The link between democracy and debt restructuring in the case of Africa is the more puzzling since it assumes a positive correlation between democracy and economic development. But such assertions are not made by the leading international organizations when commenting on and explaining the success of economic development in the newly-industrializing countries of East Asia. ... in relation to that part of the world the new orthodoxy singles out the virtue of strong, authoritarian states as the most important factor contributing to the development of the region."

Hoogvelt cites some authors who argue that the emphasis on quality of governance merely serves as an efficient means of focusing responsibility on government of developing countries, both for past ills and for implementation of reform packages. Furthermore, he argues that in many African countries, the imposition of the neo-liberal orthodoxy, including privatization of the public sector, the emasculation of the state apparatus and the insistence on electoral reform, has directly contributed to the descent into anarchy and civil wars.

Each position stated above reflects some parts of the reality. As Wangwe and Musonda put it, the African countries' responses to SAPs differ depending on a number of factors. ***However, in any sense, the impacts of the SAPs cannot be ignored with regard to the deterioration of the African economy especially during the last two decades (1980's and 1990's).***

Many authors indicate that during the past decades, the SAPs were in fact the single most important development paradigm in SSA.

- Barbara Stallings describes a crucial narrowing of development strategy choice that took place across the Third World after 1980s.
- Caroline Thomas describes how the domination of neo-liberal economic policies has affected the development of the Third World in recent years.
- Colin Parkins describes that the debt crisis of 1980s has effectively ruled out alternative responses to economic development in the Third World.
- ✚ All these authors indicate directly or indirectly that the SAPs have been the single dominant economic policy especially in SSA. And the scope and detail of the combined IMF conditionality rules and the World Bank's structural adjustment contracts have amounted to such a degree of economic intervention.
- ✚ Considering all these facts stated above, the IMF and the World Bank can never be free from responsibility on the economic deterioration of Sub-Saharan Africa. It might be said that inadequacy of their policies entailed in the packages of SAPs have greatly affected the aggravation of SSA economy and have entrenched underdevelopment there.

7.4.6. Failure of the SAPs

As we have seen earlier, the failure of the SSA economy in decades after the program is too clear to remain a room for argument over the facts themselves. The IMF and the World Bank try to evade from criticism on the misguided effects of SAPs by attributing the failure to various other factors rather than the impacts of SAPs.

The failure of SAPs in SSA is analysed by many authors. Kevin Watkins of **British** summarised the whole process simply in which SAPs have devastated SSA economy and societies as follows:

"... the application of stringent monetary policies, designed to reduce inflation through high interest rates, has undermined investment and employment. At the same time, poorly planned trade-liberalization measures have exposed local industries to extreme competition. Contrary to World Bank and IMF claims, the position of the poor and most vulnerable sections of society have all too often been undermined by the deregulation of labour markets and erosion of social welfare provisions, and by declining expenditures on health and education. Women have suffered in extreme form. The erosion of health expenditure has increased the burdens they carry as carrier, while falling real wages and rising unemployment have forced women into multiple low-wage employment in the informal sector."

Now let us look at this process more in detail.

- ✚ In the first place, the external liberalization of SAPs has brought seriously counterproductive effects in many countries. Trade liberalization has opened door to imports of formerly restricted consumer goods, leading to the destruction of already weak African industries. Cheaper imports from technologically advanced countries, especially South-East Asian countries, have made rapid inroads into the African basic consumer goods markets. With this, the devaluation of national currencies has made imports of capital goods and raw materials more expensive, thereby raising the cost of production.
- ✚ On the other hand, the export specialization policies of SAPs have caused a flooding of primary commodity markets, which forced prices downwards. During the 1980s, the terms of trade for SSA commodities fell more rapidly than any other region of the globe. The effects of financial liberalization measures have been also questionable, since foreign banks have been hesitant to finance development in Africa.
- ✚ Finally, forced privatisation and financial cutbacks on social security have seriously aggravated the problem of unemployment, and have lowered the quality of social services such as health and education.
 - ***Overall, the SAPs in Sub-Saharan Africa have produced too much negative effects rather than the positive effects they originally targeted.***

Conclusion

The SAPs sponsored by the IMF and World Bank have been criticised from the beginning due to the political intention of the packages. Recently, the focus of criticism seems to have moved to the real effects what the SAPs have brought to the economies of developing countries, especially in Sub-Saharan Africa. A lot of evidence show that they have not only failed to revive the African economy but have rather entrenched the underdevelopment and poverty there.

The negative impacts of SAPs have affected, among others, the poor and weak masses of the African societies most seriously. Abolition or reduction of various subsidies on basic necessities exacerbated the poor conditions of socially weak people. Control of wage increases, privatization, and trade liberalization measures have all brought combined negative impacts on the quality of life of the people there.

Hoogvelt argues that despite the failure of the SAPs in Africa, they have been a resounding success when measured in terms of the process of globalization. The SAPs have helped to tie the physical economic resources of the African region more tightly into servicing the global system, while oiling the financial machinery by which wealth can be transported out of Africa and into the global system. This is an ironical expression of the failure of SAPs to integrate Africa effectively into the global economy. We have already seen the aspect of marginalization of Africa from the world economy earlier in this essay. Hoogvelt seems to say that by marginalizing Africa, the global capitalist system has made it just serve the global system, rather than participate in the system as an actor.

Seeing from all the aspects we have analyzed as far, the failure of SAPs in Sub-Saharan Africa is quite clear. Then, what do these findings mean? It means that, at this point, the IMF and the World Bank should seriously reconsider the neo-liberal principle and directions of SAPs. How can the principle of free and competitive market unconditionally be applied to the poor developing countries that have no competitive industrial bases? They should not forget that even the now industrialized Western countries advocated protectionism for their infant industries during the early days of their industrialization.

And again, the target of macro-economic stabilisation at the expense of social securities should be also mediated and redirected. What are the virtues of all economic policies if they bring even more sorrow and agonies of the already poor and weak people? A serious aspect of this social impoverishment is that it does not show any sign of improvement over a prolonged time, as the overall capacity of SSA economy has consistently declined.

It seems that the IMF and the World Bank recently are more susceptible to criticism on SAPs than in the past. There are still strong criticism that these international institutions are serving the interests of Western capitalists rather than to serve the human welfare and prosperity. It might be said that this is an inevitably political aspect of international economic institutions, since the Western industrialized countries are the major shareholders of all these institutions.